

Astra

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Chapter 1

Namespace Index

1.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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Chapter 3

Class Index

3.1 Class List

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Chapter 5

Namespace Documentation

5.1 Employees Namespace Reference

[Employees](#) namespace.

Classes

- class [Delete](#)
CRUD
- class [EmployeeChecker](#)
Check the passed employees data.
- struct [EmployeeInfo](#)
EmployeeInfo struct.
- class [EmployeesUICharts](#)
- class [Insert](#)
Insert new employee.
- class [Select](#)
CRUD.
- class [Setup](#)
Generale database setup.
- class [Update](#)
CRUD.

Enumerations

- enum [EmployeeQueueFlags_integers](#) { [EMPLOYEE_ID](#) , [SALARY](#) , [PHONE_NUMBER](#) }
EmployeeQueueFlags_integers enum.
- enum [EmployeeQueueFlags_strings](#) {
[USERNAME](#) , [FIRSTNAME](#) , [LASTNAME](#) , [EMAIL](#) ,
[JOBTITLE](#) , [STATUS](#) , [PASSWORD](#) , [ADDRESS](#) ,
[DEPARTMENT](#) , [PROFILE_IMAGE_PATH](#) }
EmployeeQueueFlags_strings enum.
- enum [EmployeeQueueFlags_dates](#) { [HIRE_DATE](#) , [BIRTHDATE](#) }
EmployeeQueueFlags_dates enum.
- enum [EmployeeQueueFlags_timestamps](#) { [LAST_LOGIN](#) , [LAST_PASSWORD_RESET](#) }

- EmployeeQueueFlags_timestamps enum.*
- enum [EmployeeQueueFlags_blobs](#)
EmployeeQueueFlags_blobs enum.
- enum [EmployeeCheckerFlags](#) {
 [EMPTY](#) = -1 , [OK](#) , [REFUSED](#) , [UNIQUE_EMAIL](#) ,
 [UNIQUE_EMAIL_WITH_EXCEPTION](#) }
- EmployeeCheckerFlags enum.*
- enum [EmployeeStatusFlags](#) { [ALL](#) , [ACTIVE](#) , [INACTIVE](#) , [SUSPENDED](#) }
- enum [EmployeeDepartmentFlags](#) {
 [COMMERCIAL](#) , [SHOPS](#) , [PARTNERS](#) , [EVENTS](#) ,
 [PERSONALS](#) , [EMPLOYEES](#) }
- EmployeeDepartmentFlags enum.*

Functions

- std::string [EmployeeStatusString](#) (const [EmployeeStatusFlags](#) &)
Return the corresponding [EmployeeStatusFlags](#) string.

5.1.1 Detailed Description

[Employees](#) namespace.

5.1.2 Enumeration Type Documentation

5.1.2.1 EmployeeCheckerFlags

enum [Employees::EmployeeCheckerFlags](#)

[EmployeeCheckerFlags](#) enum.

Enumerator

EMPTY	-1
OK	0
REFUSED	1
UNIQUE_EMAIL	2
UNIQUE_EMAIL_WITH_EXCEPTION	3

5.1.2.2 EmployeeDepartmentFlags

enum [Employees::EmployeeDepartmentFlags](#)

[EmployeeDepartmentFlags](#) enum.

Enumerator

COMMERCIAL	0
SHOPS	1
PARTNERS	2
EVENTS	3
PERSONALS	4
EMPLOYEES	5

5.1.2.3 EmployeeQueueFlags_blobs

enum `Employees::EmployeeQueueFlags_blobs`

`EmployeeQueueFlags_blobs` enum.

5.1.2.4 EmployeeQueueFlags_dates

enum `Employees::EmployeeQueueFlags_dates`

`EmployeeQueueFlags_dates` enum.

Enumerator

HIRE_DATE	0
BIRTHDATE	1

5.1.2.5 EmployeeQueueFlags_integers

enum `Employees::EmployeeQueueFlags_integers`

`EmployeeQueueFlags_integers` enum.

Enumerator

EMPLOYEE_ID	0
SALARY	1
PHONE_NUMBER	2

5.1.2.6 EmployeeQueueFlags_strings

enum `Employees::EmployeeQueueFlags_strings`

`EmployeeQueueFlags_strings` enum.

Enumerator

USERNAME	0
FIRSTNAME	1
LASTNAME	2
EMAIL	3
JOBTITLE	4
STATUS	5
PASSWORD	6
ADDRESS	7
DEPARTMENT	8
PROFILE_IMAGE_PATH	9

5.1.2.7 EmployeeQueueFlags_timestamps

enum `Employees::EmployeeQueueFlags_timestamps`

`EmployeeQueueFlags_timestamps` enum.

Enumerator

LAST_LOGIN	0
LAST_PASSWORD_RESET	1

5.1.2.8 EmployeeStatusFlags

```
enum Employees::EmployeeStatusFlags
```

Enumerator

ALL	0
ACTIVE	1
INACTIVE	2
SUSPENDED	3

5.1.3 Function Documentation**5.1.3.1 EmployeeStatusString()**

```
std::string Employees::EmployeeStatusString (  
    const EmployeeStatusFlags & status)
```

Return the corresponding `EmployeeStatusFlags` string.

Parameters

<i>status</i>	{const <code>EmployeeStatusFlags</code> &}
---------------	--

Returns

std::string

5.2 Ui Namespace Reference

`Ui` namespace.

5.2.1 Detailed Description

`Ui` namespace.

Chapter 6

Class Documentation

6.1 CSVGenerator Class Reference

[CSVGenerator](#) class.

```
#include <CSVGenerator.hpp>
```

Static Public Member Functions

- static const unsigned int [generateCSV](#) (const std::string &, const std::vector< std::vector< std::string > > &)
Generate a .csv file in filePath using table

6.1.1 Detailed Description

[CSVGenerator](#) class.

6.1.2 Member Function Documentation

6.1.2.1 generateCSV()

```
const unsigned int CSVGenerator::generateCSV (  
    const std::string & filePath,  
    const std::vector< std::vector< std::string > > & table) [static]
```

Generate a .csv file in filePath using table

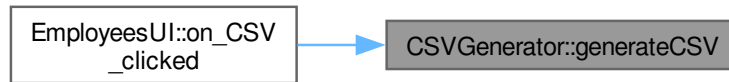
Parameters

<i>filePath</i>	{const std::string &}
<i>table</i>	{const std::vector<std::vector<std::string>> &}

Returns

{const unsigned int}

Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/CSVGenerator.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/CSVGenerator.cpp](#)

6.2 Database Class Reference

[Database](#) class.

```
#include <Database.hpp>
```

Public Member Functions

- [Database](#) ()
- [Database](#) (const char *, const char *, const char *)
- [~Database](#) ()
Destroy the [Database::Database](#) object.
- void [execute](#) (const std::string &query, const [SqlParam](#) &, int &affectedRows)
Executes INSERT|UPDATE|DELETE SQL query and update the affectedRow count.
- std::vector< [SqlParam](#) > [execute](#) (const std::string &, const [SqlParam](#) ¶ms)
Executes SELECT SQL query and return the results

Private Member Functions

- void [setSqlParams](#) (Statement *, const [SqlParam](#) &)
Set the SQL parameters into stmt

Private Attributes

- Environment * [env](#)
- Connection * [conn](#)

6.2.1 Detailed Description

[Database](#) class.

6.2.2 Constructor & Destructor Documentation

6.2.2.1 Database() [1/2]

```
Database::Database ()
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.2.2.2 Database() [2/2]

```
Database::Database (  
    const char * username,  
    const char * password,  
    const char * database)
```

6.2.2.3 ~Database()

```
Database::~Database ()
```

Destroy the [Database::Database](#) object.

Cleans up resources

6.2.3 Member Function Documentation

6.2.3.1 execute() [1/2]

```
std::vector< SqlParam > Database::execute (
    const std::string & query,
    const SqlParam & params)
```

Executes `SELECT` SQL query and return the results

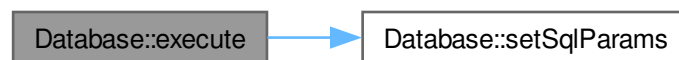
Parameters

<i>query</i>	{const std::string &}
<i>params</i>	{const SqlParam &}

Returns

std::vector<std::vector<std::string>>

Here is the call graph for this function:



6.2.3.2 execute() [2/2]

```
void Database::execute (
    const std::string & query,
    const SqlParam & params,
    int & affectedRows)
```

Executes `INSERT|UPDATE|DELETE` SQL query and update the `affectedRow` count.

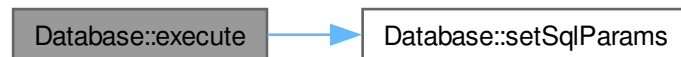
Parameters

<i>query</i>	{const std::string &}
<i>params</i>	{const SqlParam &}
<i>affectedRows</i>	{int &} -> Default:0

Returns

void

Here is the call graph for this function:

**6.2.3.3 setSqlParams()**

```
void Database::setSqlParams (  
    Statement * stmt,  
    const SqlParam & params) [inline], [private]
```

Set the SQL parameters into `stmt`

Helper function

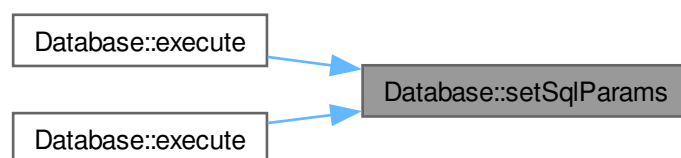
Parameters

<i>stmt</i>	{Statement *}
<i>params</i>	{const SqlParam &}

Returns

void

Here is the caller graph for this function:



6.2.4 Member Data Documentation

6.2.4.1 conn

```
Connection* Database::conn [private]
```

6.2.4.2 env

```
Environment* Database::env [private]
```

The documentation for this class was generated from the following files:

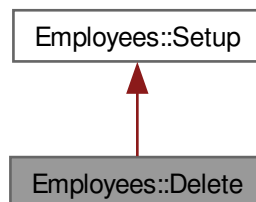
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Database.hpp
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Database.cpp

6.3 Employees::Delete Class Reference

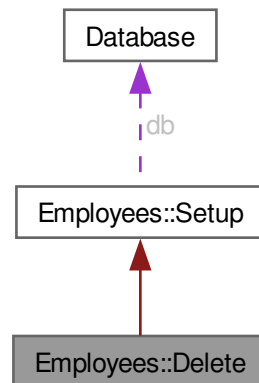
CRUD

```
#include <Employees.hpp>
```

Inheritance diagram for Employees::Delete:



Collaboration diagram for Employees::Delete:



Public Member Functions

- `int del (const Employees::EmployeeInfo< std::string > &)`
Delete user where info

Additional Inherited Members

Private Member Functions inherited from Employees::Setup

- `Setup ()`
Construct a new Employees::Setup::Setup object.
- `virtual ~Setup ()`
Destroy the Employees::Setup::Setup object.

Private Attributes inherited from Employees::Setup

- `Database * db`

6.3.1 Detailed Description

CRUD

6.3.2 Member Function Documentation

6.3.2.1 del()

```
int Employees::Delete::del (
    const Employees::EmployeeInfo< std::string > & info)
```

Delete user where info

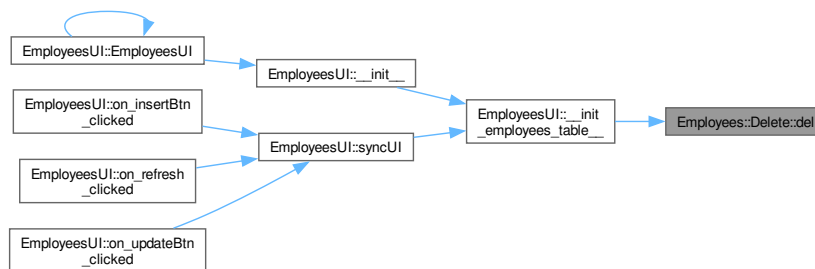
Parameters

<i>info</i>	{const Employees::EmployeeInfo &}
-------------	---

Returns

int

Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp

6.4 EmailAuth Struct Reference

[EmailAuth](#) struct.

```
#include <EmailSender.hpp>
```

Public Member Functions

- [EmailAuth](#) ()
Construct a new [EmailAuth::EmailAuth](#) object.
- [EmailAuth](#) (const std::string, const std::string, const std::string _mail_server=std::getenv("MAIL_SERVER"))

Public Attributes

- std::string [from_addr](#)
- std::string [from_app_password](#)
- std::string [mail_server](#)

6.4.1 Detailed Description

[EmailAuth](#) struct.

Examples

[Simple](#).

6.4.2 Constructor & Destructor Documentation

6.4.2.1 EmailAuth() [1/2]

```
EmailAuth::EmailAuth () [explicit]
```

Construct a new [EmailAuth::EmailAuth](#) object.

6.4.2.2 EmailAuth() [2/2]

```
EmailAuth::EmailAuth (  
    const std::string _from_addr,  
    const std::string _from_app_password,  
    const std::string _mail_server = std::getenv("MAIL_SERVER")) [explicit]
```

6.4.3 Member Data Documentation

6.4.3.1 from_addr

```
std::string EmailAuth::from_addr
```

6.4.3.2 from_app_password

```
std::string EmailAuth::from_app_password
```

6.4.3.3 mail_server

```
std::string EmailAuth::mail_server
```

The documentation for this struct was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/EmailSender.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/EmailSender.cpp](#)

6.5 EmailData Struct Reference

[EmailData](#) struct.

```
#include <EmailSender.hpp>
```

Public Member Functions

- [EmailData](#) (const std::string, const std::string, const std::string, const std::vector< std::string > _cc_addr={}, const std::vector< std::string > _attachments={})

Public Attributes

- std::string [to_addr](#)
- std::string [subject](#)
- std::string [body](#)
- std::vector< std::string > [cc_addr](#)
- std::vector< std::string > [attachments](#)

6.5.1 Detailed Description

[EmailData](#) struct.

Examples

[Simple](#).

6.5.2 Constructor & Destructor Documentation

6.5.2.1 EmailData()

```
EmailData::EmailData (
    const std::string _to_addr,
    const std::string _subject,
    const std::string _body,
    const std::vector< std::string > _cc_addr = {},
    const std::vector< std::string > _attachments = {}) [explicit]
```

6.5.3 Member Data Documentation

6.5.3.1 attachments

```
std::vector<std::string> EmailData::attachments
```

6.5.3.2 body

```
std::string EmailData::body
```


6.5.3.3 cc_addr

```
std::vector<std::string> EmailData::cc_addr
```

6.5.3.4 subject

```
std::string EmailData::subject
```

6.5.3.5 to_addr

```
std::string EmailData::to_addr
```

The documentation for this struct was generated from the following files:

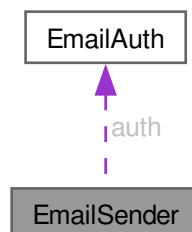
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/EmailSender.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/EmailSender.cpp](#)

6.6 EmailSender Class Reference

[EmailSender](#) class.

```
#include <EmailSender.hpp>
```

Collaboration diagram for EmailSender:



Public Member Functions

- [EmailSender](#) (const [EmailAuth](#))
Construct a new *EmailSender::EmailSender* object.
- [~EmailSender](#) ()
Destroy the *EmailSender::EmailSender* object.
- int [send](#) ([EmailData](#))
Send the given data using SMTP protocol (support it with libcurl)

Private Attributes

- [EmailAuth](#) * [auth](#)

6.6.1 Detailed Description

[EmailSender](#) class.

Examples

[Simple](#).

6.6.2 Constructor & Destructor Documentation

6.6.2.1 EmailSender()

```
EmailSender::EmailSender (  
    const EmailAuth _emailAuth) [explicit]
```

Construct a new [EmailSender::EmailSender](#) object.

Parameters

<code>_emailAuth</code>	{const EmailAuth }
-------------------------	------------------------------------

6.6.2.2 ~EmailSender()

```
EmailSender::~~EmailSender ()
```

Destroy the [EmailSender::EmailSender](#) object.

6.6.3 Member Function Documentation

6.6.3.1 send()

```
int EmailSender::send (  
    EmailData data)
```

Send the given data using SMTP protocol (support it with libcurl)

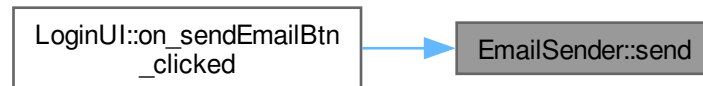
Parameters

<code>data</code>	EmailData
-------------------	---------------------------

Returns

int

Here is the caller graph for this function:



6.6.4 Member Data Documentation

6.6.4.1 auth

```
EmailAuth* EmailSender::auth [private]
```

The documentation for this class was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/[EmailSender.hpp](#)
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/[EmailSender.cpp](#)

6.7 Employees::EmployeeChecker Class Reference

Check the passed employees data.

```
#include <Employees.hpp>
```

Static Public Member Functions

- static std::pair< [Employees::EmployeeCheckerFlags](#), const char * > [is_valid](#) (const std::string &, const std::string &, const std::string &, const std::string &, std::string status, const std::pair< [Employees::EmployeeCheckerFlags](#), const char * > &flag={[Employees::EmployeeCheckerFlags::EMPTY](#), ""})
Return {[Employees::EmployeeCheckerFlags::OK](#), ""} if the passed Employee data are valid, otherwise return {[Employees::EmployeeCheckerFlags::REFUSED](#), "ERROR_MESSAGE"}

Static Private Member Functions

- static bool [is_validEmail](#) (const std::string &)
Return true if the given string email matches with the given regular expression, otherwise return false
- static bool [is_empty](#) (const std::string &, const std::string &, const std::string &, const std::string &, const std::string &)
Return true if the passed Employee data empty, otherwise return false
- static bool [is_alphaDigit](#) (const std::string &)
Return true if the passed string s is have digit characters, otherwise return false

6.7.1 Detailed Description

Check the passed employees data.

6.7.2 Member Function Documentation

6.7.2.1 is_alphaDigit()

```
bool Employees::EmployeeChecker::is_alphaDigit (
    const std::string & s) [static], [private]
```

Return `true` if the passed string `s` is have digit characters, otherwise return `false`

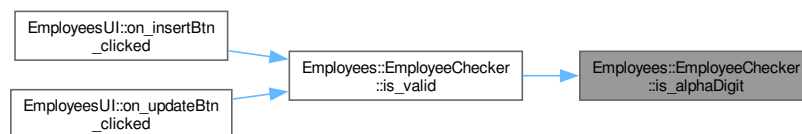
Parameters

<code>s</code>	{const std::string &}
----------------	-----------------------

Returns

`bool`

Here is the caller graph for this function:



6.7.2.2 is_empty()

```
bool Employees::EmployeeChecker::is_empty (
    const std::string & email,
    const std::string & firstname,
    const std::string & lastname,
    const std::string & password,
    const std::string & department) [static], [private]
```

Return `true` if the passed Employee data empty, otherwise return `false`

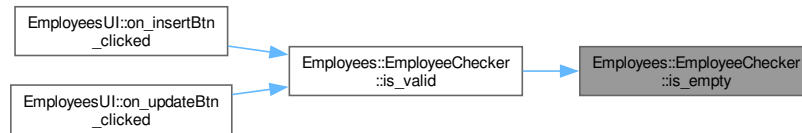
Parameters

<code>username</code>	{const std::string &}
<code>email</code>	{const std::string &}
<code>firstname</code>	{const std::string &}
<code>lastname</code>	{const std::string &}
<code>password</code>	{const std::string &}
<code>department</code>	{const std::string &}

Returns

bool

Here is the caller graph for this function:



6.7.2.3 is_valid()

```

std::pair< Employees::EmployeeCheckerFlags, const char * > Employees::EmployeeChecker::is_↵
valid (
    const std::string & email,
    const std::string & firstname,
    const std::string & lastname,
    const std::string & password,
    const std::string & department,
    std::string status,
    const std::pair< Employees::EmployeeCheckerFlags, const char * > & flag = {Employees::EmployeeCh
[static]

```

Return {`Employees::EmployeeCheckerFlags::OK`, ""} if the passed Employee data are valid, other-
wise return {`Employees::EmployeeCheckerFlags::REFUSED`, "ERROR_MESSAGE"}

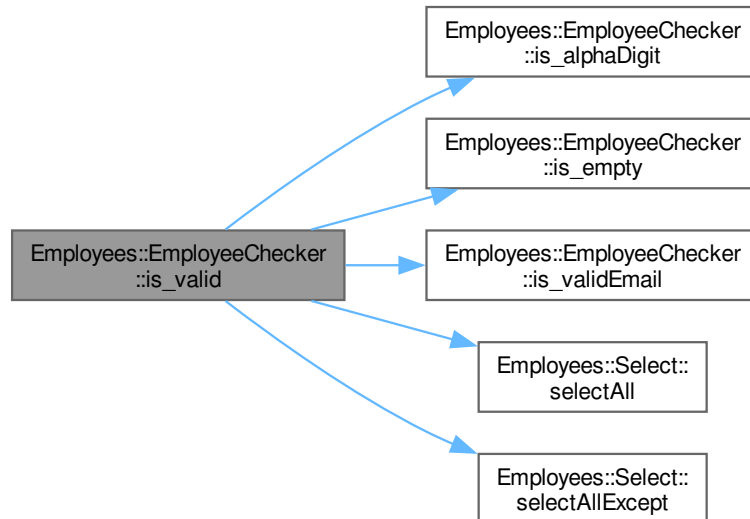
Parameters

<i>username</i>	{const std::string &}
<i>email</i>	{const std::string &}
<i>firstname</i>	{const std::string &}
<i>lastname</i>	{const std::string &}
<i>password</i>	{const std::string &}
<i>department</i>	{const std::string &}

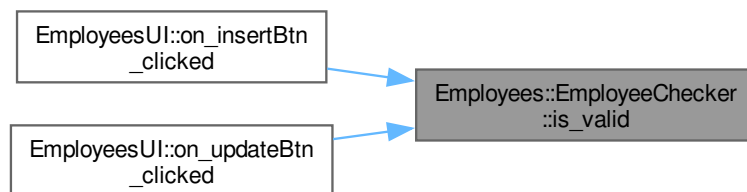
Returns

`std::pair<Employees::EmployeeCheckerFlags, const char *>`

Here is the call graph for this function:



Here is the caller graph for this function:



6.7.2.4 is_validEmail()

```

bool Employees::EmployeeChecker::is_validEmail (
    const std::string & email) [static], [private]
  
```

Return `true` if the given string email matches with the given regular expression, otherwise return `false`

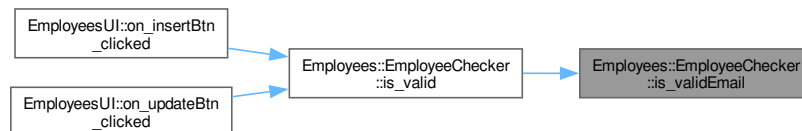
Parameters

<i>email</i>	{const std::string &}
--------------	-----------------------

Returns

bool

Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp](#)

6.8 Employees::EmployeeInfo< T > Struct Template Reference

[EmployeeInfo](#) struct.

```
#include <Employees.hpp>
```

Public Member Functions

- [EmployeeInfo](#) (const T &, const [Employees::EmployeeQueueFlags_integers](#) &)
- [EmployeeInfo](#) (const T &, const [Employees::EmployeeQueueFlags_strings](#) &)
- [EmployeeInfo](#) (const T &, const [Employees::EmployeeQueueFlags_dates](#) &)
- [EmployeeInfo](#) (const T &, const [Employees::EmployeeQueueFlags_timestamps](#) &)

Public Attributes

- T [data](#)
- std::string [arg](#)

6.8.1 Detailed Description

```
template<typename T>
struct Employees::EmployeeInfo< T >
```

[EmployeeInfo](#) struct.

6.8.2 Constructor & Destructor Documentation

6.8.2.1 EmployeeInfo() [1/4]

```
template<typename T>
Employees::EmployeeInfo< T >::EmployeeInfo (
    const T & ,
    const Employees::EmployeeQueueFlags_integers & ) [explicit]
```

6.8.2.2 EmployeeInfo() [2/4]

```
template<typename T>
Employees::EmployeeInfo< T >::EmployeeInfo (
    const T & ,
    const Employees::EmployeeQueueFlags_strings & ) [explicit]
```

6.8.2.3 EmployeeInfo() [3/4]

```
template<typename T>
Employees::EmployeeInfo< T >::EmployeeInfo (
    const T & ,
    const Employees::EmployeeQueueFlags_dates & ) [explicit]
```

6.8.2.4 EmployeeInfo() [4/4]

```
template<typename T>
Employees::EmployeeInfo< T >::EmployeeInfo (
    const T & ,
    const Employees::EmployeeQueueFlags_timestamps & ) [explicit]
```

6.8.3 Member Data Documentation

6.8.3.1 arg

```
template<typename T>
std::string Employees::EmployeeInfo< T >::arg
```

6.8.3.2 data

```
template<typename T>
T Employees::EmployeeInfo< T >::data
```

The documentation for this struct was generated from the following file:

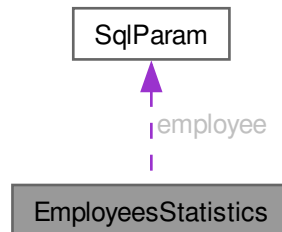
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp

6.9 EmployeesStatistics Class Reference

[EmployeesStatistics](#) class.

```
#include <EmployeesStatistics.hpp>
```

Collaboration diagram for EmployeesStatistics:



Public Member Functions

- [EmployeesStatistics](#) (const [SqlParam](#) &)
- const std::vector< unsigned int > [getStatusStats](#) ()
- const std::vector< double > [getDepartmentStats](#) ()
- const std::vector< double > [getSalaryStats](#) ()

Private Attributes

- [SqlParam](#) [employee](#)
- size_t [employees_length](#)
- std::unordered_map< std::string, unsigned int > [statuses](#)
- std::unordered_map< std::string, double > [departments](#)
- std::unordered_map< std::string, double > [salaries](#)

6.9.1 Detailed Description

[EmployeesStatistics](#) class.

6.9.2 Constructor & Destructor Documentation

6.9.2.1 EmployeesStatistics()

```
EmployeesStatistics::EmployeesStatistics (
    const SqlParam & _employee) [explicit]
```

Here is the call graph for this function:

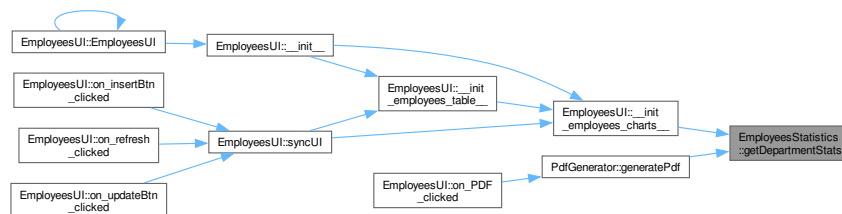


6.9.3 Member Function Documentation

6.9.3.1 getDepartmentStats()

```
const std::vector< double > EmployeesStatistics::getDepartmentStats ()
```

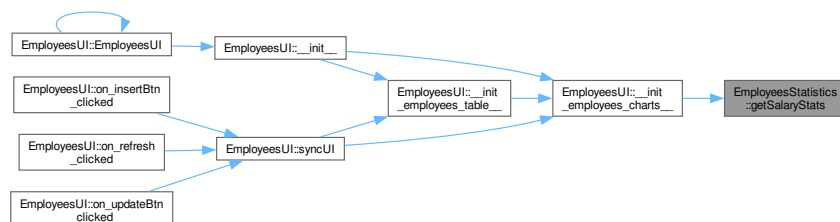
Here is the caller graph for this function:



6.9.3.2 getSalaryStats()

```
const std::vector< double > EmployeesStatistics::getSalaryStats ()
```

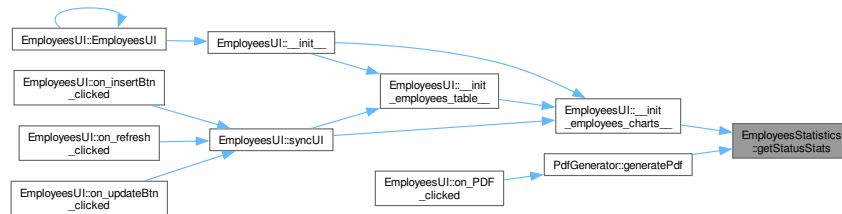
Here is the caller graph for this function:



6.9.3.3 getStatusStats()

```
const std::vector< unsigned int > EmployeesStatistics::getStatusStats ()
```

Here is the caller graph for this function:



6.9.4 Member Data Documentation

6.9.4.1 departments

```
std::unordered_map<std::string, double> EmployeesStatistics::departments [private]
```

6.9.4.2 employee

```
SqlParam EmployeesStatistics::employee [private]
```

6.9.4.3 employees_length

```
size_t EmployeesStatistics::employees_length [private]
```

6.9.4.4 salaries

```
std::unordered_map<std::string, double> EmployeesStatistics::salaries [private]
```

6.9.4.5 statuses

```
std::unordered_map<std::string, unsigned int> EmployeesStatistics::statuses [private]
```

The documentation for this class was generated from the following files:

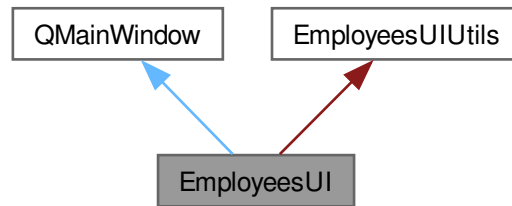
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/EmployeesStatistics.hpp
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/EmployeesStatistics.cpp

6.10 EmployeesUI Class Reference

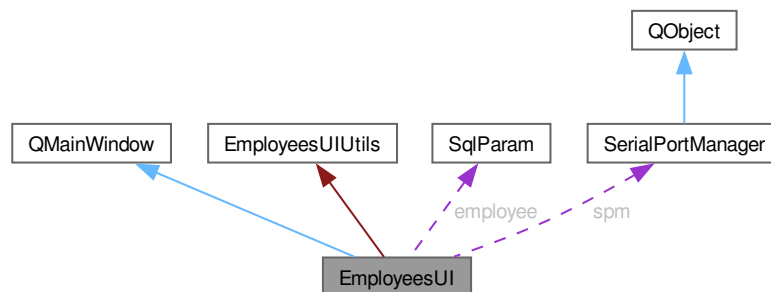
[EmployeesUI](#) class.

```
#include <EmployeesUI.hpp>
```

Inheritance diagram for EmployeesUI:



Collaboration diagram for EmployeesUI:



Public Member Functions

- [EmployeesUI](#) ([SqlParam](#), QWidget *parent=nullptr)
Construct a new [EmployeesUI::EmployeesUI](#) object.
- [~EmployeesUI](#) ()
Destroy the [EmployeesUI::EmployeesUI](#) object.
- void [set_employee](#) (const [SqlParam](#) &)
Set employee data.

Private Slots

- void [on_show_clicked](#) (bool)
Listen to "Show More/Less" button click action inside ui->Form QGroupBox.
- void [on_show_2_clicked](#) (bool)
Listen to "Show More/Less" button click action inside ui->updateForm QGroupBox.
- void [on_Add_button_clicked](#) ()
Listen to "Add" button click action.
- void [on_Cancel_form_clicked](#) ()
Listen to "Cancel" button click action inside ui->Form QGroupBox.
- void [on_Cancel_form_2_clicked](#) ()
Listen to "Cancel" button click action inside ui->updateForm QGroupBox.
- void [on_insertBtn_clicked](#) ()
Listen to "Add" employee button click action.
- void [on_updateBtn_clicked](#) ()
Listen to "Update" employee button click action.
- void [on_profileImageInsert_clicked](#) ()
Listen to "Profile Image" employee button click action.
- void [on_profileImageInsert_2_clicked](#) ()
Listen to "Profile Image" employee button click action.
- void [on_Filter_currentTextChanged](#) (const QString &)
- void [on_PDF_clicked](#) ()
Listen to ui->PDF click action.
- void [on_refresh_clicked](#) ()
Listen to ui->refresh click action.
- void [on_CSV_clicked](#) ()
- void [on_Search_bar_textChanged](#) (const QString &)

Private Member Functions

- void [__init__](#) ()
Initialize [Employees](#) first view interface.
- void [__init_current_employee_UI](#) () const
- void [__init_inset_form_group_box](#) ()
- void [__init_update_form_group_box](#) ()
- void [__init_employees_table](#) ()
Init/Async employees table view.
- void [__clear_employees_table](#) () const
Clears the contents of the [Employees](#) Table ui->EmployeesTableWidget
- void [__init_employees_charts](#) ()
Init/Async employees charts view.
- void [syncUI](#) ()
Synchronies UI data.
- void [set_pushButtonMovie](#) (QPushButton *, QMovie *) const
Set GIF as a button icon.

Private Member Functions inherited from [EmployeesUIUtils](#)

- [EmployeesUIUtils](#) ()
Construct a new [EmployeesUIUtils::EmployeesUIUtils](#) object.
- void [scaleImg](#) (const QString &, QLabel *) const
*Shape an image to be round depending of the *xRadius* and *yRadius* then put it in a label.*
- std::string [extractUsername](#) (const std::string &, const unsigned short &length=12) const
*Return the *username* from *updateForm* title.*
- std::string [strToUpper](#) (std::string) const
*Transform *s* to uppercase string.*
- void [set_shadowEffect](#) (QWidget *, QGraphicsDropShadowEffect *, const qreal xOffset=5, const qreal y↵Offset=5, const qreal blurRadius=5, const QColor color=Qt::gray)
*Set the shadow effect on *effect* and affected to *obj**
- void [filterEmployees](#) (QTableWidget *, QComboBox *, QLineEdit *)

Private Attributes

- [SqlParam](#) employee
- Ui::EmployeesUI * ui
- QMovie * [pdf_movie](#)
- QMovie * [notification_movie](#)
- QMovie * [csv_movie](#)
- QMovie * [refresh_movie](#)
- QGraphicsDropShadowEffect * [shadow_effect_components](#)
- [SerialPortManager](#) spm

Private Attributes inherited from [EmployeesUIUtils](#)

- std::string [profileImgInsertHolder](#)
- std::string [profileImgUpdateHolder](#)

Additional Inherited Members

Static Private Member Functions inherited from [EmployeesUIUtils](#)

- static std::vector< std::vector< std::string > > [extractTableData](#) (QTableWidget *)
*Extract all the data from a *QTableWidget* * into a *vector<vector<string>>**

6.10.1 Detailed Description

[EmployeesUI](#) class.

Parameters

<i>checked</i>	bool
----------------	------

Returns

void

6.10.2 Constructor & Destructor Documentation

6.10.2.1 EmployeesUI()

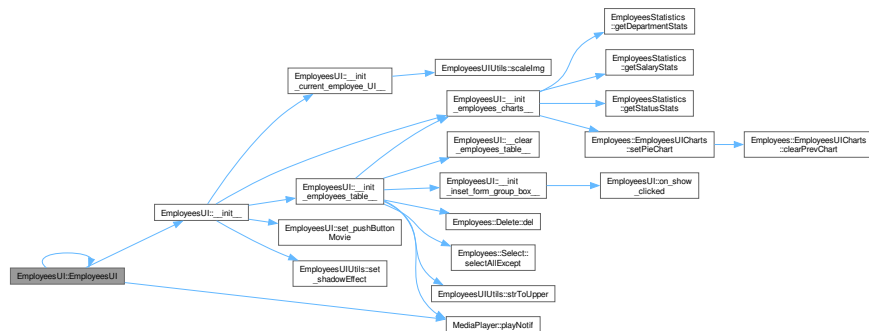
```
EmployeesUI::EmployeesUI (
    SqlParam _employee,
    QWidget * parent = nullptr) [explicit]
```

Construct a new [EmployeesUI::EmployeesUI](#) object.

Parameters

<code>_employee</code>	{ SqlParam }
<code>parent</code>	{ QWidget *}

Here is the call graph for this function:



Here is the caller graph for this function:



6.10.2.2 ~EmployeesUI()

```
EmployeesUI::~~EmployeesUI ()
```

Destroy the [EmployeesUI::EmployeesUI](#) object.

6.10.3 Member Function Documentation

6.10.3.1 __clear_employees_table__()

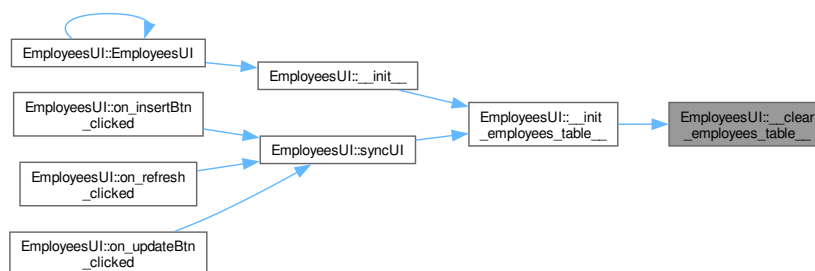
```
void EmployeesUI::__clear_employees_table__ () const [private]
```

Clears the contents of the [Employees](#) Table `ui->EmployeesTableWidget`

Returns

void

Here is the caller graph for this function:



6.10.3.2 __init__()

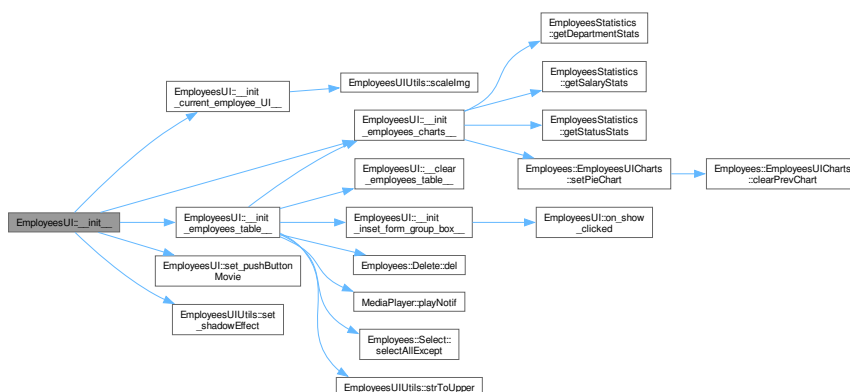
```
void EmployeesUI::__init__ () [inline], [private]
```

Initialize [Employees](#) first view interface.

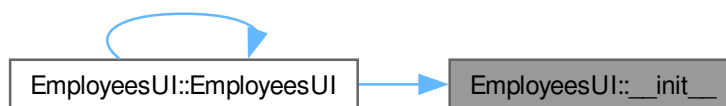
Returns

void

Here is the call graph for this function:



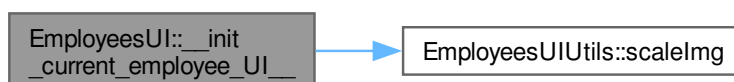
Here is the caller graph for this function:



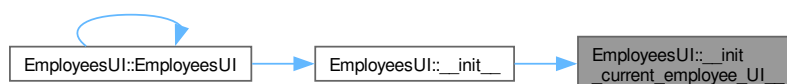
6.10.3.3 __init_current_employee_UI__()

```
void EmployeesUI::__init_current_employee_UI__ () const [private]
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.10.3.4 __init_employees_charts__()

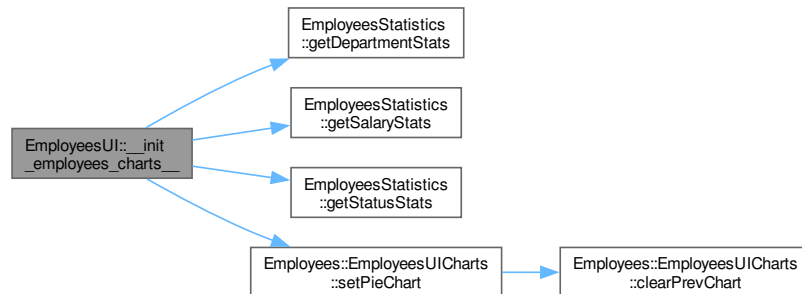
```
void EmployeesUI::__init_employees_charts__ () [inline], [private]
```

Init/Async employees charts view.

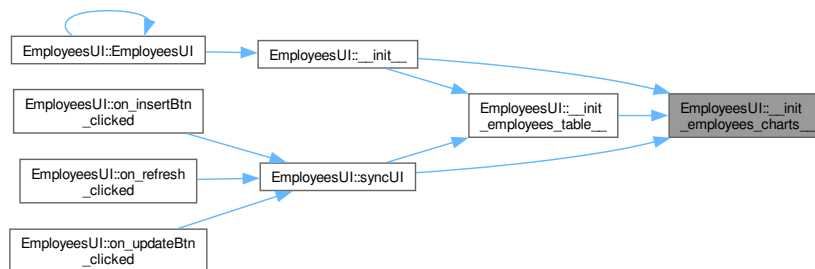
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.10.3.5 __init_employees_table__()

```
void EmployeesUI::__init_employees_table__ () [private]
```

Init/Async employees table view.

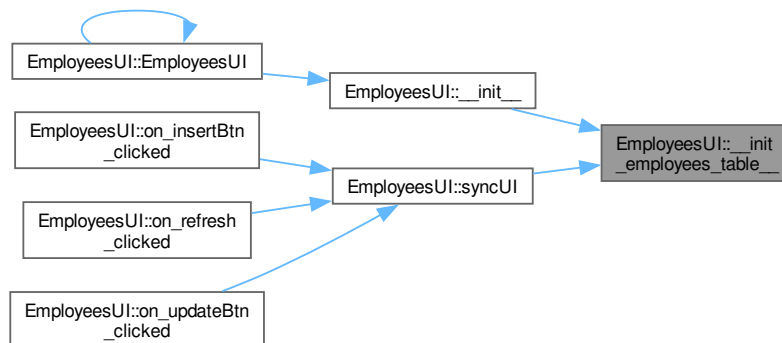
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



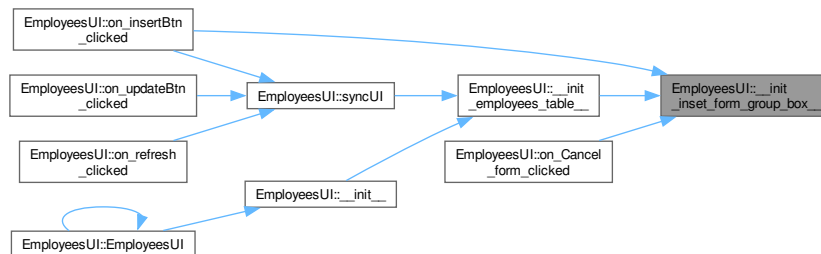
6.10.3.6 __init_inset_form_group_box__()

```
void EmployeesUI::__init_inset_form_group_box__ () [inline], [private]
```

Here is the call graph for this function:



Here is the caller graph for this function:



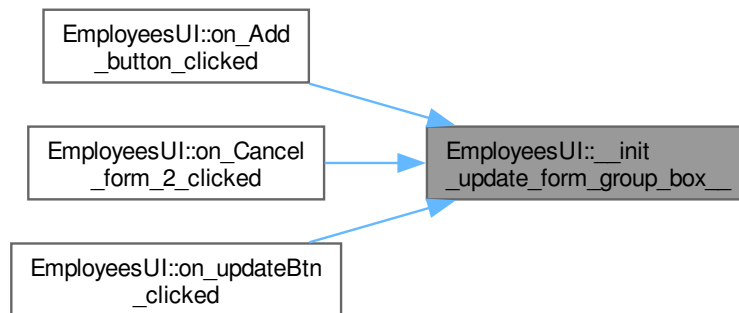
6.10.3.7 __init_update_form_group_box__()

```
void EmployeesUI::__init_update_form_group_box__ () [inline], [private]
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.10.3.8 on_Add_button_clicked

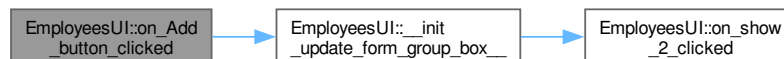
```
void EmployeesUI::on_Add_button_clicked () [private], [slot]
```

Listen to "Add" button click action.

Returns

void

Here is the call graph for this function:



6.10.3.9 on_Cancel_form_2_clicked

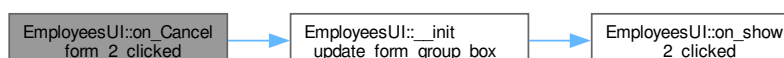
```
void EmployeesUI::on_Cancel_form_2_clicked () [private], [slot]
```

Listen to "Cancel" button click action inside ui->updateForm QGroupBox.

Returns

void

Here is the call graph for this function:



6.10.3.10 on_Cancel_form_clicked

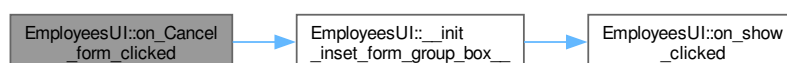
```
void EmployeesUI::on_Cancel_form_clicked () [private], [slot]
```

Listen to "Cancel" button click action inside ui->Form QGroupBox.

Returns

void

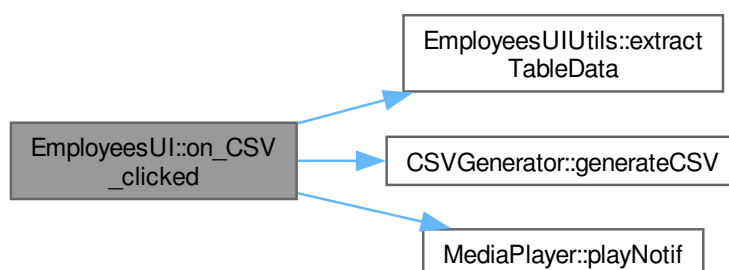
Here is the call graph for this function:



6.10.3.11 on_CSV_clicked

```
void EmployeesUI::on_CSV_clicked () [private], [slot]
```

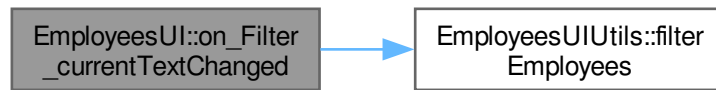
Here is the call graph for this function:



6.10.3.12 on_Filter_currentTextChanged

```
void EmployeesUI::on_Filter_currentTextChanged (  
    const QString & arg) [private], [slot]
```

Here is the call graph for this function:



6.10.3.13 on_insertBtn_clicked

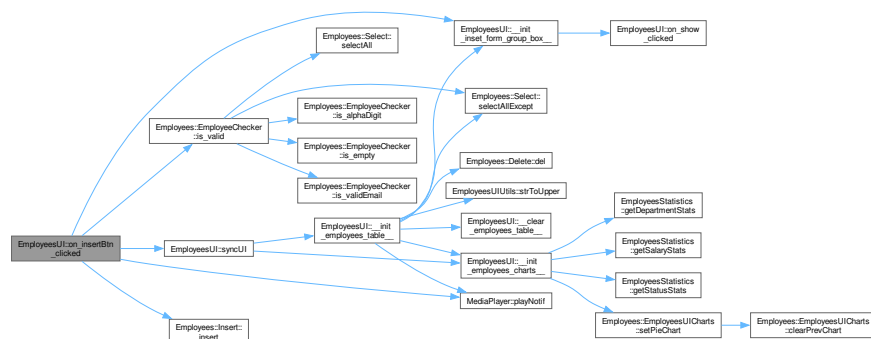
```
void EmployeesUI::on_insertBtn_clicked () [private], [slot]
```

Listen to "Add" employee button click action.

Returns

void

Here is the call graph for this function:



6.10.3.14 on_PDF_clicked

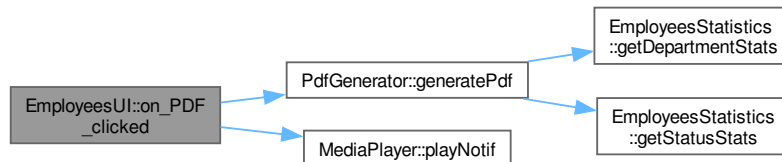
```
void EmployeesUI::on_PDF_clicked () [private], [slot]
```

Listen to ui->PDF click action.

Returns

void <https://www.qcustomplot.com/index.php/tutorials/specialcases/textdocument>

Here is the call graph for this function:

**6.10.3.15 on_profileImageInsert_2_clicked**

```
void EmployeesUI::on_profileImageInsert_2_clicked () [private], [slot]
```

Listen to "Profile Image" employee button click action.

```
ui->profileImageInsert_2 inside ui->updateForm
```

Returns

void

6.10.3.16 on_profileImageInsert_clicked

```
void EmployeesUI::on_profileImageInsert_clicked () [private], [slot]
```

Listen to "Profile Image" employee button click action.

```
ui->profileImageInsert inside ui->Form
```

Returns

void

6.10.3.17 on_refresh_clicked

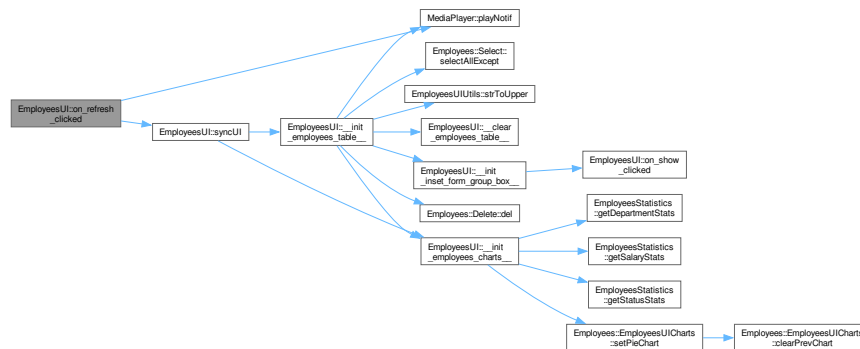
```
void EmployeesUI::on_refresh_clicked () [private], [slot]
```

Listen to ui->refresh click action.

Returns

void

Here is the call graph for this function:



6.10.3.18 on_Search_bar_textChanged

```
void EmployeesUI::on_Search_bar_textChanged (
    const QString & arg) [private], [slot]
```

Here is the call graph for this function:

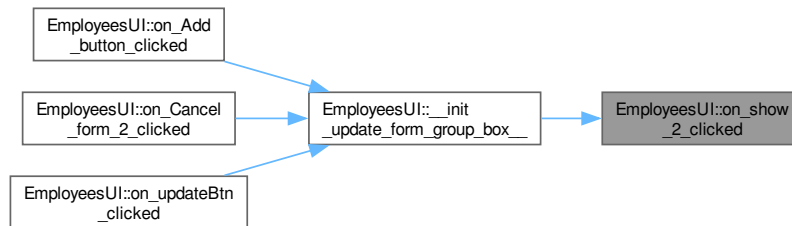


6.10.3.19 on_show_2_clicked

```
void EmployeesUI::on_show_2_clicked (
    bool checked) [private], [slot]
```

Listen to "Show More/Less" button click action inside `ui->updateForm QGroupBox`.

Here is the caller graph for this function:



6.10.3.20 on_show_clicked

```
void EmployeesUI::on_show_clicked (
    bool checked) [private], [slot]
```

Listen to "Show More/Less" button click action inside `ui->Form QGroupBox`.

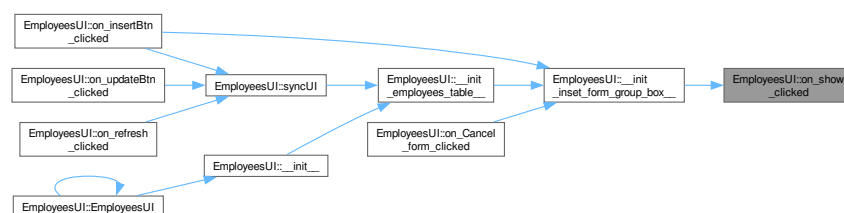
Parameters

<i>checked</i>	bool
----------------	------

Returns

void

Here is the caller graph for this function:



6.10.3.21 on_updateBtn_clicked

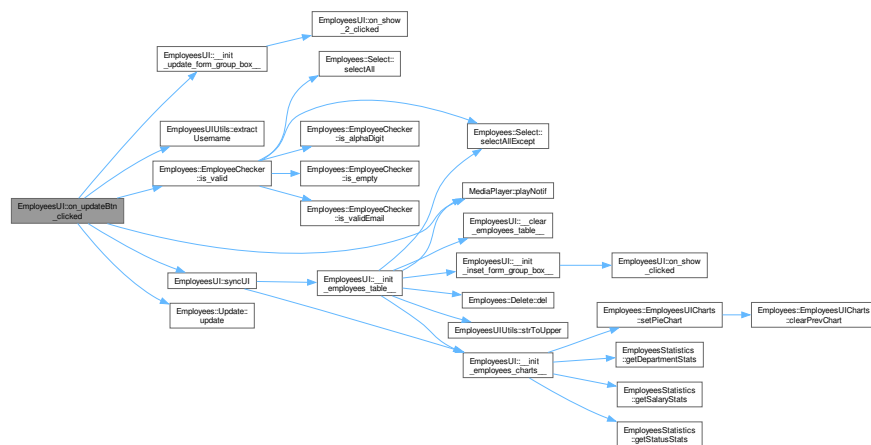
```
void EmployeesUI::on_updateBtn_clicked () [private], [slot]
```

Listen to "Update" employee button click action.

Returns

void

Here is the call graph for this function:



6.10.3.22 set_employee()

```
void EmployeesUI::set_employee (
    const SqlParam & _employee)
```

Set employee data.

Parameters

<code>_employee</code>	{const <code>SqlParam</code> &}
------------------------	---------------------------------

Returns

void

6.10.3.23 set_pushButtonMovie()

```
void EmployeesUI::set_pushButtonMovie (
    QPushButton * btn,
    QMovie * movie) const [private]
```

Set GIF as a button icon.

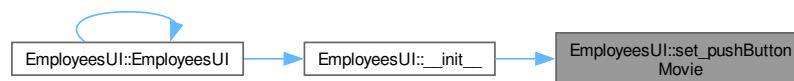
Parameters

<i>btn</i>	{QPushButton *}
<i>movie</i>	{QMovie *}

Returns

void

Here is the caller graph for this function:



6.10.3.24 syncUI()

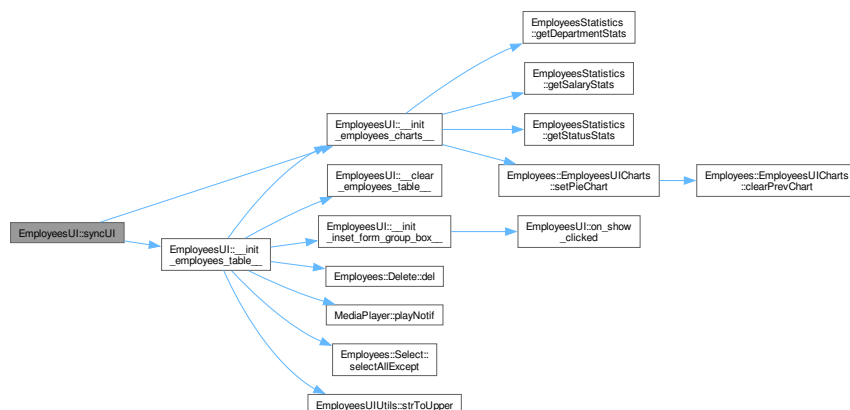
```
void EmployeesUI::syncUI () [private]
```

Synchronises UI data.

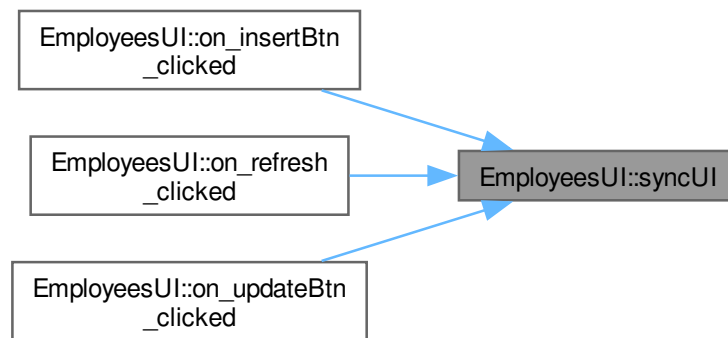
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.10.4 Member Data Documentation

6.10.4.1 csv_movie

```
QMovie * EmployeesUI::csv_movie [private]
```

6.10.4.2 employee

```
SqlParam EmployeesUI::employee [private]
```

6.10.4.3 notification_movie

```
QMovie * EmployeesUI::notification_movie [private]
```

6.10.4.4 pdf_movie

```
QMovie* EmployeesUI::pdf_movie [private]
```

6.10.4.5 refresh_movie

```
QMovie * EmployeesUI::refresh_movie [private]
```

6.10.4.6 shadow_effect_components

```
QGraphicsDropShadowEffect* EmployeesUI::shadow_effect_components [private]
```

6.10.4.7 spm

```
SerialPortManager EmployeesUI::spm [private]
```

6.10.4.8 ui

```
Ui::EmployeesUI* EmployeesUI::ui [private]
```

The documentation for this class was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/[EmployeesUI.hpp](#)
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/[EmployeesUI.cpp](#)

6.11 Employees::EmployeesUICharts Class Reference

```
#include <EmployeesUICharts.hpp>
```

Public Member Functions

- [EmployeesUICharts](#) (QMainWindow *)
Construct a new [Employees::EmployeesUICharts::EmployeesUICharts](#) object.
- void [setPieChart](#) (QHBoxLayout *, const std::vector< std::pair< std::string, double > > &)
Sets a pie chart on the provided layout with the given data.
- void [setBarChart](#) (QHBoxLayout *, const std::vector< std::pair< std::string, double > > &)
Sets a bar chart on the provided layout with the given data.
- void [setBarLineChart](#) (QHBoxLayout *, const std::vector< std::pair< std::string, double > > &)

Private Member Functions

- void [clearPrevChart](#) (QHBoxLayout *)

Private Attributes

- QMainWindow * [ui](#)

6.11.1 Constructor & Destructor Documentation

6.11.1.1 EmployeesUICharts()

```
EmployeesUICharts::EmployeesUICharts (  
    QMainWindow * _ui) [explicit]
```

Construct a new [Employees::EmployeesUICharts::EmployeesUICharts](#) object.

Parameters

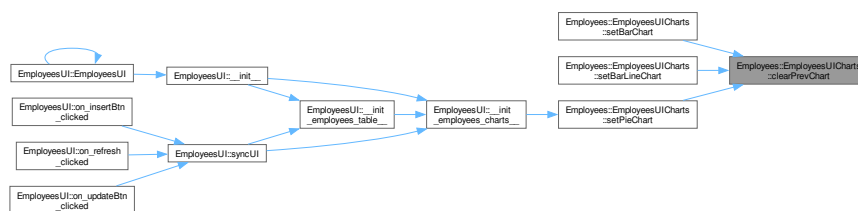
$\leftarrow$	{QMainWindow *}
$\leftarrow$	
<i>ui</i>	

6.11.2 Member Function Documentation

6.11.2.1 clearPrevChart()

```
void EmployeesUICharts::clearPrevChart (
    QHBoxLayout * frame) [private]
```

Here is the caller graph for this function:



6.11.2.2 setBarChart()

```
void EmployeesUICharts::setBarChart (
    QHBoxLayout * frame,
    const std::vector< std::pair< std::string, double > > & data)
```

Sets a bar chart on the provided layout with the given data.

This function takes a horizontal layout and a set of data, where each data entry represents a category (as a string) and its corresponding value (as a double). It creates a bar chart using Qt's `QBarSeries`, `QBarSet`, and `QChart` components. The chart is populated with the provided data, and interactive animations (hover effects) are added for better user experience.

Parameters

<i>frame</i>	The <code>QHBoxLayout</code> where the chart will be placed.
<i>data</i>	A vector of pairs where each pair consists of: - A <code>std::string</code> representing the category name. - A <code>double</code> representing the category value.

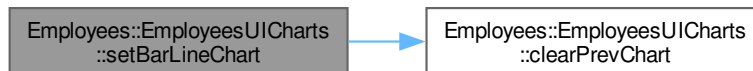
Here is the call graph for this function:



6.11.2.3 setBarLineChart()

```
void EmployeesUICharts::setBarLineChart (
    QHBoxLayout * frame,
    const std::vector< std::pair< std::string, double > > & data)
```

Here is the call graph for this function:



6.11.2.4 setPieChart()

```
void EmployeesUICharts::setPieChart (
    QHBoxLayout * frame,
    const std::vector< std::pair< std::string, double > > & data)
```

Sets a pie chart on the provided layout with the given data.

This function takes a horizontal layout and a set of data, where each data entry represents a category (as a string) and its corresponding value (as a double). It creates a pie chart using Qt's `QPieSeries` and `QChart` components. The chart is populated with the provided data, and interactive animations (hover effects and explosion) are added for better user experience.

Note

The pie chart's legend is updated with the percentage values when hovering over each slice.

Parameters

<i>frame</i>	The <code>QHBoxLayout</code> where the chart will be placed
<i>data</i>	A vector of pairs where each pair consists of:

- A `std::string` representing the category name.
- A `double` representing the category value.

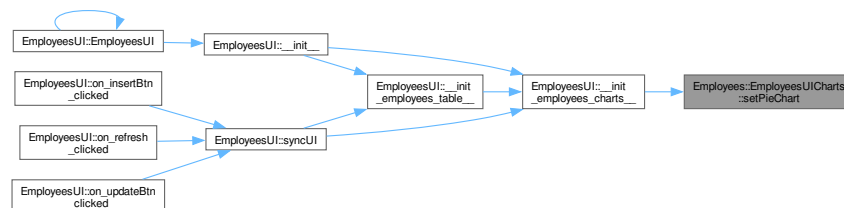
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.11.3 Member Data Documentation

6.11.3.1 ui

```
QMainWindow* Employees::EmployeesUICharts::ui [private]
```

The documentation for this class was generated from the following files:

- `/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/EmployeesUICharts.hpp`
- `/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/EmployeesUICharts.cpp`

6.12 EmployeesUICharts Class Reference

[EmployeesUICharts](#) class.

```
#include <EmployeesUICharts.hpp>
```

6.12.1 Detailed Description

[EmployeesUICharts](#) class.

The documentation for this class was generated from the following file:

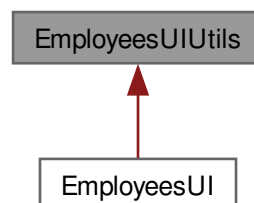
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/[EmployeesUICharts.hpp](#)

6.13 EmployeesUIUtils Class Reference

[EmployeesUIUtils](#) class.

```
#include <EmployeesUIUtils.hpp>
```

Inheritance diagram for EmployeesUIUtils:



Public Member Functions

- [EmployeesUIUtils](#) ()
Construct a new [EmployeesUIUtils::EmployeesUIUtils](#) object.

Protected Member Functions

- void [scaleImg](#) (const QString &, QLabel *) const
Shape an image to be round depending of the *xRadius* and *yRadius* then put it in a label.
- std::string [extractUsername](#) (const std::string &, const unsigned short &length=12) const
Return the *username* from *updateForm* title.
- std::string [strToUpper](#) (std::string) const
Transform *s* to uppercase string.
- void [set_shadowEffect](#) (QWidget *, QGraphicsDropShadowEffect *, const qreal xOffset=5, const qreal y↔Offset=5, const qreal blurRadius=5, const QColor color=Qt::gray)
Set the shadow effect on *effect* and affected to *obj*
- void [filterEmployees](#) (QTableWidget *, QComboBox *, QLineEdit *)

Static Protected Member Functions

- static `std::vector< std::vector< std::string > >` [extractTableData](#) (`QTableWidget *`)
*Extract all the data from a `QTableWidget` * into a `vector<vector<string>>`*

Protected Attributes

- `std::string` [profileImgInsertHolder](#)
- `std::string` [profileImgUpdateHolder](#)

6.13.1 Detailed Description

[EmployeesUIUtils](#) class.

6.13.2 Constructor & Destructor Documentation

6.13.2.1 EmployeesUIUtils()

```
EmployeesUIUtils::EmployeesUIUtils ()
```

Construct a new [EmployeesUIUtils::EmployeesUIUtils](#) object.

6.13.3 Member Function Documentation

6.13.3.1 extractTableData()

```
std::vector< std::vector< std::string > > EmployeesUIUtils::extractTableData (
    QTableWidget * tableWidget) [static], [protected]
```

Extract all the data from a `QTableWidget` * into a `vector<vector<string>>`

Parameters

<i>tableWidget</i>	{ <code>QTableWidget</code> *}
--------------------	--------------------------------

Returns

`std::vector<std::vector<std::string>>`

Here is the caller graph for this function:



6.13.3.2 extractUsername()

```
std::string EmployeesUIUtils::extractUsername (
    const std::string & title,
    const unsigned short & length = 12) const [protected]
```

Return the username from updateForm title.

Parameters

<i>title</i>	{const std::string &}
<i>length</i>	{const unsigned short &}

Returns

std::string

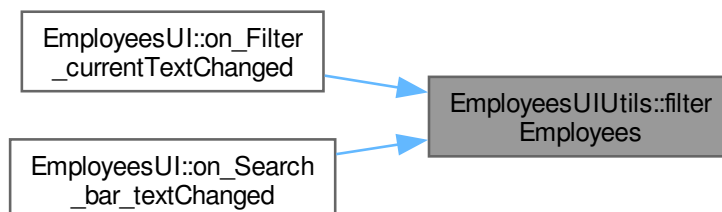
Here is the caller graph for this function:



6.13.3.3 filterEmployees()

```
void EmployeesUIUtils::filterEmployees (
    QTableWidget * tableWidget,
    QComboBox * filter,
    QLineEdit * searchBar) [protected]
```

Here is the caller graph for this function:



6.13.3.4 scaleImg()

```
void EmployeesUIUtils::scaleImg (
    const QString & path,
    QLabel * label) const [protected]
```

Shape an image to be round depending of the xRadius and yRadius then put it in a label.

Parameters

<i>path</i>	{const QString &}
<i>label</i>	{QLabel *}

Returns

void

Here is the caller graph for this function:



6.13.3.5 set_shadowEffect()

```
void EmployeesUIUtils::set_shadowEffect (
    QWidget * obj,
    QGraphicsDropShadowEffect * effect,
    const qreal xOffset = 5,
    const qreal yOffset = 5,
    const qreal blurRadius = 5,
    const QColor color = Qt::gray) [protected]
```

Set the shadow effect on *effect* and affected to *obj*

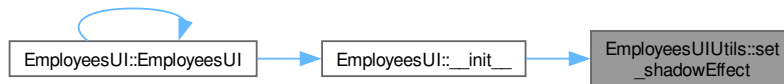
Parameters

<i>obj</i>	{QWidget *}
<i>effect</i>	{QGraphicsDropShadowEffect *}
<i>xOffset</i>	{const qreal}
<i>yOffset</i>	{const qreal}
<i>blurRadius</i>	{const qreal}
<i>color</i>	{const QColor}

Returns

void

Here is the caller graph for this function:

**6.13.3.6 strToUpper()**

```
std::string EmployeesUIUtils::strToUpper (
    std::string s) const [protected]
```

Transform *s* to uppercase string.

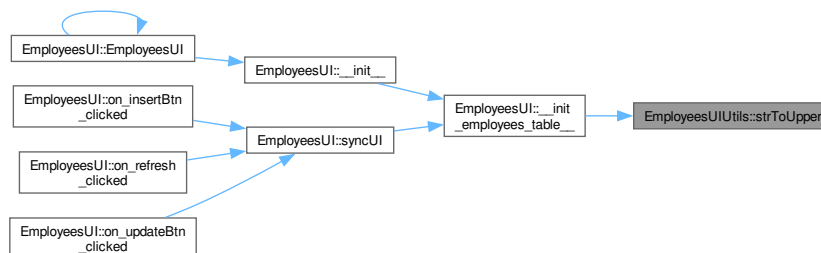
Parameters

<i>s</i>	{std::string &}
----------	-----------------

Returns

std::string

Here is the caller graph for this function:

**6.13.4 Member Data Documentation****6.13.4.1 profileImgInsertHolder**

```
std::string EmployeesUIUtils::profileImgInsertHolder [protected]
```

6.13.4.2 profileImgUpdateHolder

```
std::string EmployeesUIUtils::profileImgUpdateHolder [protected]
```

The documentation for this class was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/[EmployeesUIUtils.hpp](#)
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/[EmployeesUIUtils.cpp](#)

6.14 FaceRecognizer Class Reference

```
#include <FaceRecognizer.hpp>
```

Public Member Functions

- [~FaceRecognizer](#) ()
Destroy the [FaceRecognizer::FaceRecognizer](#) object.
- void [trainEmbeddedModel](#) (const std::string &, const std::string &, const std::string &)
- void [captureFrame](#) (QLabel *)
Capture the frame from the camera.
- std::string [recognize](#) (QLabel *)
sudo chmod 666 /dev/video0
- void [load](#) ()
load all [FaceRecognizer](#) loaders (Indexer, CascadeModel, EmbeddedModel)

Private Member Functions

- QImage [matToQImage](#) (const cv::Mat &mat) const
Convert cv::Mat to QImage
- std::pair< std::vector< cv::Mat >, std::vector< int > > [loadData](#) (const std::string &dataDir)
Load training data.
- void [loadIndexer](#) (const std::string &dataDir)
Load labels and corresponding names in indexer
- void [loadEmbeddedModel](#) (const std::string &)
Load "Embedded Model".
- void [loadCascadeModel](#) (const std::string &)
Load "Cascade Model".

Private Attributes

- cv::VideoCapture [cap](#)
- cv::Mat [currentFrame](#)
- cv::CascadeClassifier [faceCascade](#)
- cv::Ptr< cv::face::LBPHFaceRecognizer > [recognizer](#)
- std::map< int, std::string > [indexer](#)

6.14.1 Constructor & Destructor Documentation

6.14.1.1 ~FaceRecognizer()

FaceRecognizer::~~FaceRecognizer ()

Destroy the [FaceRecognizer::FaceRecognizer](#) object.

6.14.2 Member Function Documentation

6.14.2.1 captureFrame()

```
void FaceRecognizer::captureFrame (
    QLabel * cameraLabel)
```

Capture the frame from the camera.

Use it only in test mode (only show the recognized face with username label)

Parameters

<i>cameraLabel</i>	{QLabel *}
--------------------	------------

Returns

void

Here is the call graph for this function:



6.14.2.2 load()

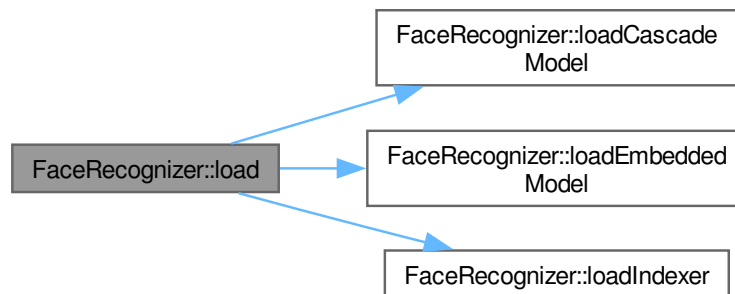
```
void FaceRecognizer::load ()
```

load all [FaceRecognizer](#) loaders (Indexer, CascadeModel, EmbeddedModel)

Returns

void

Security issue (X can change folders names and make spoofing attack)Here is the call graph for this function:

**6.14.2.3 loadCascadeModel()**

```
void FaceRecognizer::loadCascadeModel (
    const std::string & cascadeModelPath) [private]
```

Load "Cascade Model".

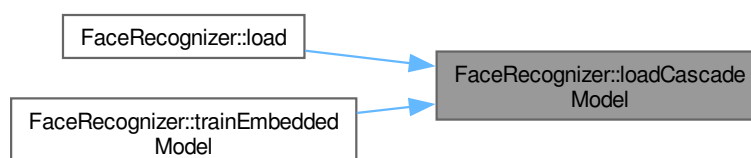
Parameters

<i>cascadeModelPath</i>	{const std::string &}
-------------------------	-----------------------

Returns

void

Here is the caller graph for this function:



6.14.2.4 loadData()

```
std::pair< std::vector< cv::Mat >, std::vector< int > > FaceRecognizer::loadData (  
    const std::string & dataDir) [private]
```

Load training data.

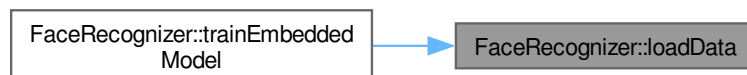
Parameters

<i>dataDir</i>	{const std::string &}
<i>flag</i>	{const FaceRecognizerFlags &}

Returns

std::pair<std::vector<cv::Mat>, std::vector<int>>>

Here is the caller graph for this function:



6.14.2.5 loadEmbeddedModel()

```
void FaceRecognizer::loadEmbeddedModel (
    const std::string & embeddedModelPath) [private]
```

Load "Embedded Model".

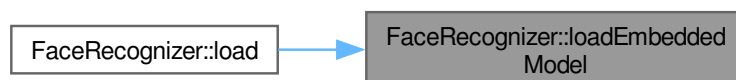
Parameters

<i>embeddedModelPath</i>	{const std::string &}
--------------------------	-----------------------

Returns

void

Here is the caller graph for this function:



6.14.2.6 loadIndexer()

```
void FaceRecognizer::loadIndexer (
    const std::string & dataDir) [private]
```

Load labels and corresponding names in `indexer`

Parameters

<i>dataDir</i>	{const std::string &}
----------------	-----------------------

Returns

void

Here is the caller graph for this function:



6.14.2.7 matToQImage()

```
QImage FaceRecognizer::matToQImage (  
    const cv::Mat & mat) const [private]
```

Convert `cv::Mat` to `QImage`

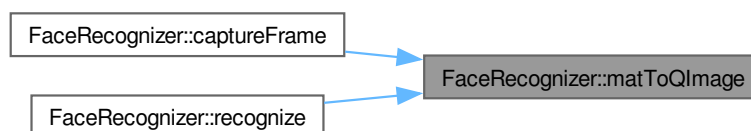
Parameters

<i>mat</i>	{const cv::Mat &}
------------	-------------------

Returns

QImage

Here is the caller graph for this function:



6.14.2.8 recognize()

```
std::string FaceRecognizer::recognize (  
    QLabel * cameraLabel)
```

```
sudo chmod 666 /dev/video0
```

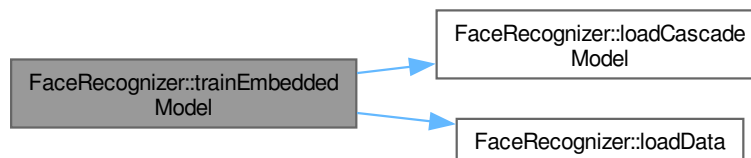
Here is the call graph for this function:



6.14.2.9 trainEmbeddedModel()

```
void FaceRecognizer::trainEmbeddedModel (  
    const std::string & modelSavePath,  
    const std::string & cascadeModelPath,  
    const std::string & dataLoadPath)
```

Here is the call graph for this function:



6.14.3 Member Data Documentation

6.14.3.1 cap

```
cv::VideoCapture FaceRecognizer::cap [private]
```

6.14.3.2 currentFrame

```
cv::Mat FaceRecognizer::currentFrame [private]
```

6.14.3.3 faceCascade

```
cv::CascadeClassifier FaceRecognizer::faceCascade [private]
```

6.14.3.4 indexer

```
std::map<int, std::string> FaceRecognizer::indexer [private]
```

6.14.3.5 recognizer

```
cv::Ptr<cv::face::LBPHFaceRecognizer> FaceRecognizer::recognizer [private]
```

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/FaceRecognizer.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/FaceRecognizer.cpp](#)

6.15 HtmlBodyFormater Class Reference

[HtmlBodyFormater](#) class.

```
#include <HtmlBodyFormatter.hpp>
```

Public Member Functions

- [HtmlBodyFormater](#) (const std::string &, const std::unordered_map< std::string, const std::string > ¶m={})
- std::string [get_inner_html](#) ()
Get your file content in inner html format.

Private Member Functions

- void [format_inner_html](#) ()
Replaces placeholders in the inner HTML with corresponding values.

Private Attributes

- std::string [inner_html](#)
- const std::unordered_map< std::string, const std::string > [param](#)

6.15.1 Detailed Description

[HtmlBodyFormater](#) class.

6.15.2 Constructor & Destructor Documentation

6.15.2.1 HtmlBodyFormatter()

```
HtmlBodyFormatter::HtmlBodyFormatter (
    const std::string & path,
    const std::unordered_map< std::string, const std::string > & _param = {}) [explicit]
```

6.15.3 Member Function Documentation

6.15.3.1 format_inner_html()

```
void HtmlBodyFormatter::format_inner_html () [private]
```

Replaces placeholders in the inner HTML with corresponding values.

This function iterates over a set of key-value pairs (`param`) and replaces occurrences of each key in `inner_html` with its associated value.

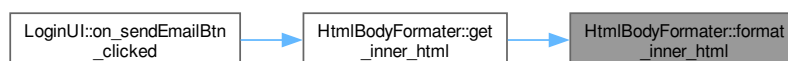
Note

Placeholders in `inner_html` should be formatted as `{{PLACEHOLDER}}` and will be replaced with their corresponding values.

Returns

void

Here is the caller graph for this function:



6.15.3.2 get_inner_html()

```
std::string HtmlBodyFormatter::get_inner_html ()
```

Get your file content in inner html format.

Returns

std::string

Here is the call graph for this function:



Here is the caller graph for this function:



6.15.4 Member Data Documentation

6.15.4.1 inner_html

```
std::string HtmlBodyFormatter::inner_html [private]
```

6.15.4.2 param

```
const std::unordered_map<std::string, const std::string> HtmlBodyFormatter::param [private]
```

The documentation for this class was generated from the following files:

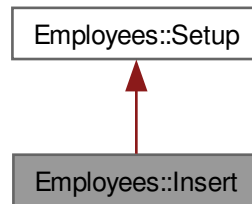
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/HtmlBodyFormatter.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/HtmlBodyFormatter.cpp](#)

6.16 Employees::Insert Class Reference

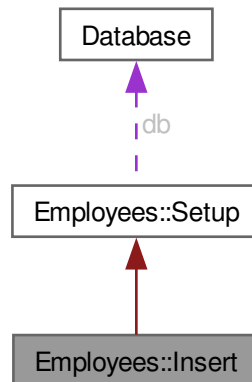
[Insert](#) new employee.

```
#include <Employees.hpp>
```

Inheritance diagram for Employees::Insert:



Collaboration diagram for Employees::Insert:



Public Member Functions

- int [insert](#) (const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &)
- int [insert](#) (const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< std::string > &, const [Employees::EmployeeInfo](#)< int > &, const [Employees::EmployeeInfo](#)< int > &, const [Employees::EmployeeInfo](#)< oracle::occi::Date > &)

Additional Inherited Members

Private Member Functions inherited from `Employees::Setup`

- `Setup ()`
Construct a new `Employees::Setup::Setup` object.
- virtual `~Setup ()`
Destroy the `Employees::Setup::Setup` object.

Private Attributes inherited from `Employees::Setup`

- `Database * db`

6.16.1 Detailed Description

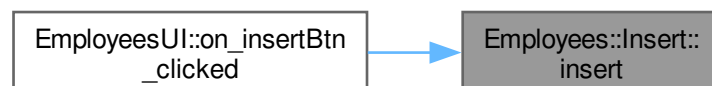
`Insert` new employee.

6.16.2 Member Function Documentation

6.16.2.1 `insert()` [1/2]

```
int Employees::Insert::insert (
    const Employees::EmployeeInfo< std::string > & email,
    const Employees::EmployeeInfo< std::string > & firstName,
    const Employees::EmployeeInfo< std::string > & lastName,
    const Employees::EmployeeInfo< std::string > & password,
    const Employees::EmployeeInfo< std::string > & department)
```

Here is the caller graph for this function:



6.16.2.2 insert() [2/2]

```
int Employees::Insert::insert (
    const Employees::EmployeeInfo< std::string > & email,
    const Employees::EmployeeInfo< std::string > & firstName,
    const Employees::EmployeeInfo< std::string > & lastName,
    const Employees::EmployeeInfo< std::string > & password,
    const Employees::EmployeeInfo< std::string > & department,
    const Employees::EmployeeInfo< std::string > & jobTitle,
    const Employees::EmployeeInfo< std::string > & address,
    const Employees::EmployeeInfo< std::string > & imagePath,
    const Employees::EmployeeInfo< int > & salary,
    const Employees::EmployeeInfo< int > & tel,
    const Employees::EmployeeInfo< oracle::occi::Date > & birthdate)
```

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp](#)

6.17 Login Class Reference

[Login](#) class.

```
#include <LoginUI.hpp>
```

6.17.1 Detailed Description

[Login](#) class.

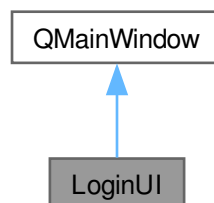
The documentation for this class was generated from the following file:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/LoginUI.hpp](#)

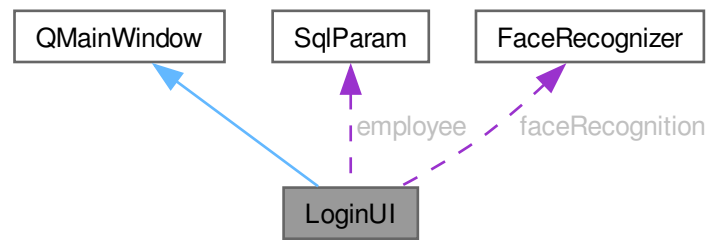
6.18 LoginUI Class Reference

```
#include <LoginUI.hpp>
```

Inheritance diagram for LoginUI:



Collaboration diagram for LoginUI:



Signals

- void [loginSuccessful](#) ()

Public Member Functions

- [LoginUI](#) (QWidget *parent=nullptr)
Construct a new [LoginUI::LoginUI](#) object.
- [~LoginUI](#) ()
Destroy the [LoginUI::LoginUI](#) object.
- const [SqlParam](#) [get_employee](#) ()
Get verified employee data.

Private Slots

- void [on_pushButton_clicked](#) ()
Listen to [Login](#) button click action.
- void [on_hide_show_btn_clicked](#) ()
Listen to [hide_show_btn](#) click action.
- void [on_reset_clicked](#) ()
Verify the taped code and new/confirm password.
- void [on_faceId_clicked](#) ()
Listen to [FaceID](#) button click action.
- void [on_fp_returnBtn_clicked](#) ()
- void [on_hide_show_btn_2_clicked](#) ()
Listen to [hide_show_btn](#) click action.
- void [on_returnBtn_clicked](#) ()
Listen to [Return](#) button click action.
- void [on_sendEmailBtn_clicked](#) ()
Verify the given mail and send a 6-chars reset code to reset the password.

Private Member Functions

- bool [eventFilter](#) (QObject *, QEvent *) override
Manage global events in login interface.
- bool [forget_password_events](#) (QObject *, QEvent *)
Manage forget password events.
- bool [login_btn_events](#) (QObject *, QLabel *, QEvent *)
Manage login button in the login/forgetPassword QGroups.
- void [updateGif](#) ()
Update the Gif every `__LOGIN_GIF_ANIMATION__`
- void [QGroupBoxFadeOutEffect](#) (QGroupBox *, QGroupBox *)
*Make a "fade out" effect to `group1` using opacity transition from `__FULL_VISIBLE__` to `__FULL__`↔
`TRANSPARENT__` and hide `group1`, otherwise for `group2`*
- QPropertyAnimation * [FadeEffect](#) (QGroupBox *, const QVariant, const QVariant)
Make fade In/Out for `group`
- void [change_hideShowBtnIcon](#) (QLineEdit *, QPushButton *, const QLineEdit::EchoMode mode=QLineEdit::Password, const QString path="/home/zouari_omar/Documents/Daily/Projects/Astra/project/assets/login
imgs/closedEye.png")
Helper function.
- void [enableResetPassword](#) (const bool &)
Reset the forget password QGroupBox to the default view (enable/disable proccess)
- void [clearResetPassword](#) ()
Reset all forget password QLineEdit's to the default state (empty)

Private Attributes

- Ui::Login * [ui](#)
- QTimer * [gifTimer](#)
- QTimer * [camTimer](#)
- QMovie * [currentMovie](#)
- int [currentGifIndex](#)
- const QStringList [gifPaths](#)
- std::string [generated_password](#)
- QSqlParam [employee](#)
- FaceRecognizer * [faceRecognition](#)

6.18.1 Constructor & Destructor Documentation

6.18.1.1 LoginUI()

```
LoginUI::LoginUI (
    QWidget * parent = nullptr) [explicit]
```

Construct a new [LoginUI::LoginUI](#) object.

Parameters

<i>parent</i>	{QWidget *}
---------------	-------------

6.18.1.2 ~LoginUI()

LoginUI::~~LoginUI ()

Destroy the [LoginUI::LoginUI](#) object.

6.18.2 Member Function Documentation

6.18.2.1 change_hideShowBtnIcon()

```
void LoginUI::change_hideShowBtnIcon (
    QLineEdit * lineEdit,
    QPushButton * btn,
    const QLineEdit::EchoMode mode = QLineEdit::Password,
    const QString path = "/home/zouari_omar/Documents/Daily/Projects/Astra/project/assets/login_imgs/
[private]
```

Helper function.

this->[on_hide_show_btn_clicked\(\)](#) | this->[on_hide_show_btn_clicked\(\)](#)

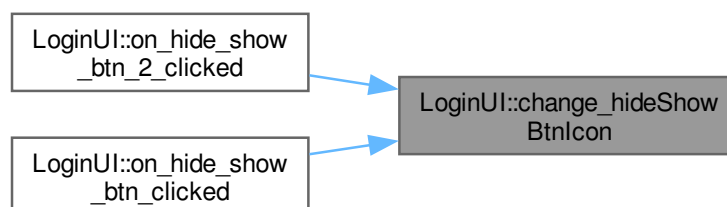
Parameters

<i>lineEdit</i>	{QLineEdit *}
<i>btn</i>	{QPushButton *}
<i>mode</i>	{const QLineEdit::EchoMode}
<i>path</i>	{const QString}

Returns

void

Here is the caller graph for this function:



6.18.2.2 clearResetPassword()

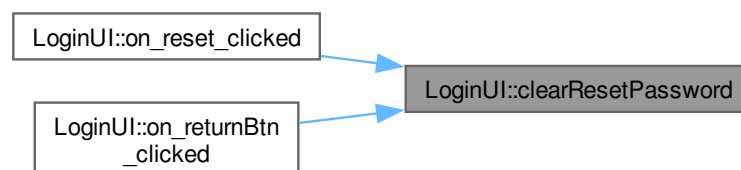
```
void LoginUI::clearResetPassword () [private]
```

Reset all forget password QLineEdit's to the default state (empty)

Returns

void

Here is the caller graph for this function:



6.18.2.3 enableResetPassword()

```
void LoginUI::enableResetPassword (
    const bool & status) [private]
```

Reset the forget password QGroupBox to the default view (enable/disable process)

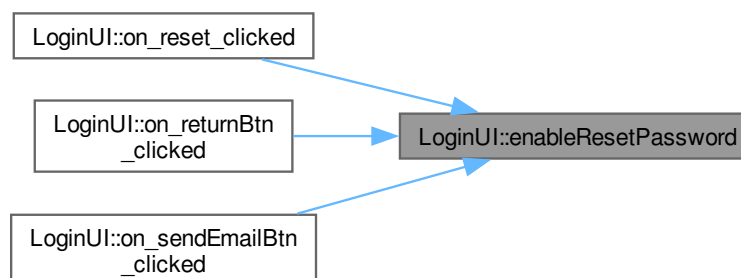
Parameters

<i>status</i>	{const bool &}
---------------	----------------

Returns

void

Here is the caller graph for this function:



6.18.2.4 eventFilter()

```
bool LoginUI::eventFilter (
    QObject * obj,
    QEvent * event) [override], [private]
```

Manage global events in login interface.

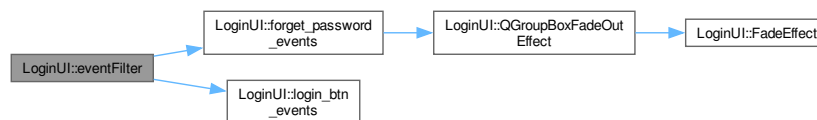
Parameters

<i>obj</i>	{QObject *}
<i>event</i>	{QEvent *}

Returns

bool

Here is the call graph for this function:



6.18.2.5 FadeEffect()

```
QPropertyAnimation * LoginUI::FadeEffect (
    QGroupBox * group,
    const QVariant startVal,
    const QVariant endVal) [private]
```

Make fade In/Out for group

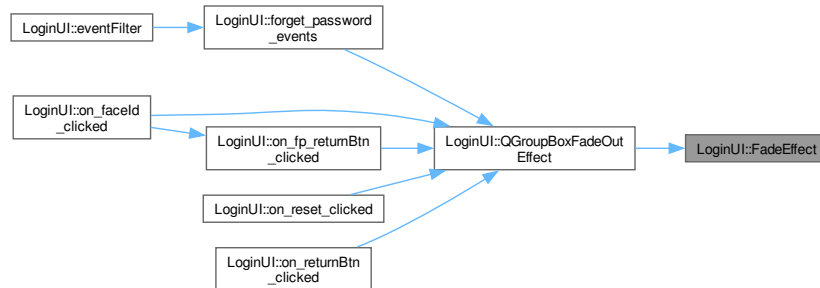
Parameters

<i>group</i>	{QGroupBox *}
<i>startVal</i>	{const QVariant}
<i>endVal</i>	{const QVariant}

Returns

{QPropertyAnimation *}

Here is the caller graph for this function:



6.18.2.6 forget_password_events()

```
bool LoginUI::forget_password_events (
    QObject * obj,
    QEvent * event) [private]
```

Manage forget password events.

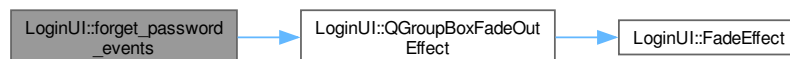
Parameters

<i>obj</i>	{QObject *}
<i>event</i>	{QEvent *}

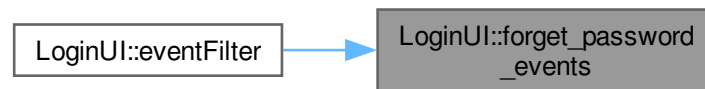
Returns

bool

Here is the call graph for this function:



Here is the caller graph for this function:



6.18.2.7 get_employee()

```
const SqlParam LoginUI::get_employee ()
```

Get verified employee data.

Returns

```
{const SqlParam}
```

6.18.2.8 login_btn_events()

```
bool LoginUI::login_btn_events (
    QObject * obj,
    QLabel * icon,
    QEvent * event) [private]
```

Manage login button in the login/forgetPassword QGroups.

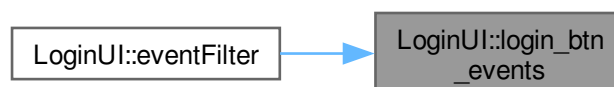
Parameters

<i>obj</i>	{QObject *}
<i>icon</i>	{QLabel *}
<i>event</i>	{QEvent *}

Returns

```
bool
```

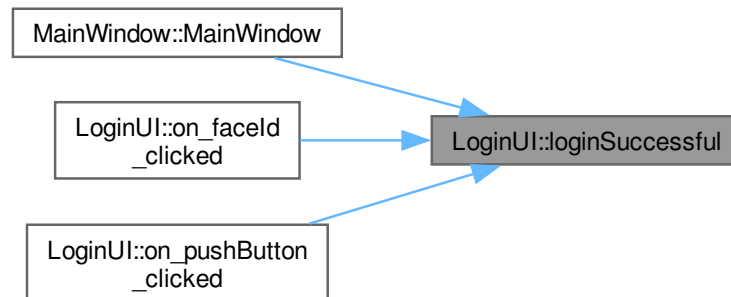
Here is the caller graph for this function:



6.18.2.9 loginSuccessful

```
void LoginUI::loginSuccessful () [signal]
```

Here is the caller graph for this function:



6.18.2.10 on_faceld_clicked

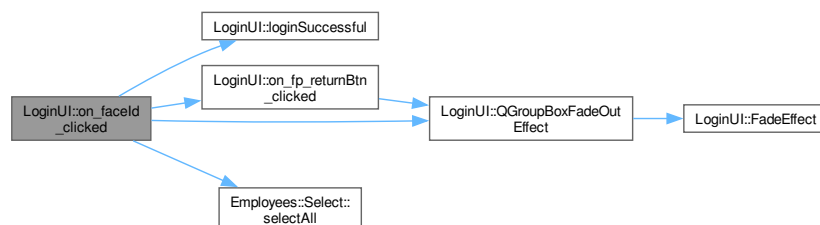
```
void LoginUI::on_faceId_clicked () [private], [slot]
```

Listen to FaceID button click action.

Returns

void

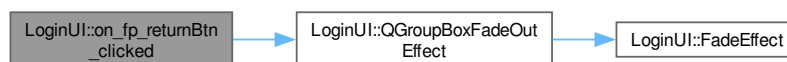
Here is the call graph for this function:



6.18.2.11 on_fp_returnBtn_clicked

```
void LoginUI::on_fp_returnBtn_clicked () [private], [slot]
```

Here is the call graph for this function:



Here is the caller graph for this function:



6.18.2.12 on_hide_show_btn_2_clicked

```
void LoginUI::on_hide_show_btn_2_clicked () [private], [slot]
```

Listen to `hide_show_btn` click action.

Change the echomode of the `QLineEdit` & the button icon of `QPushButton`

Returns

void

Here is the call graph for this function:



6.18.2.13 on_hide_show_btn_clicked

```
void LoginUI::on_hide_show_btn_clicked () [private], [slot]
```

Listen to hide_show_btn click action.

Change the echomode of the QLineEdit & the button icon of QPushButton

Returns

void

Here is the call graph for this function:



6.18.2.14 on_pushButton_clicked

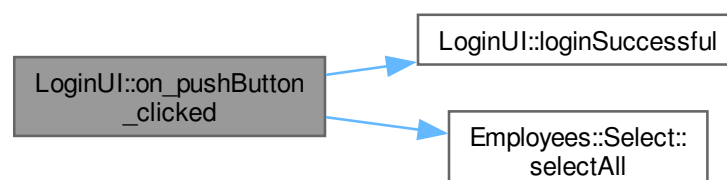
```
void LoginUI::on_pushButton_clicked () [private], [slot]
```

Listen to [Login](#) button click action.

Returns

void

Here is the call graph for this function:



6.18.2.15 on_reset_clicked

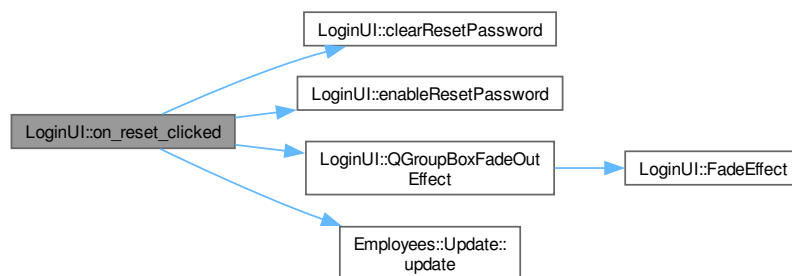
```
void LoginUI::on_reset_clicked () [private], [slot]
```

Verify the taped code and new/confirm password.

Returns

void

Here is the call graph for this function:



6.18.2.16 on_returnBtn_clicked

```
void LoginUI::on_returnBtn_clicked () [private], [slot]
```

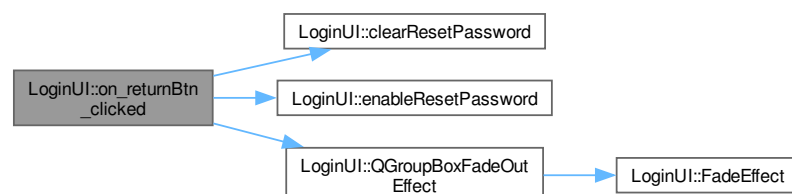
Listen to Return button click action.

This button is included inside the `f_pwd_interface` `QGroupBox`

Returns

void

Here is the call graph for this function:



6.18.2.17 on_sendEmailBtn_clicked

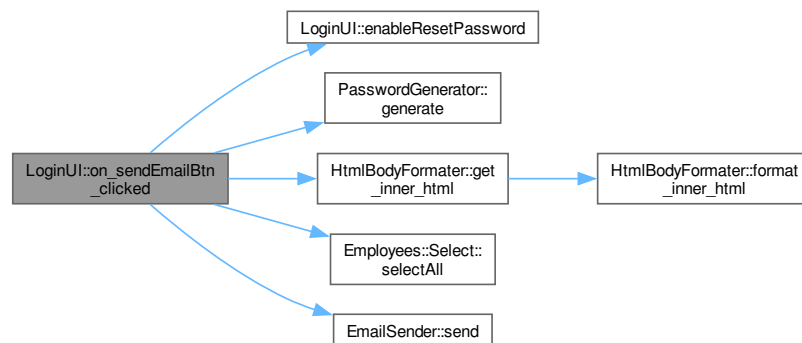
```
void LoginUI::on_sendEmailBtn_clicked () [private], [slot]
```

Verify the given mail and send a 6-chars reset code to reset the password.

Returns

void

Here is the call graph for this function:



6.18.2.18 QGroupBoxFadeOutEffect()

```
void LoginUI::QGroupBoxFadeOutEffect (
    QGroupBox * group1,
    QGroupBox * group2) [private]
```

Make a "fade out" effect to `group1` using opacity transition from `__FULL_VISIBLE__` to `__FULL__` `TRANSPARENT__` and hide `group1`, otherwise for `group2`

This effect work only with `QGroupBox`

| `QPropertyAnimation`

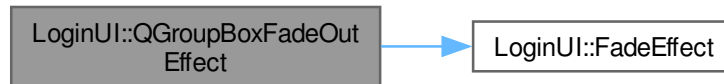
Parameters

<i>group1</i>	{QGroupBox *}
<i>group2</i>	{QGroupBox *}

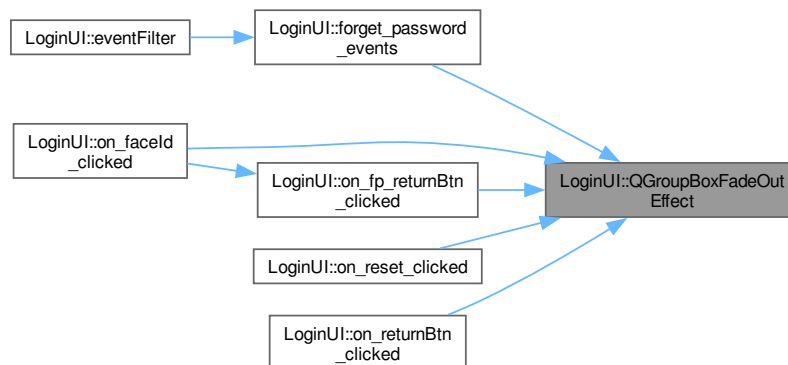
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:

**6.18.2.19 updateGif()**

```
void LoginUI::updateGif () [private]
```

Update the Gif every `__LOGIN_GIF_ANIMATION__`

This change make a beautiful animation

Returns

void

6.18.3 Member Data Documentation**6.18.3.1 camTimer**

```
QTimer * LoginUI::camTimer [private]
```


6.18.3.2 currentGifIndex

```
int LoginUI::currentGifIndex [private]
```

6.18.3.3 currentMovie

```
QMovie* LoginUI::currentMovie [private]
```

6.18.3.4 employee

```
SqlParam LoginUI::employee [private]
```

6.18.3.5 faceRecognition

```
FaceRecognizer* LoginUI::faceRecognition [private]
```

6.18.3.6 generated_password

```
std::string LoginUI::generated_password [private]
```

6.18.3.7 gifPaths

```
const QStringList LoginUI::gifPaths [private]
```

6.18.3.8 gifTimer

```
QTimer* LoginUI::gifTimer [private]
```

6.18.3.9 ui

```
Ui::Login* LoginUI::ui [private]
```

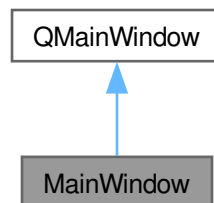
The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/LoginUI.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/LoginUI.cpp](#)

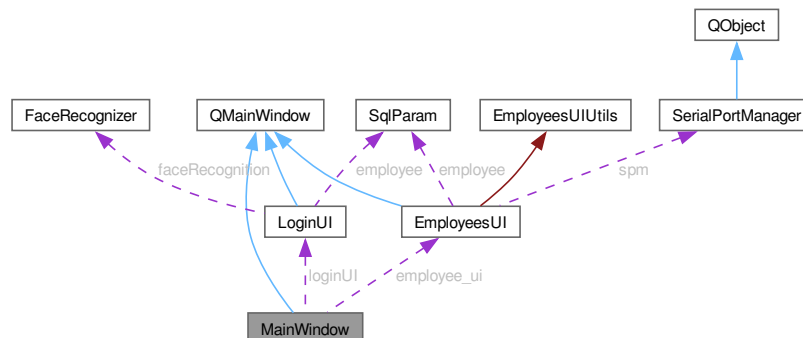
6.19 MainWindow Class Reference

```
#include <MainWindow.hpp>
```

Inheritance diagram for MainWindow:



Collaboration diagram for MainWindow:



Public Member Functions

- [MainWindow](#) ()
Construct a new `MainWindow::MainWindow` object.
- [~MainWindow](#) ()
Destroy the `MainWindow::MainWindow` object.

Private Attributes

- `QStackedWidget *` [stackedWidget](#)
- `LoginUI *` [loginUI](#)
- `EmployeesUI *` [employee_ui](#)

6.19.1 Constructor & Destructor Documentation

6.19.1.1 MainWindow()

`MainWindow::MainWindow ()`

Construct a new `MainWindow::MainWindow` object.

Here is the call graph for this function:



6.19.1.2 ~MainWindow()

`MainWindow::~~MainWindow ()`

Destroy the `MainWindow::MainWindow` object.

6.19.2 Member Data Documentation

6.19.2.1 employee_ui

`EmployeesUI* MainWindow::employee_ui [private]`

6.19.2.2 loginUI

`LoginUI* MainWindow::loginUI [private]`

6.19.2.3 stackedWidget

`QStackedWidget* MainWindow::stackedWidget [private]`

The documentation for this class was generated from the following files:

- `/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/MainWindow.hpp`
- `/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/MainWindow.cpp`

6.20 MediaPlayer Class Reference

[MediaPlayer](#) class.

```
#include <MediaPlayer.hpp>
```

Static Public Member Functions

- static void [playNotif](#) (const QString &, const int &volume=1)
Play notification sound.

6.20.1 Detailed Description

[MediaPlayer](#) class.

6.20.2 Member Function Documentation

6.20.2.1 playNotif()

```
void MediaPlayer::playNotif (
    const QString & path,
    const int & volume = 1) [static]
```

Play notification sound.

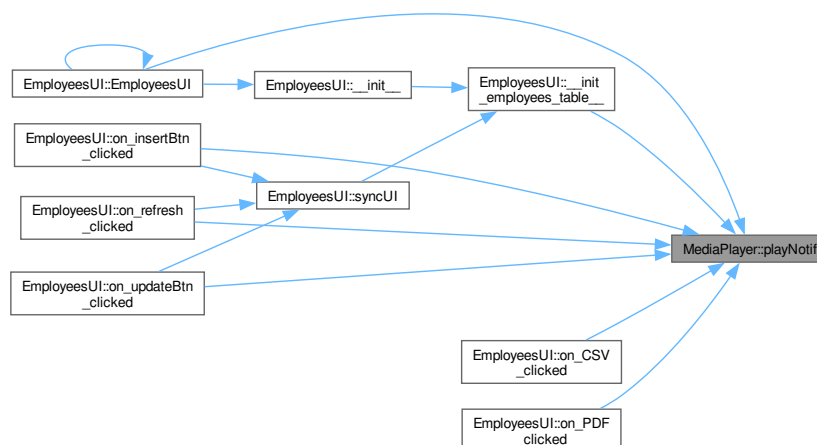
Parameters

<i>path</i>	{const QString &}
<i>volume</i>	{const int &}

Returns

void

Here is the caller graph for this function:



The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/MediaPlayer.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/MediaPlayer.cpp](#)

6.21 Melody Class Reference

[Melody](#) class.

```
#include <Melody.hpp>
```

Static Public Member Functions

- `template<unsigned int N>`
`static void play (const int(&sound)[2][N], unsigned int volume=100, const unsigned int &buzzer_↵`
`pin=BUZZER\_PIN)`

Static Public Attributes

- static constexpr int [succesSound](#) [2][3]
- static constexpr int [failSound](#) [2][3]

6.21.1 Detailed Description

[Melody](#) class.

6.21.2 Member Function Documentation

6.21.2.1 `play()`

```
template<unsigned int N>
static void Melody::play (
    const int(&) sound[2][N],
    unsigned int volume = 100,
    const unsigned int & buzzer_pin = BUZZER\_PIN) [inline], [static]
```

AnalogWrite doesn't take frequency, only duty cycleHere is the caller graph for this function:



6.21.3 Member Data Documentation

6.21.3.1 failSound

```
int Melody::failSound [static], [constexpr]
```

Initial value:

```
= {
    {NOTE_DS3, NOTE_D3, NOTE_CS3},
    {200, 200, 300}}
```

6.21.3.2 succesSound

```
int Melody::succesSound [static], [constexpr]
```

Initial value:

```
= {
    {NOTE_C5, NOTE_E5, NOTE_G5},
    {150, 150, 150}}
```

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/Melody.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/Melody.cpp](#)

6.22 PasswordGenerator Class Reference

```
#include <PasswordGenerator.hpp>
```

Static Public Member Functions

- static std::string [generate](#) (const int &length=6, const std::string &pattern=[PASSWORD_GENERATOR_DEFAULT_PATTERN](#))
Generate a random password using the system seed and "Mersenne twister algorithm".
- static std::string [sha256sum](#) (const std::string &)
Hash the given data on 32-bit format using sha256 encryption protocol.

6.22.1 Member Function Documentation

6.22.1.1 generate()

```
std::string PasswordGenerator::generate (
    const int & length = 6,
    const std::string & pattern = PASSWORD\_GENERATOR\_DEFAULT\_PATTERN) [static]
```

Generate a random password using the system seed and "Mersenne twister algorithm".

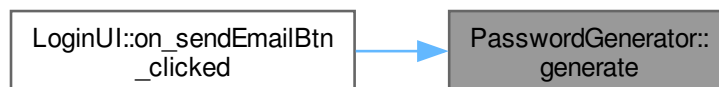
Parameters

<i>pattern</i>	{const std::string}
<i>length</i>	{const int}

Returns

std::string

Here is the caller graph for this function:



6.22.1.2 sha256sum()

```
std::string PasswordGenerator::sha256sum (  
    const std::string & data) [static]
```

Hash the given `data` on 32-bit format using sha256 encryption protocol.

SHA-256 is used for secure password hashing

Parameters

<i>data</i>	{const std::string}
-------------	---------------------

Returns

std::string

Note

Test with:

```
$printf "Terminal Root" | sha256sum
```

The documentation for this class was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/PasswordGenerator.hpp
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/PasswordGenerator.cpp

6.23 PdfGenerator Class Reference

```
#include <PdfGenerator.hpp>
```

Static Public Member Functions

- static const unsigned int [generatePdf](#) (const QString &filePath, const [SqlParam](#) &)
Generate a .pdf file from filePath, it return EXIT_SUCCESS if the .pdf file generated successfully, otherwise return EXIT_FAILURE

6.23.1 Member Function Documentation

6.23.1.1 generatePdf()

```
const unsigned int PdfGenerator::generatePdf (
    const QString & filePath,
    const SqlParam & emp) [static]
```

Generate a .pdf file from filePath, it return EXIT_SUCCESS if the .pdf file generated successfully, otherwise return EXIT_FAILURE

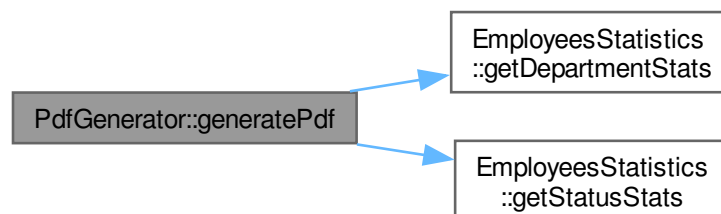
Parameters

<i>filePath</i>	{const QString &}
<i>emp</i>	{const SqlParam &} - Main user/employee data

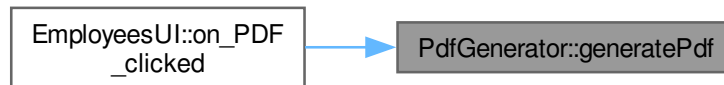
Returns

const unsigned int BlankLocalImageIssue

Here is the call graph for this function:



Here is the caller graph for this function:



The documentation for this class was generated from the following files:

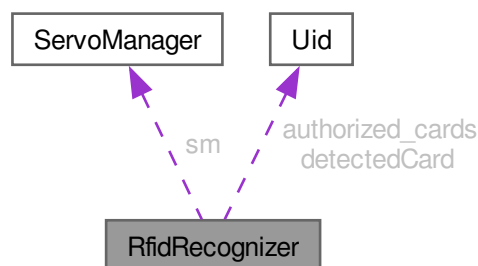
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/PdfGenerator.hpp
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/PdfGenerator.cpp

6.24 RfidRecognizer Class Reference

[RfidRecognizer](#) class.

```
#include <RfidRecognizer.hpp>
```

Collaboration diagram for RfidRecognizer:



Public Member Functions

- [RfidRecognizer](#) ()
Construct new `RfidRecognizer::RfidRecognizer` object.
- void [setup](#) ()
Setup `RfidRecognizer` object.
- void [recognize](#) ()
Listen to the `RFID-RC522` detector and verify thte existence of the `detectedCard`

Private Member Functions

- void `get_UID ()`
Store the detected UID in `detectedCard` @retrn void.
- const bool `is_exist ()`
Return *true* if the UID of the `detectedCard` exist in `authorized_cards`, otherwise return *false*
- const bool `listenFromExternal ()`
- const String `listenFromInternal ()` const
- void `notify (const uint8_t &) const`
Notify the user/employee physically.

Private Attributes

- MFRC522 `rc`
- `ServoManager` `sm`
- `Uid` `detectedCard`
- `Uid` `authorized_cards [MAX_CARDS]`

6.24.1 Detailed Description

`RfidRecognizer` class.

6.24.2 Constructor & Destructor Documentation

6.24.2.1 `RfidRecognizer()`

```
RfidRecognizer::RfidRecognizer () [explicit]
```

Construct new `RfidRecognizer::RfidRecognizer` object.

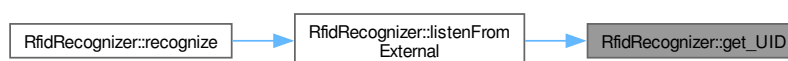
6.24.3 Member Function Documentation

6.24.3.1 `get_UID()`

```
void RfidRecognizer::get_UID () [inline], [private]
```

Store the detected UID in `detectedCard` @retrn void.

Here is the caller graph for this function:



6.24.3.2 is_exist()

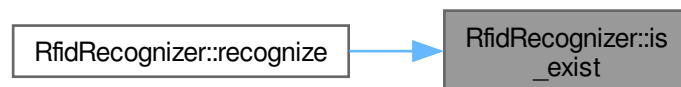
```
const bool RfidRecognizer::is_exist () [private]
```

Return `true` if the UID of the `detectedCard` exist in `authorized_cards`, otherwise return `false`

Returns

{const bool}

Here is the caller graph for this function:



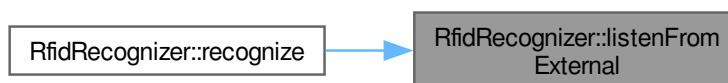
6.24.3.3 listenFromExternal()

```
const bool RfidRecognizer::listenFromExternal () [private]
```

Here is the call graph for this function:



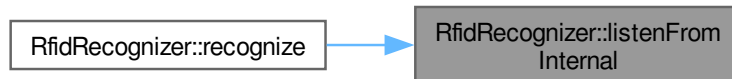
Here is the caller graph for this function:



6.24.3.4 listenFromInternal()

```
const String RfidRecognizer::listenFromInternal () const [private]
```

Here is the caller graph for this function:



6.24.3.5 notify()

```
void RfidRecognizer::notify (
    const uint8_t & led_pin) const [private]
```

Notify the user/employee physically.

Parameters

<i>msg</i>	{const String &}
<i>led_pin</i>	{const uint8_t &}

Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.24.3.6 recognize()

```
void RfidRecognizer::recognize ()
```

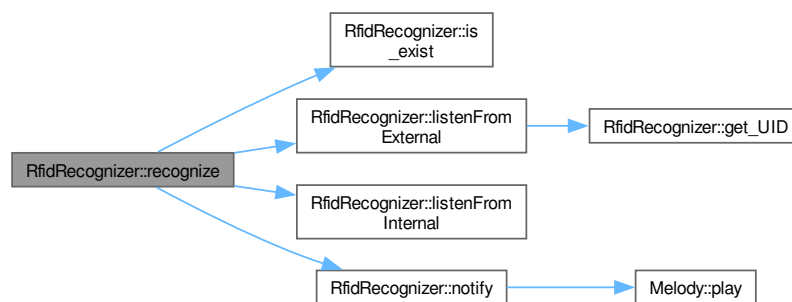
Listen to the RFID-RC522 detector and verify thte existence of the detectedCard

The main [RfidRecognizer](#) lancer function!

Returns

void

Here is the call graph for this function:



6.24.3.7 setup()

```
void RfidRecognizer::setup ()
```

Setup [RfidRecognizer](#) object.

The main [RfidRecognizer](#) setup function!

Returns

void

6.24.4 Member Data Documentation

6.24.4.1 authorized_cards

```
UId RfidRecognizer::authorized_cards[MAX_CARDS] [private]
```

6.24.4.2 detectedCard

```
UId RfidRecognizer::detectedCard [private]
```

6.24.4.3 rc

```
MFRC522 RfidRecognizer::rc [private]
```

6.24.4.4 sm

```
ServoManager RfidRecognizer::sm [private]
```

The documentation for this class was generated from the following files:

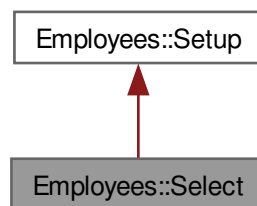
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/RfidRecognizer.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/RfidRecognizer.cpp](#)

6.25 Employees::Select Class Reference

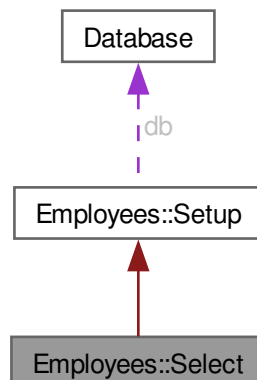
CRUD.

```
#include <Employees.hpp>
```

Inheritance diagram for Employees::Select:



Collaboration diagram for Employees::Select:



Public Member Functions

- `std::vector< SqlParam > selectAll ()`
Return all users without any exceptions.
- `std::vector< SqlParam > selectAll (const Employees::EmployeeInfo< std::string > &)`
- `std::vector< SqlParam > selectAll (const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &)`
- `std::vector< SqlParam > selectAllExcept (const Employees::EmployeeInfo< std::string > &)`
- `std::vector< SqlParam > selectAllExcept (const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &)`
- `std::vector< SqlParam > selectLastInsertedRow ()`
Select the last inserted employee from the [Employees](#) table.

Additional Inherited Members

Private Member Functions inherited from [Employees::Setup](#)

- `Setup ()`
Construct a new [Employees::Setup::Setup](#) object.
- `virtual ~Setup ()`
Destroy the [Employees::Setup::Setup](#) object.

Private Attributes inherited from [Employees::Setup](#)

- `Database * db`

6.25.1 Detailed Description

CRUD.

6.25.2 Member Function Documentation

6.25.2.1 `selectAll()` [1/3]

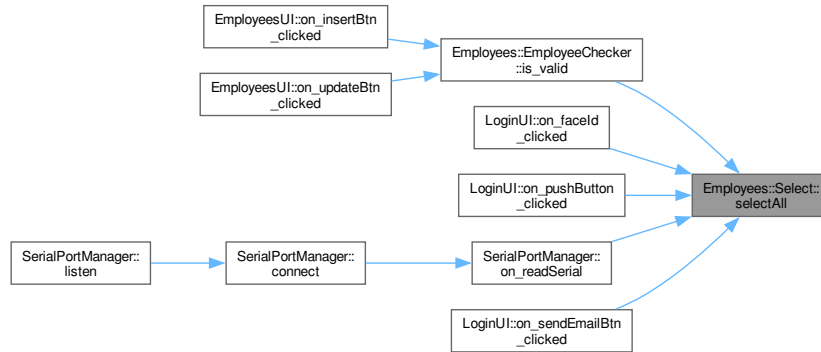
```
std::vector< SqlParam > Employees::Select::selectAll ()
```

Return all users without any exceptions.

Returns

`std::vector<SqlParam>`

Here is the caller graph for this function:



6.25.2.2 selectAll() [2/3]

```
std::vector< SqlParam > Employees::Select::selectAll (
    const Employees::EmployeeInfo< std::string > & info)
```

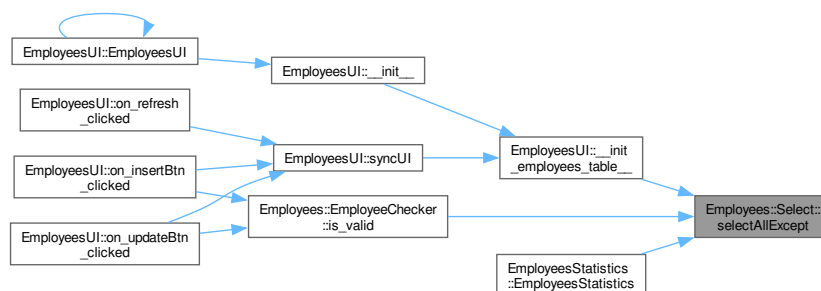
6.25.2.3 selectAll() [3/3]

```
std::vector< SqlParam > Employees::Select::selectAll (
    const Employees::EmployeeInfo< std::string > & info_00,
    const Employees::EmployeeInfo< std::string > & info_01)
```

6.25.2.4 selectAllExcept() [1/2]

```
std::vector< SqlParam > Employees::Select::selectAllExcept (
    const Employees::EmployeeInfo< std::string > & info)
```

Here is the caller graph for this function:



6.25.2.5 selectAllExcept() [2/2]

```
std::vector< SqlParam > Employees::Select::selectAllExcept (
    const Employees::EmployeeInfo< std::string > & info_00,
    const Employees::EmployeeInfo< std::string > & info_01)
```

6.25.2.6 selectLastInsertedRow()

```
std::vector< SqlParam > Employees::Select::selectLastInsertedRow ()
```

Select the last inserted employee from the [Employees](#) table.

Returns

```
std::vector<SqlParam>
```

The documentation for this class was generated from the following files:

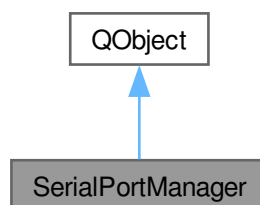
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp](#)

6.26 SerialPortManager Class Reference

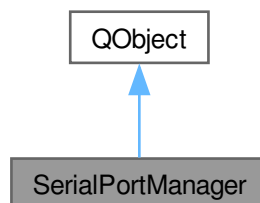
[SerialPortManager](#) class.

```
#include <SerialPortManager.hpp>
```

Inheritance diagram for SerialPortManager:



Collaboration diagram for SerialPortManager:



Public Member Functions

- [SerialPortManager](#) (QObject *)
Construct a new [SerialPortManager::SerialPortManager](#) object.
- [~SerialPortManager](#) ()
Destruct [SerialPortManager::SerialPortManager](#) object.
- void [details](#) () const
Prints the description, vendor id, and product id of all ports.
- void [listen](#) ()
The lancer [SerialPortManager](#) function.

Private Slots

- void [on_readSerial](#) ()
Read all available data from the port every `\n\r`

Private Member Functions

- void [connect](#) ()
Connect `ino` serial port to [EmployeesUI](#) on `SerialPort::readyRead` signal.
- const bool [is_inoAvailable](#) ()
Return `true` if the arduino uno connect Successfully (physically), otherwise retrun `false`
- void [setupSerialPort](#) ()
Setup the serial monitor reading port.
- void [openSerialPort](#) ()
Open the serial monitor reading port.

Private Attributes

- QObject * [ui](#)
- QSerialPort * [ino](#)
- const quint16 [inoVendorId](#)
- const quint16 [inoProductId](#)
- QString [serialBuffer](#)
- QString [parsed_data](#)
- QByteArray [serialData](#)
- QString [inoPortName](#)

6.26.1 Detailed Description

[SerialPortManager](#) class.

6.26.2 Constructor & Destructor Documentation

6.26.2.1 SerialPortManager()

```
SerialPortManager::SerialPortManager (
    QObject * parent)
```

Construct a new [SerialPortManager::SerialPortManager](#) object.

Parameters

<i>parent</i>	{QMainWindow *}
---------------	-----------------

6.26.2.2 ~SerialPortManager()

```
SerialPortManager::~~SerialPortManager ()
```

Destruct [SerialPortManager::SerialPortManager](#) object.

6.26.3 Member Function Documentation

6.26.3.1 connect()

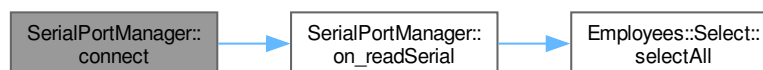
```
void SerialPortManager::connect () [private]
```

Connect ino serial port to [EmployeesUI](#) on `SerialPort::readyRead` signal.

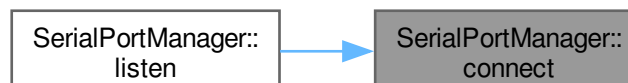
Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.3.2 details()

```
void SerialPortManager::details () const
```

Prints the description, vendor id, and product id of all ports.

[TESTING MODE] - Used it to determine the values for the arduino uno.

Returns

void

6.26.3.3 is_inoAvailable()

```
const bool SerialPortManager::is_inoAvailable () [private]
```

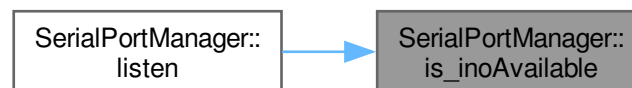
Return `true` if the arduino uno connect Successfully (physically), otherwise retrun `false`

In true result, we take the arduino connected port name

Returns

{const bool}

Here is the caller graph for this function:



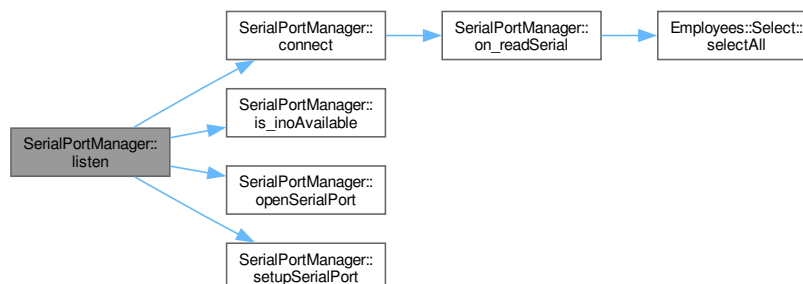
6.26.3.4 listen()

```
void SerialPortManager::listen ()
```

The lancer `SerialPortManager` function.

this function listen to serial monitor buffer and take action dependant on the recived data.

@retrun void Here is the call graph for this function:



6.26.3.5 on_readSerial

```
void SerialPortManager::on_readSerial () [private], [slot]
```

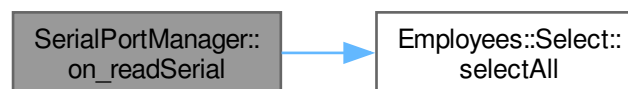
Read all available data from the port every `\n\r`

Append incoming data to `serialData` buffer, and split the buffer at the first `completeMessage`. Then we split the message using the " - " separator into `uid|part[0]` and `username|part[1]` (extracting process). After that we verify the `status` of the employee/user and depending on it we repend the serial monitor.

Returns

void

Here is the call graph for this function:



Here is the caller graph for this function:



6.26.3.6 openSerialPort()

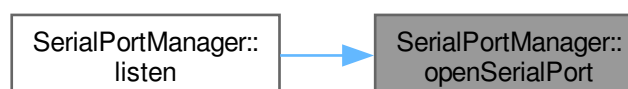
```
void SerialPortManager::openSerialPort () [inline], [private]
```

Open the serial monitor reading port.

Returns

void

Here is the caller graph for this function:



6.26.3.7 setupSerialPort()

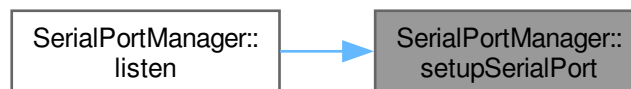
```
void SerialPortManager::setupSerialPort () [private]
```

Setup the serial monitor reading port.

Returns

void

Here is the caller graph for this function:



6.26.4 Member Data Documentation

6.26.4.1 ino

```
QSerialPort* SerialPortManager::ino [private]
```

6.26.4.2 inoPortName

```
QString SerialPortManager::inoPortName [private]
```

6.26.4.3 inoProductId

```
const quint16 SerialPortManager::inoProductId [private]
```

6.26.4.4 inoVendorId

```
const quint16 SerialPortManager::inoVendorId [private]
```

6.26.4.5 parsed_data

```
QString SerialPortManager::parsed_data [private]
```

6.26.4.6 serialBuffer

```
QString SerialPortManager::serialBuffer [private]
```

6.26.4.7 serialData

```
QByteArray SerialPortManager::serialData [private]
```

6.26.4.8 ui

```
QObject* SerialPortManager::ui [private]
```

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/ServoManager.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/ServoManager.cpp](#)

6.27 ServoManager Class Reference

```
#include <ServoManager.hpp>
```

Public Member Functions

- [ServoManager](#) ()
- [ServoManager](#) (int)
- void [setup](#) ()
- void [goToAngle](#) (const int &)

Private Attributes

- Servo [servo](#)
- int [angle](#)

6.27.1 Constructor & Destructor Documentation

6.27.1.1 ServoManager() [1/2]

```
ServoManager::ServoManager ()
```

6.27.1.2 ServoManager() [2/2]

```
ServoManager::ServoManager (
    int _angle)
```

6.27.2 Member Function Documentation

6.27.2.1 goToAngle()

```
void ServoManager::goToAngle (
    const int & newAngle)
```

6.27.2.2 setup()

```
void ServoManager::setup ()
```

6.27.3 Member Data Documentation

6.27.3.1 angle

```
int ServoManager::angle [private]
```

6.27.3.2 servo

```
Servo ServoManager::servo [private]
```

The documentation for this class was generated from the following files:

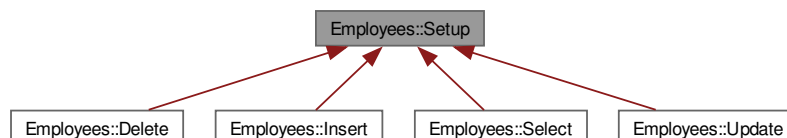
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/ServoManager.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/ServoManager.cpp](#)

6.28 Employees::Setup Class Reference

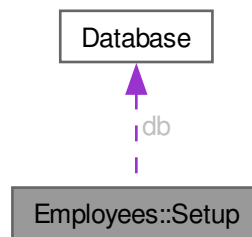
Generale database setup.

```
#include <Employees.hpp>
```

Inheritance diagram for Employees::Setup:



Collaboration diagram for Employees::Setup:



Public Member Functions

- [Setup](#) ()
Construct a new [Employees::Setup::Setup](#) object.
- virtual [~Setup](#) ()
Destroy the [Employees::Setup::Setup](#) object.

Protected Attributes

- [Database](#) * [db](#)

6.28.1 Detailed Description

Generale database setup.

6.28.2 Constructor & Destructor Documentation

6.28.2.1 Setup()

```
Employees::Setup::Setup ()
```

Construct a new [Employees::Setup::Setup](#) object.

6.28.2.2 ~Setup()

```
Employees::Setup::~Setup () [virtual]
```

Destroy the [Employees::Setup::Setup](#) object.

6.28.3 Member Data Documentation

6.28.3.1 db

`Database* Employees::Setup::db` [protected]

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp](#)

6.29 SqlParam Struct Reference

[SqlParam](#) struct ?

```
#include <Database.hpp>
```

Public Member Functions

- [SqlParam](#) (const std::vector< std::pair< unsigned int, std::string > > &strParams={}, const std::vector< std::pair< unsigned int, int > > &intParams={}, const std::vector< std::pair< unsigned int, oracle::occi::Date > > &dateParams={}, const std::vector< std::pair< unsigned int, oracle::occi::Timestamp > > ×tampsParams={})

Construct a new [SqlParam::SqlParam](#) object.

Public Attributes

- std::vector< std::pair< unsigned int, std::string > > [strings](#)
- std::vector< std::pair< unsigned int, int > > [integers](#)
- std::vector< std::pair< unsigned int, oracle::occi::Date > > [dates](#)
- std::vector< std::pair< unsigned int, oracle::occi::Timestamp > > [timestamps](#)

6.29.1 Detailed Description

[SqlParam](#) struct ?

We use param.first as the parameter name ? And param.second as the passed data

6.29.2 Constructor & Destructor Documentation

6.29.2.1 SqlParam()

```
SqlParam::SqlParam (
    const std::vector< std::pair< unsigned int, std::string > > & strParams = {},
    const std::vector< std::pair< unsigned int, int > > & intParams = {},
    const std::vector< std::pair< unsigned int, oracle::occi::Date > > & dateParams
= {},
    const std::vector< std::pair< unsigned int, oracle::occi::Timestamp > > & timestampsParams = {})) [explicit]
```

Construct a new [SqlParam::SqlParam](#) object.

Parameters

<i>strParams</i>	{const std::vector<std::pair<unsigned int, std::string>> &}
<i>intParams</i>	{const std::vector<std::pair<unsigned int, int>> &}
<i>dateParams</i>	{const std::vector<std::pair<unsigned int, oracle::occi::Date>> &}
<i>timestampsParams</i>	{const std::vector<std::pair<unsigned int, oracle::occi::Timestamp>> &}

6.29.3 Member Data Documentation

6.29.3.1 dates

```
std::vector<std::pair<unsigned int, oracle::occi::Date> > SqlParam::dates
```

6.29.3.2 integers

```
std::vector<std::pair<unsigned int, int> > SqlParam::integers
```

6.29.3.3 strings

```
std::vector<std::pair<unsigned int, std::string> > SqlParam::strings
```

6.29.3.4 timestamps

```
std::vector<std::pair<unsigned int, oracle::occi::Timestamp> > SqlParam::timestamps
```

The documentation for this struct was generated from the following files:

- /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/[Database.hpp](#)
- /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/[Database.cpp](#)

6.30 Uid Class Reference

[Uid](#) class.

```
#include <Uid.hpp>
```

Public Member Functions

- [Uid](#) ()
Construct new [Uid::Uid](#) object.
- [Uid](#) (const byte *, const String &)
Construct new [Uid::Uid](#) object.
- void [print](#) () const
Print the UID (name + data)
- void [print_name](#) () const
Print the name UID.
- void [print_data](#) () const
Print the data in hex format.
- void [clear](#) ()
Reset data to zeros and name to empty String.
- const byte * [get_data](#) () const
Get data (getter function)
- const String [get_name](#) () const
- bool [operator==](#) (const [Uid](#) &) const
Compare only [Uid](#)'s data.
- [Uid](#) & [operator<<](#) (const byte *)
Assign otherData to this->data
- [Uid](#) & [operator<<](#) (const String &)
Assign otherName to this->name

Private Attributes

- byte [data](#) [4]
- String [name](#)

6.30.1 Detailed Description

[Uid](#) class.

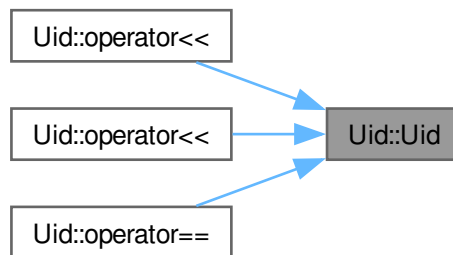
6.30.2 Constructor & Destructor Documentation

6.30.2.1 [Uid](#)() [1/2]

```
Uid::Uid () [explicit]
```

Construct new [Uid::Uid](#) object.

Here is the caller graph for this function:



6.30.2.2 Uid() [2/2]

```

Uid::Uid (
    const byte * _data,
    const String & _name) [explicit]
  
```

Construct new [Uid::Uid](#) object.

Parameters

<code>_data()</code>	{const byte *}
<code>_name</code>	{const String &}

6.30.3 Member Function Documentation

6.30.3.1 clear()

```
void Uid::clear ()
```

Reset `data` to zeros and `name` to empty String.

Returns

void

6.30.3.2 get_data()

```
const byte * Uid::get_data () const
```

Get `data` (getter function)

Returns

{const byte *}

6.30.3.3 get_name()

```
const String Uid::get_name () const
```

6.30.3.4 operator<<() [1/2]

```
Uid & Uid::operator<< (
    const byte * otherData)
```

Assign otherData to this->data

Parameters

<i>other</i>	{const byte *}
--------------	----------------

Returns

{Uid &}

Here is the call graph for this function:

**6.30.3.5 operator<<() [2/2]**

```
Uid & Uid::operator<< (
    const String & otherName)
```

Assign otherName to this->name

Parameters

<i>otherName</i>	{const String &}
------------------	------------------

Returns

{Uid &}

Here is the call graph for this function:



6.30.3.6 operator==()

```
bool Uid::operator== (
    const Uid & other) const
```

Compare only `Uid`'s data.

Parameters

<i>other</i>	{const <code>Uid</code> &}
--------------	----------------------------

Returns

bool

Here is the call graph for this function:



6.30.3.7 print()

```
void Uid::print () const
```

Print the UID (name + data)

Returns

void

6.30.3.8 print_data()

```
void Uid::print_data () const
```

Print the data in hex format.

Returns

void

6.30.3.9 print_name()

```
void Uid::print_name () const
```

Print the name UID.

Get name (getter function)

Returns

```
void  
{const byte *}
```

6.30.4 Member Data Documentation

6.30.4.1 data

```
byte Uid::data[4] [private]
```

6.30.4.2 name

```
String Uid::name [private]
```

The documentation for this class was generated from the following files:

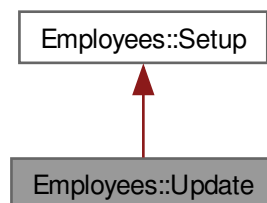
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/Uid.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/Uid.cpp](#)

6.31 Employees::Update Class Reference

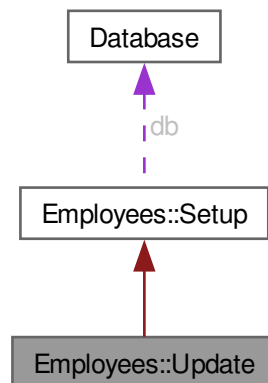
CRUD.

```
#include <Employees.hpp>
```

Inheritance diagram for Employees::Update:



Collaboration diagram for Employees::Update:



Public Member Functions

- `int update (const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &)`
- `int update (const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &)`
- `int update (const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< std::string > &, const Employees::EmployeeInfo< int > &, const Employees::EmployeeInfo< int > &, const Employees::EmployeeInfo< oracle::occi::Date > &)`

Update all employee data where username

Additional Inherited Members

Private Member Functions inherited from Employees::Setup

- `Setup ()`
Construct a new Employees::Setup::Setup object.
- `virtual ~Setup ()`
Destroy the Employees::Setup::Setup object.

Private Attributes inherited from Employees::Setup

- `Database * db`

6.31.1 Detailed Description

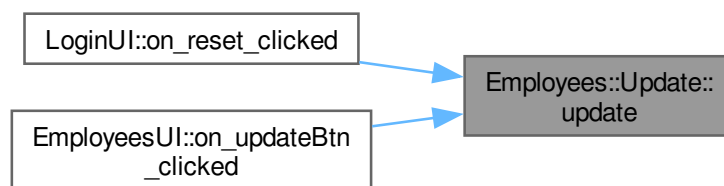
CRUD.

6.31.2 Member Function Documentation

6.31.2.1 update() [1/3]

```
int Employees::Update::update (
    const Employees::EmployeeInfo< std::string > & info_00,
    const Employees::EmployeeInfo< std::string > & info_01)
```

Here is the caller graph for this function:



6.31.2.2 update() [2/3]

```
int Employees::Update::update (
    const Employees::EmployeeInfo< std::string > & email,
    const Employees::EmployeeInfo< std::string > & firstname,
    const Employees::EmployeeInfo< std::string > & lastname,
    const Employees::EmployeeInfo< std::string > & password,
    const Employees::EmployeeInfo< std::string > & department,
    const Employees::EmployeeInfo< std::string > & username)
```

6.31.2.3 update() [3/3]

```
int Employees::Update::update (
    const Employees::EmployeeInfo< std::string > & username,
    const Employees::EmployeeInfo< std::string > & email,
    const Employees::EmployeeInfo< std::string > & firstName,
    const Employees::EmployeeInfo< std::string > & lastName,
    const Employees::EmployeeInfo< std::string > & password,
    const Employees::EmployeeInfo< std::string > & department,
    const Employees::EmployeeInfo< std::string > & jobTitle,
    const Employees::EmployeeInfo< std::string > & address,
    const Employees::EmployeeInfo< std::string > & imagePath,
    const Employees::EmployeeInfo< std::string > & status,
    const Employees::EmployeeInfo< int > & salary,
    const Employees::EmployeeInfo< int > & tel,
    const Employees::EmployeeInfo< oracle::occi::Date > & birthdate)
```

[Update](#) all employee data where username

Parameters

<i>username</i>	{const Employees::EmployeeInfo<std::string> &}
<i>email</i>	{const Employees::EmployeeInfo<std::string> &}
<i>firstname</i>	{const Employees::EmployeeInfo<std::string> &}
<i>lastname</i>	{const Employees::EmployeeInfo<std::string> &}
<i>password</i>	{const Employees::EmployeeInfo<std::string> &}
<i>department</i>	{const Employees::EmployeeInfo<std::string> &}
<i>jobTitle</i>	{const Employees::EmployeeInfo<std::string> &}
<i>address</i>	{const Employees::EmployeeInfo<std::string> &}
<i>imagePath</i>	{const Employees::EmployeeInfo<std::string> &}
<i>status</i>	{const Employees::EmployeeInfo<std::string> &}
<i>salary</i>	{const Employees::EmployeeInfo<int> &}
<i>tel</i>	{const Employees::EmployeeInfo<int> &}
<i>birthdate</i>	{const Employees::EmployeeInfo<oracle::occi::Date> &}

Returns

int

The documentation for this class was generated from the following files:

- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/Employees.hpp](#)
- [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Employees.cpp](#)

Chapter 7

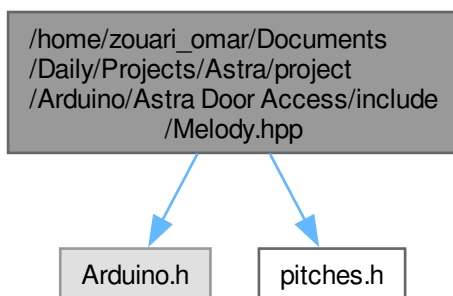
File Documentation

7.1 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/Melody.hpp File Reference

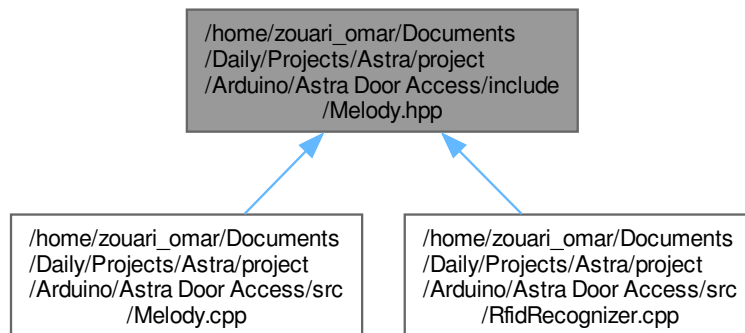
Melody header file.

```
#include <Arduino.h>
#include "pitches.h"
```

Include dependency graph for Melody.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `Melody`
Melody class.

Macros

- `#define BUZZER_PIN 3`

7.1.1 Detailed Description

`Melody` header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-26

Copyright

Copyright (c) 2025 Melody.hpp

7.1.2 Macro Definition Documentation

7.1.2.1 BUZZER_PIN

```
#define BUZZER_PIN 3
```

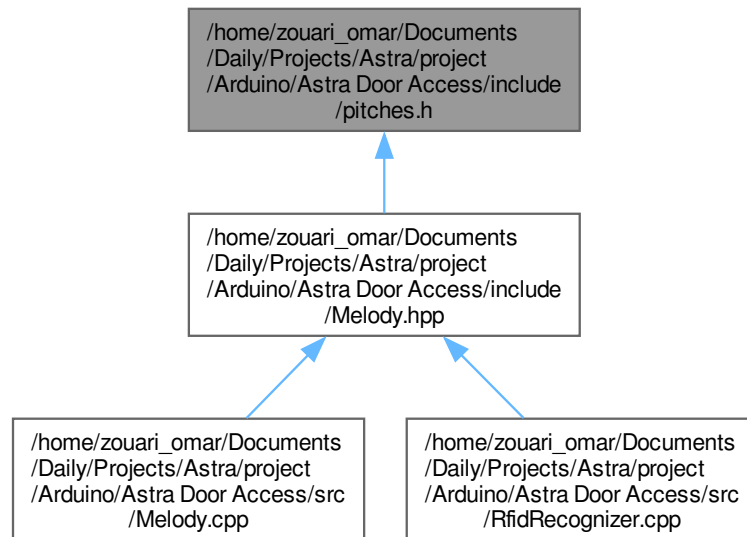
7.2 Melody.hpp

[Go to the documentation of this file.](#)

```
00001
00010
00011 /// Pre-processor prototype declaration part
00012 #ifndef __MELODY_HPP__
00013 #define __MELODY_HPP__
00014 #define BUZZER_PIN 3
00015
00016 /// Include prototype declaration part
00017 /// Include std Arduino header(s)
00018 #include <Arduino.h>
00019
00020 /// Include custom header(s)
00021 #include "pitches.h"
00022
00023 /// Class prototype declaration part
00024
00029 class Melody {
00030 public:
00031     static constexpr int succesSound[2][3] = {
00032         {NOTE_C5, NOTE_E5, NOTE_G5}, /// `succesSound` melody
00033         {150, 150, 150}}; /// `succesSound` durations
00034
00035     static constexpr int failSound[2][3] = {
00036         {NOTE_DS3, NOTE_D3, NOTE_CS3}, /// `failSound` melody
00037         {200, 200, 300}}; /// `failSound` durations
00038
00049     template <unsigned int N>
00050     static void play(const int (&sound)[2][N], unsigned int volume = 100, const unsigned int &buzzer_pin
= BUZZER_PIN) {
00051         volume = constrain(volume, 0, 100); /// Ensure the volume is between 0 and 100
00052
00053         for (unsigned int i{}; i < N; ++i) {
00054             unsigned int note(sound[0][i]), duration(sound[1][i]);
00055
00056             /// Convert frequency (note) to period (in microseconds)
00057             unsigned long period = 1000000L / note, /// Period of the note in microseconds
00058             pulseWidth = period * volume / 100; /// Duty cycle to simulate volume
00059
00060             /// Play the note with PWM
00062             for (unsigned long j{}; j < duration * 1000L; j += period) {
00063                 analogWrite(buzzer_pin, 127); /// Set to half the maximum value for a 50% duty cycle
00064                 delayMicroseconds(pulseWidth); /// High for `pulseWidth` time
00065                 analogWrite(buzzer_pin, 0); /// Turn off buzzer
00066                 delayMicroseconds(period - pulseWidth); /// Low for the rest of the period
00067             }
00068
00069             noTone(buzzer_pin); /// Make sure no tone is playing
00070             delay(duration * 1.3); /// Extra delay to prevent overlapping sounds
00071         }
00072     }
00073 }; /// Melody classs
00074
00075 #endif /// __MELODY_HPP__
```

7.3 /home/zouari_omar/Documents/Daily/Projects/Astra/project/↔ Arduino/Astra Door Access/include/pitches.h File Reference

This graph shows which files directly or indirectly include this file:



Macros

- #define NOTE_B0 31
- #define NOTE_C1 33
- #define NOTE_CS1 35
- #define NOTE_D1 37
- #define NOTE_DS1 39
- #define NOTE_E1 41
- #define NOTE_F1 44
- #define NOTE_FS1 46
- #define NOTE_G1 49
- #define NOTE_GS1 52
- #define NOTE_A1 55
- #define NOTE_AS1 58
- #define NOTE_B1 62
- #define NOTE_C2 65
- #define NOTE_CS2 69
- #define NOTE_D2 73
- #define NOTE_DS2 78
- #define NOTE_E2 82
- #define NOTE_F2 87
- #define NOTE_FS2 93
- #define NOTE_G2 98
- #define NOTE_GS2 104
- #define NOTE_A2 110

- `#define NOTE_AS2` 117
- `#define NOTE_B2` 123
- `#define NOTE_C3` 131
- `#define NOTE_CS3` 139
- `#define NOTE_DB3` 139
- `#define NOTE_D3` 147
- `#define NOTE_DS3` 156
- `#define NOTE_EB3` 156
- `#define NOTE_E3` 165
- `#define NOTE_F3` 175
- `#define NOTE_FS3` 185
- `#define NOTE_G3` 196
- `#define NOTE_GS3` 208
- `#define NOTE_A3` 220
- `#define NOTE_AS3` 233
- `#define NOTE_B3` 247
- `#define NOTE_C4` 262
- `#define NOTE_CS4` 277
- `#define NOTE_D4` 294
- `#define NOTE_DS4` 311
- `#define NOTE_E4` 330
- `#define NOTE_F4` 349
- `#define NOTE_FS4` 370
- `#define NOTE_G4` 392
- `#define NOTE_GS4` 415
- `#define NOTE_A4` 440
- `#define NOTE_AS4` 466
- `#define NOTE_B4` 494
- `#define NOTE_C5` 523
- `#define NOTE_CS5` 554
- `#define NOTE_D5` 587
- `#define NOTE_DS5` 622
- `#define NOTE_E5` 659
- `#define NOTE_F5` 698
- `#define NOTE_FS5` 740
- `#define NOTE_G5` 784
- `#define NOTE_GS5` 831
- `#define NOTE_A5` 880
- `#define NOTE_AS5` 932
- `#define NOTE_B5` 988
- `#define NOTE_C6` 1047
- `#define NOTE_CS6` 1109
- `#define NOTE_D6` 1175
- `#define NOTE_DS6` 1245
- `#define NOTE_E6` 1319
- `#define NOTE_F6` 1397
- `#define NOTE_FS6` 1480
- `#define NOTE_G6` 1568
- `#define NOTE_GS6` 1661
- `#define NOTE_A6` 1760
- `#define NOTE_AS6` 1865
- `#define NOTE_B6` 1976
- `#define NOTE_C7` 2093
- `#define NOTE_CS7` 2217
- `#define NOTE_D7` 2349

- `#define` [NOTE_DS7](#) 2489
- `#define` [NOTE_E7](#) 2637
- `#define` [NOTE_F7](#) 2794
- `#define` [NOTE_FS7](#) 2960
- `#define` [NOTE_G7](#) 3136
- `#define` [NOTE_GS7](#) 3322
- `#define` [NOTE_A7](#) 3520
- `#define` [NOTE_AS7](#) 3729
- `#define` [NOTE_B7](#) 3951
- `#define` [NOTE_C8](#) 4186
- `#define` [NOTE_CS8](#) 4435
- `#define` [NOTE_D8](#) 4699
- `#define` [NOTE_DS8](#) 4978
- `#define` [REST](#) 0

7.3.1 Macro Definition Documentation

7.3.1.1 NOTE_A1

```
#define NOTE_A1 55
```

7.3.1.2 NOTE_A2

```
#define NOTE_A2 110
```

7.3.1.3 NOTE_A3

```
#define NOTE_A3 220
```

7.3.1.4 NOTE_A4

```
#define NOTE_A4 440
```

7.3.1.5 NOTE_A5

```
#define NOTE_A5 880
```

7.3.1.6 NOTE_A6

```
#define NOTE_A6 1760
```

7.3.1.7 NOTE_A7

```
#define NOTE_A7 3520
```

7.3.1.8 NOTE_AS1

```
#define NOTE_AS1 58
```

7.3.1.9 NOTE_AS2

```
#define NOTE_AS2 117
```

7.3.1.10 NOTE_AS3

```
#define NOTE_AS3 233
```

7.3.1.11 NOTE_AS4

```
#define NOTE_AS4 466
```

7.3.1.12 NOTE_AS5

```
#define NOTE_AS5 932
```

7.3.1.13 NOTE_AS6

```
#define NOTE_AS6 1865
```

7.3.1.14 NOTE_AS7

```
#define NOTE_AS7 3729
```

7.3.1.15 NOTE_B0

```
#define NOTE_B0 31
```

7.3.1.16 NOTE_B1

```
#define NOTE_B1 62
```

7.3.1.17 NOTE_B2

```
#define NOTE_B2 123
```

7.3.1.18 NOTE_B3

```
#define NOTE_B3 247
```

7.3.1.19 NOTE_B4

```
#define NOTE_B4 494
```

7.3.1.20 NOTE_B5

```
#define NOTE_B5 988
```

7.3.1.21 NOTE_B6

```
#define NOTE_B6 1976
```

7.3.1.22 NOTE_B7

```
#define NOTE_B7 3951
```

7.3.1.23 NOTE_C1

```
#define NOTE_C1 33
```

7.3.1.24 NOTE_C2

```
#define NOTE_C2 65
```

7.3.1.25 NOTE_C3

```
#define NOTE_C3 131
```

7.3.1.26 NOTE_C4

```
#define NOTE_C4 262
```

7.3.1.27 NOTE_C5

```
#define NOTE_C5 523
```

7.3.1.28 NOTE_C6

```
#define NOTE_C6 1047
```

7.3.1.29 NOTE_C7

```
#define NOTE_C7 2093
```

7.3.1.30 NOTE_C8

```
#define NOTE_C8 4186
```

7.3.1.31 NOTE_CS1

```
#define NOTE_CS1 35
```

7.3.1.32 NOTE_CS2

```
#define NOTE_CS2 69
```

7.3.1.33 NOTE_CS3

```
#define NOTE_CS3 139
```

7.3.1.34 NOTE_CS4

```
#define NOTE_CS4 277
```

7.3.1.35 NOTE_CS5

```
#define NOTE_CS5 554
```

7.3.1.36 NOTE_CS6

```
#define NOTE_CS6 1109
```

7.3.1.37 NOTE_CS7

```
#define NOTE_CS7 2217
```

7.3.1.38 NOTE_CS8

```
#define NOTE_CS8 4435
```

7.3.1.39 NOTE_D1

```
#define NOTE_D1 37
```

7.3.1.40 NOTE_D2

```
#define NOTE_D2 73
```

7.3.1.41 NOTE_D3

```
#define NOTE_D3 147
```

7.3.1.42 NOTE_D4

```
#define NOTE_D4 294
```

7.3.1.43 NOTE_D5

```
#define NOTE_D5 587
```

7.3.1.44 NOTE_D6

```
#define NOTE_D6 1175
```

7.3.1.45 NOTE_D7

```
#define NOTE_D7 2349
```

7.3.1.46 NOTE_D8

```
#define NOTE_D8 4699
```

7.3.1.47 NOTE_DB3

```
#define NOTE_DB3 139
```

7.3.1.48 NOTE_DS1

```
#define NOTE_DS1 39
```

7.3.1.49 NOTE_DS2

```
#define NOTE_DS2 78
```

7.3.1.50 NOTE_DS3

```
#define NOTE_DS3 156
```

7.3.1.51 NOTE_DS4

```
#define NOTE_DS4 311
```

7.3.1.52 NOTE_DS5

```
#define NOTE_DS5 622
```

7.3.1.53 NOTE_DS6

```
#define NOTE_DS6 1245
```

7.3.1.54 NOTE_DS7

```
#define NOTE_DS7 2489
```

7.3.1.55 NOTE_DS8

```
#define NOTE_DS8 4978
```

7.3.1.56 NOTE_E1

```
#define NOTE_E1 41
```

7.3.1.57 NOTE_E2

```
#define NOTE_E2 82
```

7.3.1.58 NOTE_E3

```
#define NOTE_E3 165
```

7.3.1.59 NOTE_E4

```
#define NOTE_E4 330
```

7.3.1.60 NOTE_E5

```
#define NOTE_E5 659
```

7.3.1.61 NOTE_E6

```
#define NOTE_E6 1319
```

7.3.1.62 NOTE_E7

```
#define NOTE_E7 2637
```

7.3.1.63 NOTE_EB3

```
#define NOTE_EB3 156
```

7.3.1.64 NOTE_F1

```
#define NOTE_F1 44
```

7.3.1.65 NOTE_F2

```
#define NOTE_F2 87
```

7.3.1.66 NOTE_F3

```
#define NOTE_F3 175
```

7.3.1.67 NOTE_F4

```
#define NOTE_F4 349
```


7.3.1.68 NOTE_F5

```
#define NOTE_F5 698
```

7.3.1.69 NOTE_F6

```
#define NOTE_F6 1397
```

7.3.1.70 NOTE_F7

```
#define NOTE_F7 2794
```

7.3.1.71 NOTE_FS1

```
#define NOTE_FS1 46
```

7.3.1.72 NOTE_FS2

```
#define NOTE_FS2 93
```

7.3.1.73 NOTE_FS3

```
#define NOTE_FS3 185
```

7.3.1.74 NOTE_FS4

```
#define NOTE_FS4 370
```

7.3.1.75 NOTE_FS5

```
#define NOTE_FS5 740
```

7.3.1.76 NOTE_FS6

```
#define NOTE_FS6 1480
```

7.3.1.77 NOTE_FS7

```
#define NOTE_FS7 2960
```

7.3.1.78 NOTE_G1

```
#define NOTE_G1 49
```

7.3.1.79 NOTE_G2

```
#define NOTE_G2 98
```

7.3.1.80 NOTE_G3

```
#define NOTE_G3 196
```

7.3.1.81 NOTE_G4

```
#define NOTE_G4 392
```

7.3.1.82 NOTE_G5

```
#define NOTE_G5 784
```

7.3.1.83 NOTE_G6

```
#define NOTE_G6 1568
```

7.3.1.84 NOTE_G7

```
#define NOTE_G7 3136
```

7.3.1.85 NOTE_GS1

```
#define NOTE_GS1 52
```

7.3.1.86 NOTE_GS2

```
#define NOTE_GS2 104
```

7.3.1.87 NOTE_GS3

```
#define NOTE_GS3 208
```

7.3.1.88 NOTE_GS4

```
#define NOTE_GS4 415
```

7.3.1.89 NOTE_GS5

```
#define NOTE_GS5 831
```

7.3.1.90 NOTE_GS6

```
#define NOTE_GS6 1661
```

7.3.1.91 NOTE_GS7

```
#define NOTE_GS7 3322
```

7.3.1.92 REST

```
#define REST 0
```

7.4 pitches.h

[Go to the documentation of this file.](#)

```
00001 // -----
00002 // Copyright (c) 2021 HiBit <https://www.hibit.dev>
00003 // -----
00004
00005 #ifndef pitches_h
00006 #define pitches_h
00007
00008 #define NOTE_B0 31
00009 #define NOTE_C1 33
00010 #define NOTE_CS1 35
00011 #define NOTE_D1 37
00012 #define NOTE_DS1 39
00013 #define NOTE_E1 41
00014 #define NOTE_F1 44
00015 #define NOTE_FS1 46
00016 #define NOTE_G1 49
00017 #define NOTE_GS1 52
00018 #define NOTE_A1 55
00019 #define NOTE_AS1 58
00020 #define NOTE_B1 62
00021 #define NOTE_C2 65
00022 #define NOTE_CS2 69
00023 #define NOTE_D2 73
00024 #define NOTE_DS2 78
00025 #define NOTE_E2 82
00026 #define NOTE_F2 87
00027 #define NOTE_FS2 93
00028 #define NOTE_G2 98
00029 #define NOTE_GS2 104
00030 #define NOTE_A2 110
00031 #define NOTE_AS2 117
00032 #define NOTE_B2 123
00033 #define NOTE_C3 131
00034 #define NOTE_CS3 139
00035 #define NOTE_DB3 139
00036 #define NOTE_D3 147
00037 #define NOTE_DS3 156
```

```

00038 #define NOTE_EB3 156
00039 #define NOTE_E3 165
00040 #define NOTE_F3 175
00041 #define NOTE_FS3 185
00042 #define NOTE_G3 196
00043 #define NOTE_GS3 208
00044 #define NOTE_A3 220
00045 #define NOTE_AS3 233
00046 #define NOTE_B3 247
00047 #define NOTE_C4 262
00048 #define NOTE_CS4 277
00049 #define NOTE_D4 294
00050 #define NOTE_DS4 311
00051 #define NOTE_E4 330
00052 #define NOTE_F4 349
00053 #define NOTE_FS4 370
00054 #define NOTE_G4 392
00055 #define NOTE_GS4 415
00056 #define NOTE_A4 440
00057 #define NOTE_AS4 466
00058 #define NOTE_B4 494
00059 #define NOTE_C5 523
00060 #define NOTE_CS5 554
00061 #define NOTE_D5 587
00062 #define NOTE_DS5 622
00063 #define NOTE_E5 659
00064 #define NOTE_F5 698
00065 #define NOTE_FS5 740
00066 #define NOTE_G5 784
00067 #define NOTE_GS5 831
00068 #define NOTE_A5 880
00069 #define NOTE_AS5 932
00070 #define NOTE_B5 988
00071 #define NOTE_C6 1047
00072 #define NOTE_CS6 1109
00073 #define NOTE_D6 1175
00074 #define NOTE_DS6 1245
00075 #define NOTE_E6 1319
00076 #define NOTE_F6 1397
00077 #define NOTE_FS6 1480
00078 #define NOTE_G6 1568
00079 #define NOTE_GS6 1661
00080 #define NOTE_A6 1760
00081 #define NOTE_AS6 1865
00082 #define NOTE_B6 1976
00083 #define NOTE_C7 2093
00084 #define NOTE_CS7 2217
00085 #define NOTE_D7 2349
00086 #define NOTE_DS7 2489
00087 #define NOTE_E7 2637
00088 #define NOTE_F7 2794
00089 #define NOTE_FS7 2960
00090 #define NOTE_G7 3136
00091 #define NOTE_GS7 3322
00092 #define NOTE_A7 3520
00093 #define NOTE_AS7 3729
00094 #define NOTE_B7 3951
00095 #define NOTE_C8 4186
00096 #define NOTE_CS8 4435
00097 #define NOTE_D8 4699
00098 #define NOTE_DS8 4978
00099 #define REST 0
00100
00101 #endif

```

7.5 /home/zouari_omar/Documents/Daily/Projects/Astra/project/↵ Arduino/Astra Door Access/include/RfidRecognizer.hpp File Reference

[RfidRecognizer](#) header file.

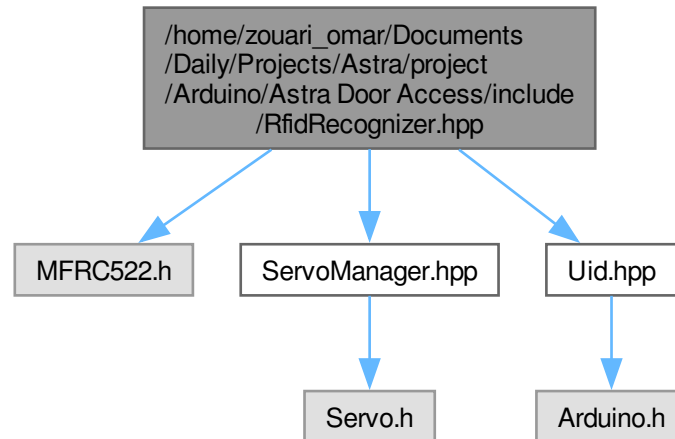
```

#include <MFRC522.h>
#include "ServoManager.hpp"

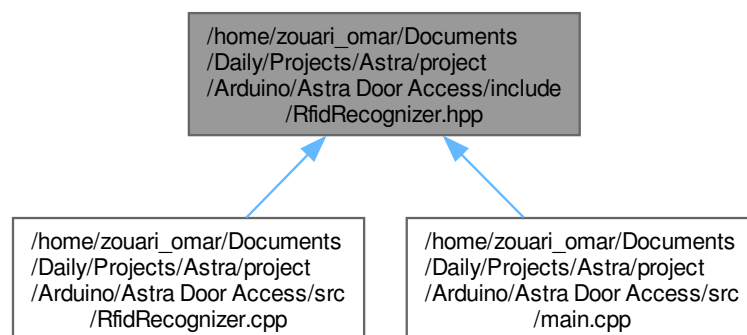
```

```
#include "Uid.hpp"
```

Include dependency graph for RfidRecognizer.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `RfidRecognizer`
RfidRecognizer class.

Macros

- `#define MAX_CARDS 2`
- `#define ACCESS_SUCCESS_LED_PIN 6`

- #define ACCESS_DENIED_LED_PIN 7
- #define RSTPIN 9
- #define SSPIN 10
- #define ACCESS_SUCCESS_LED_PIN 6
- #define ACCESS_DENIED_LED_PIN 7
- #define AUTHORIZATION_DENIED_MSG "CARD NOT AUTHORISED"
- #define AUTHORIZATION_SUCCESS_MSG "CARD AUTHORISED"

7.5.1 Detailed Description

RfidRecognizer header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-23

Copyright

Copyright (c) 2025 RfidRecognizer.hpp

7.5.2 Macro Definition Documentation

7.5.2.1 ACCESS_DENIED_LED_PIN [1/2]

```
#define ACCESS_DENIED_LED_PIN 7
```

7.5.2.2 ACCESS_DENIED_LED_PIN [2/2]

```
#define ACCESS_DENIED_LED_PIN 7
```

7.5.2.3 ACCESS_SUCCESS_LED_PIN [1/2]

```
#define ACCESS_SUCCESS_LED_PIN 6
```

7.5.2.4 ACCESS_SUCCESS_LED_PIN [2/2]

```
#define ACCESS_SUCCESS_LED_PIN 6
```

7.5.2.5 AUTHORIZATION_DENIED_MSG

```
#define AUTHORIZATION_DENIED_MSG "CARD NOT AUTHORISED"
```

7.5.2.6 AUTHORIZATION_SUCCESS_MSG

```
#define AUTHORIZATION_SUCCESS_MSG "CARD AUTHORISED"
```

7.5.2.7 MAX_CARDS

```
#define MAX_CARDS 2
```

7.5.2.8 RSTPIN

```
#define RSTPIN 9
```

7.5.2.9 SSPIN

```
#define SSPIN 10
```

7.6 RfidRecognizer.hpp

[Go to the documentation of this file.](#)

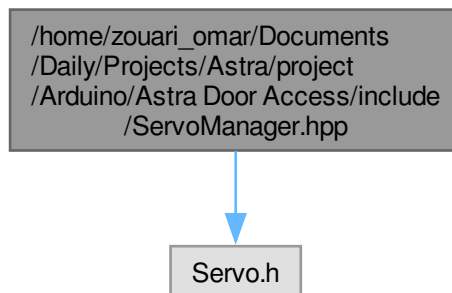
```
00001
00010
00011 /// Pre-processor prototype declaration part
00012 #ifndef __RFID_RECOGNIZER_HPP__
00013 #define __RFID_RECOGNIZER_HPP__
00014 #define MAX_CARDS 2
00015 #define ACCESS_SUCCESS_LED_PIN 6
00016 #define ACCESS_DENIED_LED_PIN 7
00017 #define RSTPIN 9
00018 #define SSPIN 10
00019 #define ACCESS_SUCCESS_LED_PIN 6
00020 #define ACCESS_DENIED_LED_PIN 7
00021 #define AUTHORIZATION_DENIED_MSG "CARD NOT AUTHORISED"
00022 #define AUTHORIZATION_SUCCESS_MSG "CARD AUTHORISED"
00023
00024 /// Include prototype declaration part
00025 /// Include std Arduino header(s)
00026 #include <MFRC522.h>
00027
00028 /// Include custom header(s)
00029 #include "ServoManager.hpp"
00030 #include "Uid.hpp"
00031
00032 /// Class prototype declaration part
00033
00034 class RfidRecognizer {
00035 public:
00040     explicit RfidRecognizer();
00041     void setup();
00042     void recognize();
00043
00044 private:
00045     MFRC522 rc;
00046     ServoManager sm;
00047     Uid detectedCard, // A single UID object to hold the detected card
00048         authorized_cards[MAX_CARDS]; // Store the authorized UIDs and their associated names
00049     inline void get_UID();
00050     const bool is_exist();
00051     const bool listenFromExternal();
00052     const String listenFromInternal() const;
00053     void notify(const uint8_t &) const;
00054 }; // RfidRecognizer class
00055
00056 #endif // __RFID_RECOGNIZER_HPP__
```

7.7 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/ServoManager.hpp File Reference

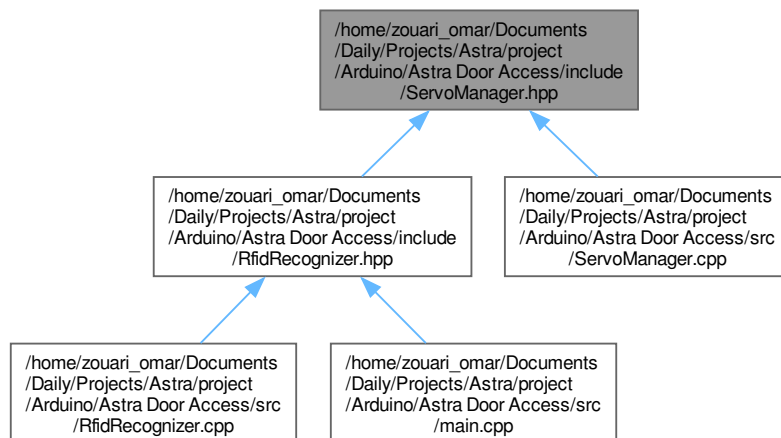
[ServoManager](#) header file.

```
#include <Servo.h>
```

Include dependency graph for ServoManager.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [ServoManager](#)

Macros

- #define `SERVO_PIN` 4

7.7.1 Detailed Description

`ServoManager` header file.

Author

@ZouariOmar (`zouariomar20@gmail.com`)

Version

0.1

Date

2025-04-26

Copyright

Copyright (c) 2025

VCC(Red wire) - 5V SIG(yellow/orange) - `SERVO_PIN` GND(Black/Brown) - GND SG90 `ServoManager.hpp`

7.7.2 Macro Definition Documentation**7.7.2.1 `SERVO_PIN`**

```
#define SERVO_PIN 4
```

7.8 `ServoManager.hpp`

[Go to the documentation of this file.](#)

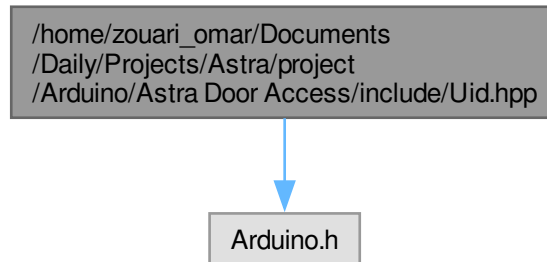
```
00001
00014
00015 /// Pre-processor prototype declaration part
00016 #ifndef __SERVO_MANAGER_HPP__
00017 #define __SERVO_MANAGER_HPP__
00018 #define SERVO_PIN 4
00019
00020 /// Include prototype declaration part
00021 /// Include std Arduino header(s)
00022 #include <Servo.h>
00023
00024 /// Class prototype declaration part
00025
00026 class ServoManager {
00027 private:
00028     Servo servo;
00029     int angle;
00030
00031 public:
00032     ServoManager();
00033     ServoManager(int);
00034     void setup();
00035     void goToAngle(const int &);
00036 }; // ServoManager class
00037
00038 #endif // __SERVO_MANAGER_HPP__
```

7.9 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/include/Uid.hpp File Reference

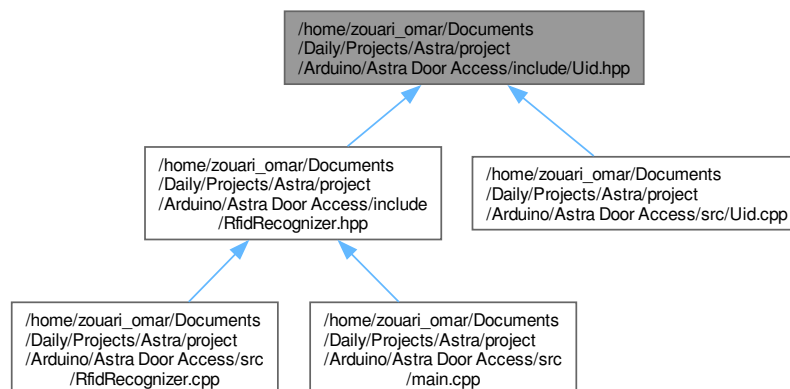
Uid header file.

```
#include <Arduino.h>
```

Include dependency graph for Uid.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [Uid](#)
Uid class.

7.9.1 Detailed Description

[Uid](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-25

Copyright

Copyright (c) 2025 Uid.hpp

7.10 Uid.hpp

[Go to the documentation of this file.](#)

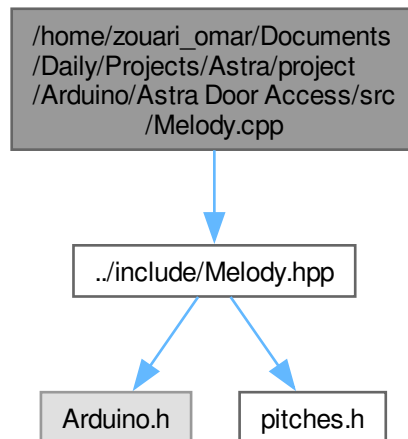
```
00001
00010
00011 /// Pre-processor prototype declaration part
00012 #ifndef __UID_HPP__
00013 #define __UID_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include std Arduino header(s)
00017 #include <Arduino.h>
00018
00019 /// Class prototype declaration part
00020
00025 class Uid {
00026 private:
00027     byte data[4];
00028     String name;
00029
00030 public:
00031     explicit Uid();
00032     explicit Uid(const byte *, const String &);
00033     void print() const;
00034     void print_name() const;
00035     void print_data() const;
00036     void clear();
00037     const byte *get_data() const;
00038     const String get_name() const;
00039     bool operator==(const Uid &) const;
00040     Uid &operator<<(const byte *);
00041     Uid &operator<<(const String &);
00042 }; // Uid class
00043
00044 #endif // __UID_HPP__
```

7.11 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/Melody.cpp File Reference

[Melody](#) source file.

```
#include "../include/Melody.hpp"
```

Include dependency graph for Melody.cpp:



7.11.1 Detailed Description

[Melody](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-26

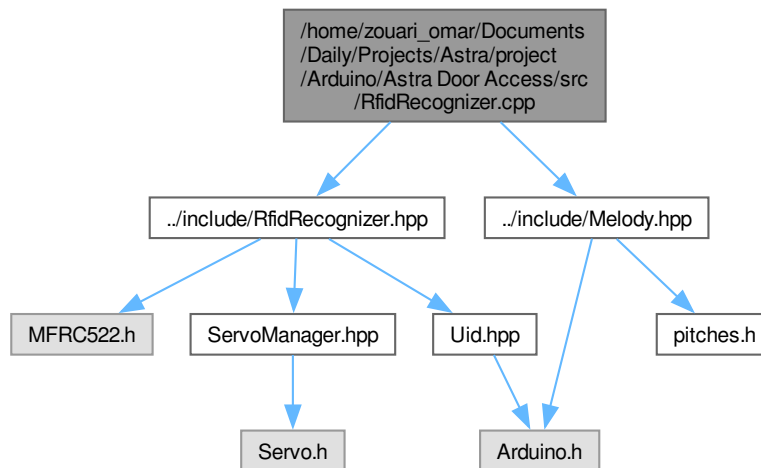
Copyright

Copyright (c) 2025 ZouariOmar

7.12 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/RfidRecognizer.cpp File Reference

[RfidRecognizer](#) source file.

```
#include "../include/RfidRecognizer.hpp"  
#include "../include/Melody.hpp"  
Include dependency graph for RfidRecognizer.cpp:
```



7.12.1 Detailed Description

[RfidRecognizer](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-23

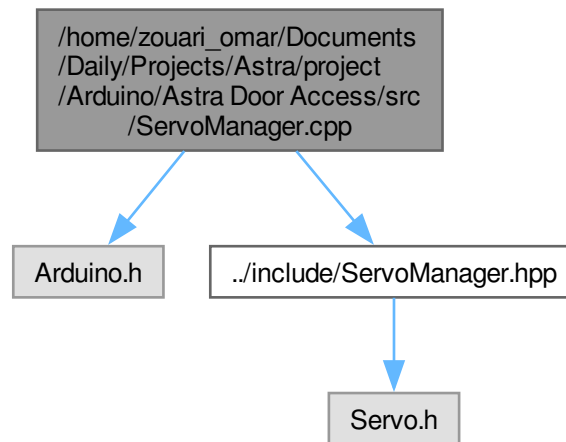
Copyright

Copyright (c) 2025 RfidRecognizer.cpp

7.13 /home/zouari_omar/Documents/Daily/Projects/Astra/project/↵ Arduino/Astra Door Access/src/ServoManager.cpp File Reference

[ServoManager](#) source file.

```
#include <Arduino.h>
#include "../include/ServoManager.hpp"
Include dependency graph for ServoManager.cpp:
```



7.13.1 Detailed Description

[ServoManager](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-26

Copyright

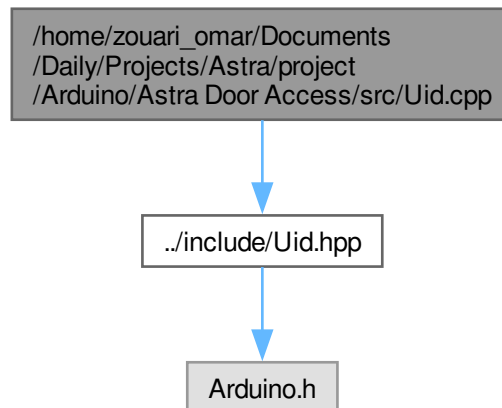
Copyright (c) 2025 ServoManager.cpp

7.14 /home/zouari_omar/Documents/Daily/Projects/Astra/project/Arduino/Astra Door Access/src/Uid.cpp File Reference ↩

Uid source file.

```
#include "../include/Uid.hpp"
```

Include dependency graph for Uid.cpp:



7.14.1 Detailed Description

Uid source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-25

Copyright

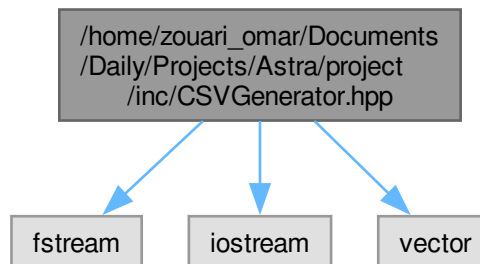
Copyright (c) 2025 Uid.cpp

7.15 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ CSVGenerator.hpp File Reference

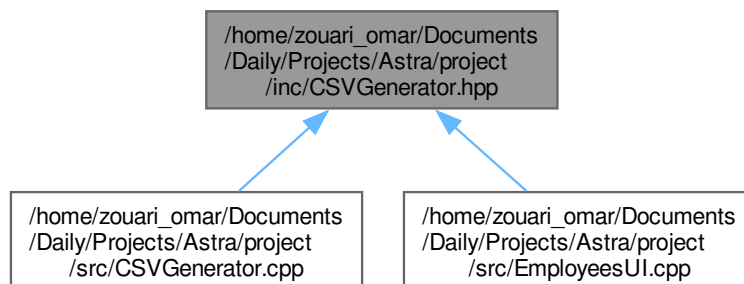
[CSVGenerator](#) header file.

```
#include <fstream>
#include <iostream>
#include <vector>
```

Include dependency graph for CSVGenerator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [CSVGenerator](#)
CSVGenerator class.

7.15.1 Detailed Description

[CSVGenerator](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-16

Copyright

Copyright (c) 2025 CSVGenerator.hpp

7.16 CSVGenerator.hpp

[Go to the documentation of this file.](#)

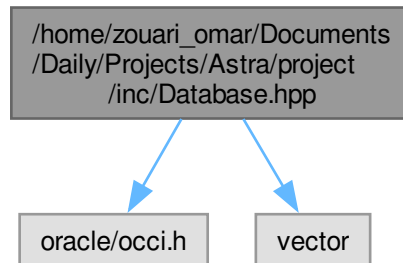
```
00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __CSVGENERATOR_HPP__
00013 #define __CSVGENERATOR_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include the stander c++ header(s)
00017 #include <fstream>
00018 #include <iostream>
00019 #include <vector>
00020
00021 /// Classes prototype declaration part
00022
00027 class CSVGenerator {
00028 public:
00029     static const unsigned int generateCSV(const std::string &, const
        std::vector<std::vector<std::string>> &);
00030 }; // CSVGenerator class
00031
00032 #endif // __CSVGENERATOR_HPP__
```

7.17 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ Database.hpp File Reference

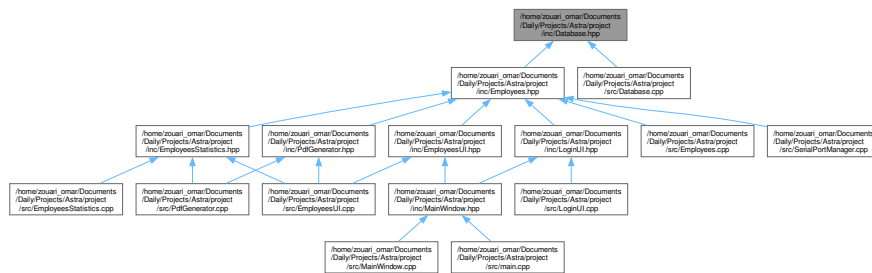
[Database](#) header file.

```
#include <oracle/occi.h>
#include <vector>
```

Include dependency graph for Database.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [SqlParam](#)
SqlParam struct ?
- class [Database](#)
Database class.

7.17.1 Detailed Description

[Database](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-13

Copyright

Copyright (c) 2025 oracle Database.hpp

7.18 Database.hpp

[Go to the documentation of this file.](#)

```

00001
00012
00013 /// Pre-Processor prototype declaration part
00014 #ifndef __DATABASE_HPP__
00015 #define __DATABASE_HPP__
00016
00017 /// Include prototype declaration part
00018 /// Include std c++ header(s)
00019 #include <oracle/occi.h>
00020 #include <vector>
00021
00022 /// Use workspaces
00023 using namespace oracle::occi;
00024
00025 /// Struct prototype declaration part
00026
00027 struct SqlParam {
00028     std::vector<std::pair<unsigned int, std::string>
00029         strings; /// Parameters of type string
00030     std::vector<std::pair<unsigned int, int> integers; /// Parameters of type int
00031     std::vector<std::pair<unsigned int, oracle::occi::Date>
00032         dates; /// Parameters of type dates
00033     std::vector<std::pair<unsigned int, oracle::occi::Timestamp>
00034         timestamps; /// Parameters of type timestamp
00035     explicit SqlParam(
00036         const std::vector<std::pair<unsigned int, std::string> &strParams = {},
00037         const std::vector<std::pair<unsigned int, int> &intParams = {},
00038         const std::vector<std::pair<unsigned int, oracle::occi::Date>
00039             &dateParams = {},
00040         const std::vector<std::pair<unsigned int, oracle::occi::Timestamp>
00041             &timestampsParams = {});
00042 }; /// SqlParam struct
00043
00044 /// Classes prototype declaration part
00045
00046 class Database {
00047 private:
00048     /// Database private vars
00049     Environment *env;
00050     Connection *conn;
00051
00052     /// Database private functions
00053     inline void setSqlParams(Statement *, const SqlParam &);
00054
00055 public:
00056     /// Database constructors and destructor
00057     Database();
00058     Database(const char *, const char *, const char *);
00059     ~Database();
00060
00061     /// Database public functions
00062     void execute(const std::string &query, const SqlParam &, int &affectedRows);
00063     std::vector<SqlParam> execute(const std::string &, const SqlParam &params);
00064 }; /// Database class
00065
00066 #endif /// __DATABASE_HPP__

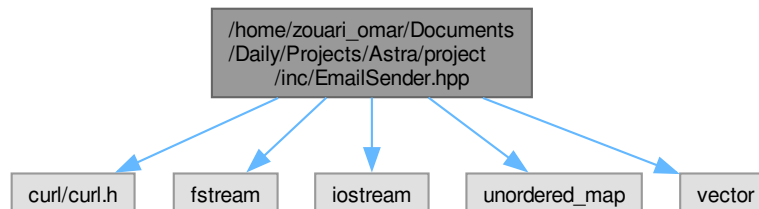
```

7.19 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/EmailSender.hpp File Reference

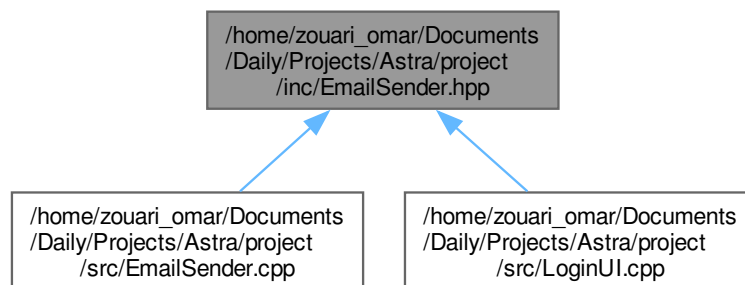
[EmailSender](#) header file.

```
#include <curl/curl.h>
#include <fstream>
#include <iostream>
#include <unordered_map>
#include <vector>
```

Include dependency graph for EmailSender.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [EmailAuth](#)
EmailAuth struct.
- struct [EmailData](#)
EmailData struct.
- class [EmailSender](#)
EmailSender class.

7.19.1 Detailed Description

[EmailSender](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-23

Copyright

Copyright (c) 2025 EmailSender.hpp smtp-mail.c

7.20 EmailSender.hpp

[Go to the documentation of this file.](#)

```

00001
00011
00024
00025 /// Pre-Processor prototype declaration part
00026 #ifndef __EMAIL_SENDER_HPP__
00027 #define __EMAIL_SENDER_HPP__
00028
00029 /// Include prototype declaration part
00030 // * Include std C++ header
00031 #include <curl/curl.h>
00032 #include <fstream>
00033 #include <iostream>
00034 #include <unordered_map>
00035 #include <vector>
00036
00037 /// Structres prototype declaration part
00038
00043 struct EmailAuth {
00044     std::string from_addr, from_app_password, mail_server;
00045     explicit EmailAuth();
00046     explicit EmailAuth(const std::string, const std::string, const std::string _mail_server =
std::getenv("MAIL_SERVER"));
00047 }; // EmailAuth struct
00048
00053 struct EmailData {
00054     std::string to_addr, subject, body;
00055     std::vector<std::string> cc_addr, attachments;
00056     explicit EmailData(const std::string, const std::string, const std::string, const
std::vector<std::string> _cc_addr = {}, const std::vector<std::string> _attachments = {});
00057 }; // EmailData struct
00058
00059 /// Classes prototype declaration part
00060
00065 class EmailSender {
00066 public:
00067     explicit EmailSender(const EmailAuth);
00068     ~EmailSender();
00069     int send(EmailData);
00070
00071 private:
00072     EmailAuth *auth;
00073 }; // EmailSender class
00074
00075 #endif // __EMAIL_SENDER_HPP__

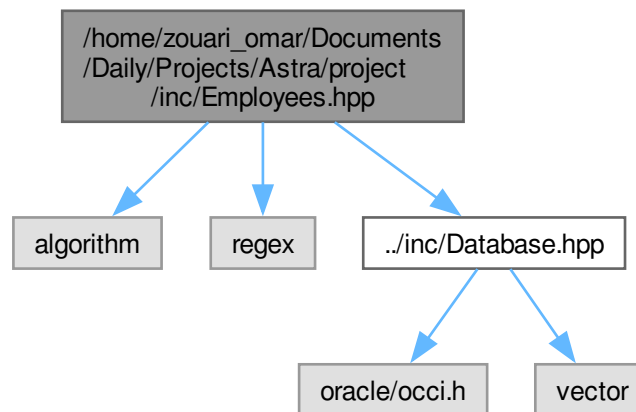
```

7.21 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ Employees.hpp File Reference

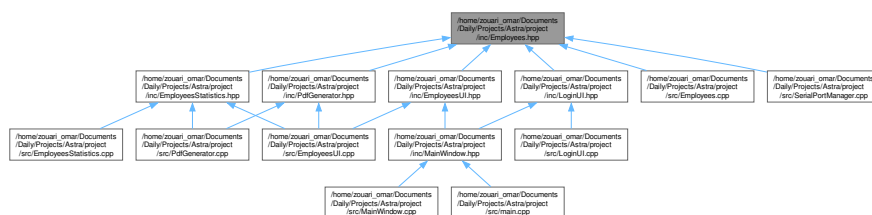
[Employees](#) header file.

```
#include <algorithm>
#include <regex>
#include "../inc/Database.hpp"
```

Include dependency graph for Employees.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- struct [Employees::EmployeeInfo< T >](#)
EmployeeInfo struct.
- class [Employees::Setup](#)
Generale database setup.
- class [Employees::Select](#)
CRUD.
- class [Employees::Insert](#)
Insert new employee.

- class [Employees::Update](#)
CRUD.
- class [Employees::Delete](#)
CRUD
- class [Employees::EmployeeChecker](#)
Check the passed employees data.

Namespaces

- namespace [Employees](#)
[Employees](#) namespace.

Macros

- #define [EMAIL_MATCHER_PATTERN](#) `"(\\w+)(\\.|_)?(\\w*)@(\\w+)(\\.|(\\w+))+"`

Enumerations

- enum [Employees::EmployeeQueueFlags_integers](#) { [Employees::EMPLOYEE_ID](#) , [Employees::SALARY](#) , [Employees::PHONE_NUMBER](#) }
EmployeeQueueFlags_integers enum.
- enum [Employees::EmployeeQueueFlags_strings](#) { [Employees::USERNAME](#) , [Employees::FIRSTNAME](#) , [Employees::LASTNAME](#) , [Employees::EMAIL](#) , [Employees::JOBTITLE](#) , [Employees::STATUS](#) , [Employees::PASSWORD](#) , [Employees::ADDRESS](#) , [Employees::DEPARTMENT](#) , [Employees::PROFILE_IMAGE_PATH](#) }
EmployeeQueueFlags_strings enum.
- enum [Employees::EmployeeQueueFlags_dates](#) { [Employees::HIRE_DATE](#) , [Employees::BIRTHDATE](#) }
EmployeeQueueFlags_dates enum.
- enum [Employees::EmployeeQueueFlags_timestamps](#) { [Employees::LAST_LOGIN](#) , [Employees::LAST_PASSWORD_RESET](#) }
EmployeeQueueFlags_timestamps enum.
- enum [Employees::EmployeeQueueFlags_blobs](#)
EmployeeQueueFlags_blobs enum.
- enum [Employees::EmployeeCheckerFlags](#) { [Employees::EMPTY](#) = -1 , [Employees::OK](#) , [Employees::REFUSED](#) , [Employees::UNIQUE_EMAIL](#) , [Employees::UNIQUE_EMAIL_WITH_EXCEPTION](#) }
EmployeeCheckerFlags enum.
- enum [Employees::EmployeeStatusFlags](#) { [Employees::ALL](#) , [Employees::ACTIVE](#) , [Employees::INACTIVE](#) , [Employees::SUSPENDED](#) }
- enum [Employees::EmployeeDepartmentFlags](#) { [Employees::COMMERCIAL](#) , [Employees::SHOPS](#) , [Employees::PARTNERS](#) , [Employees::EVENTS](#) , [Employees::PERSONALS](#) , [Employees::EMPLOYEES](#) }
EmployeeDepartmentFlags enum.

Functions

- std::string [Employees::EmployeeStatusString](#) (const [EmployeeStatusFlags](#) &)
Return the corresponding [EmployeeStatusFlags](#) string.

7.21.1 Detailed Description

[Employees](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-01

Copyright

Copyright (c) 2025 Employees.hpp

7.21.2 Macro Definition Documentation

7.21.2.1 EMAIL_MATCHER_PATTERN

```
#define EMAIL_MATCHER_PATTERN " (\\w+) (\\.|_|_)? (\\w*) @ (\\w+) (\\. (\\w+)) +"
```

7.22 Employees.hpp

[Go to the documentation of this file.](#)

```
00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __EMPLOYEES_HPP__
00013 #define __EMPLOYEES_HPP__
00014 #define EMAIL_MATCHER_PATTERN " (\\w+) (\\.|_|_)? (\\w*) @ (\\w+) (\\. (\\w+)) +"
00015
00016 /// Include prototype declaration part
00017 /// Include std C++ header
00018 #include <algorithm>
00019 #include <regex>
00020
00021 /// Include custom header(s)
00022 #include "../inc/Database.hpp"
00023
00024 /// Namespaces declaration part
00025
00030 namespace Employees {
00031
00036 enum EmployeeQueueFlags_integers {
00037     EMPLOYEE_ID,
00038     SALARY,
00039     PHONE_NUMBER
00040 }; // EmployeeQueueFlags_integers enum
00041
00046 enum EmployeeQueueFlags_strings {
00047     USERNAME,
00048     FIRSTNAME,
00049     LASTNAME,
00050     EMAIL,
00051     JOBTITLE,
00052     STATUS,
```



```

00053     PASSWORD,
00054     ADDRESS,
00055     DEPARTMENT,
00056     PROFILE_IMAGE_PATH
00057 }; // EmployeeQueueFlags_strings enum
00058
00063 enum EmployeeQueueFlags_dates {
00064     HIRE_DATE,
00065     BIRTHDATE
00066 }; // EmployeeQueueFlags_dates enum
00067
00072 enum EmployeeQueueFlags_timestamps {
00073     LAST_LOGIN,
00074     LAST_PASSWORD_RESET
00075 }; // EmployeeQueueFlags_timestamps enum
00076
00081 enum EmployeeQueueFlags_blobs {
00082 }; // EmployeeQueueFlags_blobs enum
00083
00088 enum EmployeeCheckerFlags {
00089     EMPTY = -1,
00090     OK,
00091     REFUSED,
00092     UNIQUE_EMAIL,
00093     UNIQUE_EMAIL_WITH_EXCEPTION
00094 }; // EmployeeCheckerFlags enum
00095
00100 enum EmployeeStatusFlags {
00101     ALL,
00102     ACTIVE,
00103     INACTIVE,
00104     SUSPENDED
00105 }; // EmployeesStatusFlags
00106
00111 enum EmployeeDepartmentFlags {
00112     COMMERCIAL,
00113     SHOPS,
00114     PARTNERS,
00115     EVENTS,
00116     PERSONALS,
00117     EMPLOYEES
00118 }; // EmployeeDepartmentFlags
00119
00120 std::string EmployeeStatusString(const EmployeeStatusFlags &);
00121 template <typename T>
00122 struct EmployeeInfo;
00123 class Setup;
00124 class Select;
00125 class Insert;
00126 class Update;
00127 class Delete;
00128 class EmployeeChecker;
00129 } // namespace Employees
00130
00131 /* Structure declaration part
00132
00137 template <typename T>
00138 struct Employees::EmployeeInfo {
00139     T data;
00140     std::string arg;
00141     explicit EmployeeInfo(const T &, const Employees::EmployeeQueueFlags_integers &);
00142     explicit EmployeeInfo(const T &, const Employees::EmployeeQueueFlags_strings &);
00143     explicit EmployeeInfo(const T &, const Employees::EmployeeQueueFlags_dates &);
00144     explicit EmployeeInfo(const T &, const Employees::EmployeeQueueFlags_timestamps &);
00145 }; // EmployeeInfo struct
00146
00147 /* Class(es) prototype declaration part
00148
00153 class Employees::Setup {
00154 protected:
00155     Database *db;
00156
00157 public:
00158     Setup();
00159     virtual ~Setup();
00160 }; // Employees::Setup class
00161
00166 class Employees::Select : private Employees::Setup {
00167 public:
00168     /* Employees::Select functions
00169     std::vector<SqlParameter> selectAll();
00170     std::vector<SqlParameter> selectAll(const Employees::EmployeeInfo<std::string> &);
00171     std::vector<SqlParameter> selectAll(const Employees::EmployeeInfo<std::string> &, const
Employees::EmployeeInfo<std::string> &);
00172     std::vector<SqlParameter> selectAllExcept(const Employees::EmployeeInfo<std::string> &);
00173     std::vector<SqlParameter> selectAllExcept(const Employees::EmployeeInfo<std::string> &, const
Employees::EmployeeInfo<std::string> &);

```

```

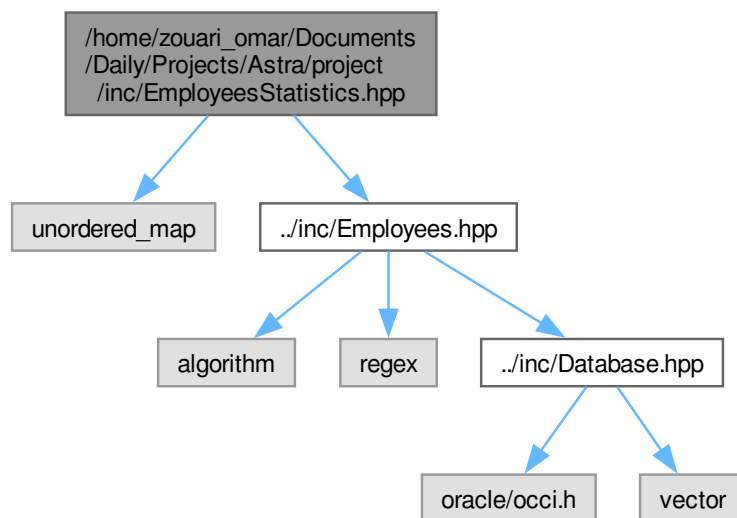
00174     std::vector<SqlParameter> selectLastInsertedRow();
00175 }; // Employees::Select class
00176
00181 class Employees::Insert : private Employees::Setup {
00182 public:
00183     /* Employees::Insert function(s)
00184     int insert(const Employees::EmployeeInfo<std::string> &,
00185               const Employees::EmployeeInfo<std::string> &,
00186               const Employees::EmployeeInfo<std::string> &,
00187               const Employees::EmployeeInfo<std::string> &,
00188               const Employees::EmployeeInfo<std::string> &);
00189
00190     int insert(const Employees::EmployeeInfo<std::string> &,
00191               const Employees::EmployeeInfo<std::string> &,
00192               const Employees::EmployeeInfo<std::string> &,
00193               const Employees::EmployeeInfo<std::string> &,
00194               const Employees::EmployeeInfo<std::string> &,
00195               const Employees::EmployeeInfo<std::string> &,
00196               const Employees::EmployeeInfo<std::string> &,
00197               const Employees::EmployeeInfo<std::string> &,
00198               const Employees::EmployeeInfo<int> &,
00199               const Employees::EmployeeInfo<int> &,
00200               const Employees::EmployeeInfo<oracle::occi::Date> &);
00201 }; // Employees::Insert class
00202
00207 class Employees::Update : private Employees::Setup {
00208 public:
00209     /* Employees::Update functions
00210     int update(const Employees::EmployeeInfo<std::string> &,
00211               const Employees::EmployeeInfo<std::string> &);
00212
00213     int update(const Employees::EmployeeInfo<std::string> &,
00214               const Employees::EmployeeInfo<std::string> &,
00215               const Employees::EmployeeInfo<std::string> &,
00216               const Employees::EmployeeInfo<std::string> &,
00217               const Employees::EmployeeInfo<std::string> &,
00218               const Employees::EmployeeInfo<std::string> &);
00219
00220     int update(const Employees::EmployeeInfo<std::string> &,
00221               const Employees::EmployeeInfo<std::string> &,
00222               const Employees::EmployeeInfo<std::string> &,
00223               const Employees::EmployeeInfo<std::string> &,
00224               const Employees::EmployeeInfo<std::string> &,
00225               const Employees::EmployeeInfo<std::string> &,
00226               const Employees::EmployeeInfo<std::string> &,
00227               const Employees::EmployeeInfo<std::string> &,
00228               const Employees::EmployeeInfo<std::string> &,
00229               const Employees::EmployeeInfo<std::string> &,
00230               const Employees::EmployeeInfo<int> &,
00231               const Employees::EmployeeInfo<int> &,
00232               const Employees::EmployeeInfo<oracle::occi::Date> &);
00233 ;
00234 }; // Employees::Update class
00235
00240 class Employees::Delete : private Employees::Setup {
00241 public:
00242     /* Employees::Delete functions
00243     int del(const Employees::EmployeeInfo<std::string> &);
00244 }; // Employees::Delete class
00245
00250 class Employees::EmployeeChecker {
00251 private:
00252     static bool is_validEmail(const std::string &);
00253     static bool is_empty(const std::string &, const std::string &, const std::string &, const
std::string &, const std::string &);
00254     static bool is_alphaDigit(const std::string &);
00255
00256 public:
00257     static std::pair<Employees::EmployeeCheckerFlags, const char *> is_valid(const std::string &,
00258                                     const std::string &,
00259                                     const std::string &,
00260                                     const std::string &,
00261                                     const std::string &,
00262                                     const std::string &,
00263                                     const
std::pair<Employees::EmployeeCheckerFlags, const char *>
00264                                     &flag =
{Employees::EmployeeCheckerFlags::EMPTY, ""});
00265 }; // EmployeeChecker class
00266
00267 #if __has_include("../templates/Employees.tpp")
00268 #include "../templates/Employees.tpp"
00269 #else
00270 #error "Employees.tpp not found!"
00271 #endif
00272
00273 #endif // __EMPLOYEES_HPP__

```

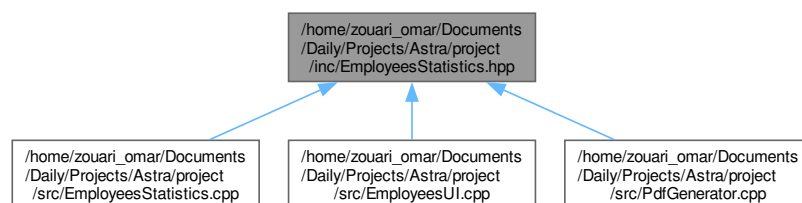
7.23 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/ EmployeesStatistics.hpp File Reference

[EmployeesStatistics](#) header file.

```
#include <unordered_map>
#include "../inc/Employees.hpp"
Include dependency graph for EmployeesStatistics.hpp:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [EmployeesStatistics](#)
[EmployeesStatistics](#) class.

7.23.1 Detailed Description

[EmployeesStatistics](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-01

Copyright

Copyright (c) 2025 EmployeesStatistics.hpp

7.24 EmployeesStatistics.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 // ? Pre-Processor prototype declaration part
00012 #ifndef __EMPLOYEES_STATISTICS__
00013 #define __EMPLOYEES_STATISTICS__
00014
00015 // ? Include(s) prototype declaration part
00016 // * Include std c++ header(s)
00017 #include <unordered_map>
00018
00019 /* Include generated ui_.h file(s)
00020 #include "../inc/Employees.hpp"
00021
00022 // ? Class prototype declaration part
00023
00028 class EmployeesStatistics {
00029 private:
00030     SqlParam employee;
00031     size_t employees_length;
00032     std::unordered_map<std::string, unsigned int> statuses;
00033     std::unordered_map<std::string, double> departments;
00034     std::unordered_map<std::string, double> salaries;
00035
00036 public:
00037     explicit EmployeesStatistics(const SqlParam &);
00038     const std::vector<unsigned int> getStatusStats();
00039     const std::vector<double> getDepartmentStats();
00040     const std::vector<double> getSalaryStats();
00041 }; // EmployeesStatistics class
00042
00043 #endif // __EMPLOYEES_STATISTICS__

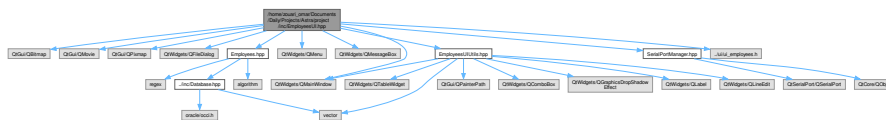
```

7.25 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ EmployeesUI.hpp File Reference

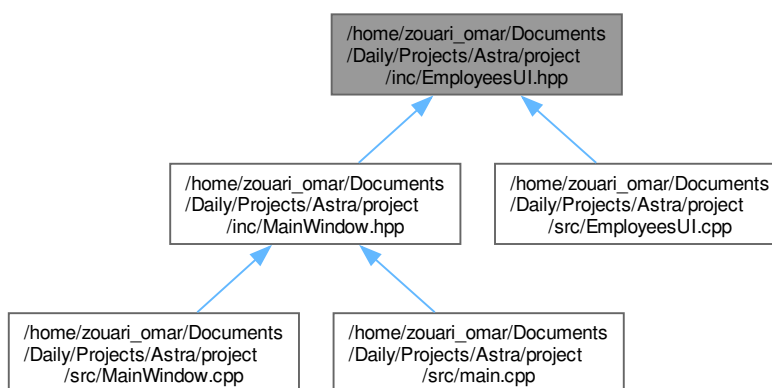
EmployeesUI header file.

```
#include <QtGui/QBitmap>
#include <QtGui/QMovie>
#include <QtGui/QPixmap>
#include <QtWidgets/QFileDialog>
#include <QtWidgets/QMainWindow>
#include <QtWidgets/QMenu>
#include <QtWidgets/QMessageBox>
#include "Employees.hpp"
#include "EmployeesUIUtils.hpp"
#include "SerialPortManager.hpp"
#include "../ui/ui_employees.h"
```

Include dependency graph for EmployeesUI.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class `EmployeesUI`
EmployeesUI class.

Namespaces

- namespace **Ui**
Ui namespace.

Macros

- `#define SHADOWS_EFFECT_COMBO_NUMBERS 11`

7.25.1 Detailed Description

[EmployeesUI](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-08

Copyright

Copyright (c) 2025 EmployeesUI.hpp

7.25.2 Macro Definition Documentation

7.25.2.1 SHADOWS_EFFECT_COMBO_NUMBERS

```
#define SHADOWS_EFFECT_COMBO_NUMBERS 11
```

7.26 EmployeesUI.hpp

[Go to the documentation of this file.](#)

```
00001
00011
00012 /// Pre-Processor prototype declaration part
00013 #ifndef __EMPLOYEES_UI_HPP__
00014 #define __EMPLOYEES_UI_HPP__
00015 #define SHADOWS_EFFECT_COMBO_NUMBERS 11
00016
00017 /// Include prototype declaration part
00018 /// Include std Qt header(s)
00019 #include <QtGui/QBitmap>
00020 #include <QtGui/QMovie>
00021 #include <QtGui/QPixmap>
00022 #include <QtWidgets/QFileDialog>
00023 #include <QtWidgets/QMainWindow>
00024 #include <QtWidgets/QMenu>
00025 #include <QtWidgets/QMessageBox>
00026 // #include <QtMultimedia/QAudioOutput>
00027 // #include <QtMultimedia/QMediaPlayer>
00028
00029 /// Include custom header(s)
00030 #include "Employees.hpp"
00031 #include "EmployeesUIUtils.hpp"
00032 #include "SerialPortManager.hpp"
00033
```

```

00034 /** Include generated ui_.h file(s)
00035 #include "../ui/ui_employees.h"
00036
00037 /** Classes prototype declaration part
00038
00043 QT_BEGIN_NAMESPACE
00044 namespace Ui {
00045 class EmployeesUI;
00046 }; // Namespace Ui
00047 QT_END_NAMESPACE
00048
00053 class EmployeesUI : public QMainWindow, private EmployeesUIUtils {
00054     Q_OBJECT
00055
00056 public:
00057     explicit EmployeesUI(SqlParam, QWidget *parent = nullptr);
00058     ~EmployeesUI();
00059     void set_employee(const SqlParam &);
00060
00061 private: /// Private EmployeesUI vars
00062     SqlParam employee;
00063     Ui::EmployeesUI *ui;
00064     QMovie *pdf_movie, *notification_movie, *csv_movie, *refresh_movie;
00065     QGraphicsDropShadowEffect *shadow_effect_components;
00066     SerialPortManager spm;
00067
00068 private: /// Private EmployeesUI function
00069     inline void __init__();
00070     void __init_current_employee_UI__() const;
00071     inline void __init_inset_form_group_box__();
00072     inline void __init_update_form_group_box__();
00073     void __init_employees_table__();
00074     void __clear_employees_table__() const;
00075     inline void __init_employees_charts__();
00076     // void insertRow_employees_table();
00077     void syncUI();
00078     void set_pushButtonMovie(QPushButton *, QMovie *) const;
00079
00080 private slots:
00081     void on_show_clicked(bool);
00082     void on_show_2_clicked(bool);
00083     void on_Add_button_clicked();
00084     void on_Cancel_form_clicked();
00085     void on_Cancel_form_2_clicked();
00086     void on_insertBtn_clicked();
00087     void on_updateBtn_clicked();
00088     void on_profileImageInsert_clicked();
00089     void on_profileImageInsert_2_clicked();
00090     void on_Filter_currentTextChanged(const QString &);
00091     void on_PDF_clicked();
00092     void on_refresh_clicked();
00093     void on_CSV_clicked();
00094     void on_Search_bar_textChanged(const QString &);
00095 }; // EmployeesUI class
00096
00097 #endif // __EMPLOYEES_UI_HPP__

```

7.27 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/ EmployeesUICharts.hpp File Reference

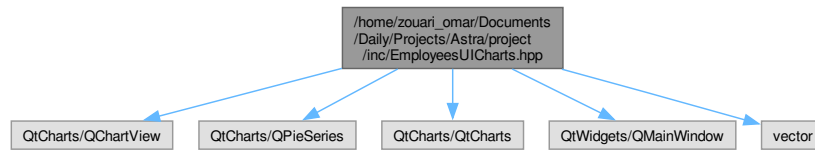
[EmployeesUICharts](#) header file.

```

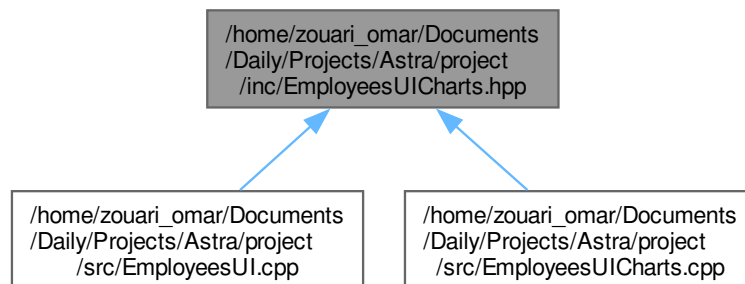
#include <QtCharts/QChartView>
#include <QtCharts/QPieSeries>
#include <QtCharts/QtCharts>
#include <QtWidgets/QMainWindow>
#include <vector>

```

Include dependency graph for EmployeesUICharts.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [Employees::EmployeesUICharts](#)

Namespaces

- namespace [Employees](#)
[Employees](#) namespace.

7.27.1 Detailed Description

[EmployeesUICharts](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-05

Copyright

Copyright (c) 2025 EmployeesUICharts.hpp

7.28 EmployeesUICharts.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __EMPLOYEES_UI_CHARTS__
00013 #define __EMPLOYEES_UI_CHARTS__
00014
00015 /// Include prototype declaration part
00016 /// Include std Qt header(s)
00017 #include <QtCharts/QChartView>
00018 #include <QtCharts/QPieSeries>
00019 #include <QtCharts/QtCharts>
00020 #include <QtWidgets/QMainWindow>
00021
00022 /// Include std C++ header(s)
00023 #include <vector>
00024
00029 namespace Employees {
00030 class EmployeesUICharts;
00031 } // Namespace Employees
00032
00033 /// Classes prototype declaration part
00034
00039 class Employees::EmployeesUICharts {
00040 private:
00041     QMainWindow *ui;
00042     void clearPrevChart(QHBoxLayout *);
00043
00044 public:
00045     explicit EmployeesUICharts(QMainWindow *);
00046     void setPieChart(QHBoxLayout *, const std::vector<std::pair<std::string, double> &);
00047     void setBarChart(QHBoxLayout *, const std::vector<std::pair<std::string, double> &);
00048     void setBarLineChart(QHBoxLayout *, const std::vector<std::pair<std::string, double> &);
00049 }; // EmployeesUICharts class
00050
00051 #endif // __EMPLOYEES_UI_CHARTS__

```

7.29 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/ EmployeesUIUtils.hpp File Reference

[EmployeesUIUtils](#) header file.

```

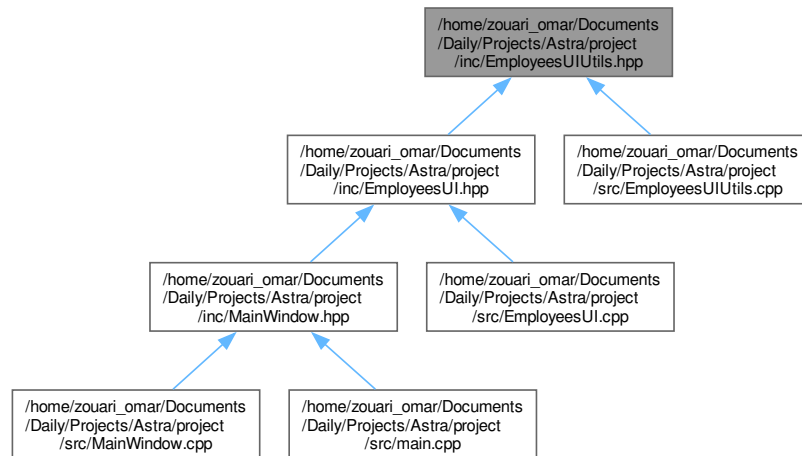
#include <QtGui/QPainterPath>
#include <QtWidgets/QComboBox>
#include <QtWidgets/QGraphicsDropShadowEffect>
#include <QtWidgets/QLabel>
#include <QtWidgets/QLineEdit>
#include <QtWidgets/QMainWindow>
#include <QtWidgets/QTableWidget>
#include <vector>

```

Include dependency graph for EmployeesUIUtils.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [EmployeesUIUtils](#)
[EmployeesUIUtils](#) class.

7.29.1 Detailed Description

[EmployeesUIUtils](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-11

Copyright

Copyright (c) 2025 EmployeesUIUtils.hpp

7.30 EmployeesUIUtils.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __EMPLOYEES_UI_UTILS_HPP__
00013 #define __EMPLOYEES_UI_UTILS_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include std Qt header(s)
00017 #include <QtGui/QPainterPath>
00018 #include <QtWidgets/QComboBox>
00019 #include <QtWidgets/QGraphicsDropShadowEffect>
00020 #include <QtWidgets/QLabel>
00021 #include <QtWidgets/QLineEdit>
00022 #include <QtWidgets/QMainWindow>
00023 #include <QtWidgets/QTableWidget>
00024
00025 /// Include std c++ header(s)
00026 #include <vector>
00027
00028 /// Class prototype declaration part
00029
00034 class EmployeesUIUtils {
00035 public:
00036     EmployeesUIUtils();
00037
00038 protected:
00039     std::string profileImgInsertHolder, profileImgUpdateHolder;
00040     void scaleImg(const QString &, QLabel *) const;
00041     std::string extractUsername(const std::string &, const unsigned short &length = 12) const;
00042     std::string strToUpper(std::string) const;
00043     void set_shadowEffect(QWidget *, QGraphicsDropShadowEffect *, const qreal xOffset = 5, const qreal
yOffset = 5, const qreal blurRadius = 5, const QColor color = Qt::gray);
00044     static std::vector<std::vector<std::string> > extractTableData(QTableWidget *);
00045     void filterEmployees(QTableWidget *, QComboBox *, QLineEdit *);
00046 }; // EmployeesUIUtils class
00047
00048 #endif // __EMPLOYEES_UI_UTILS_HPP__

```

7.31 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ FaceRecognizer.hpp File Reference

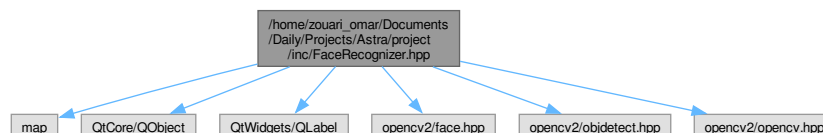
FaceRecognizer header file.

```

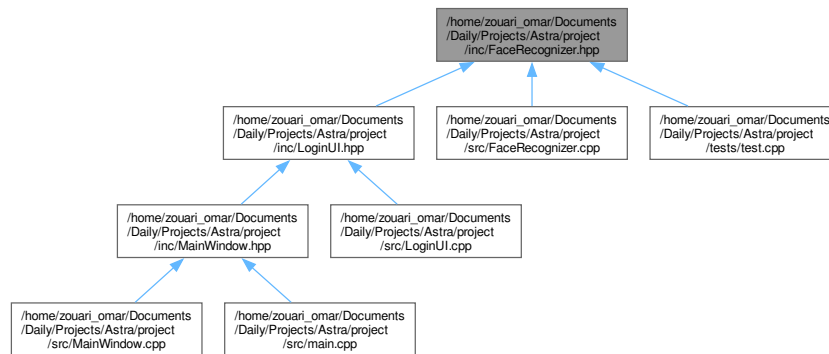
#include <map>
#include <QtCore/QObject>
#include <QtWidgets/QLabel>
#include <opencv2/face.hpp>
#include <opencv2/objdetect.hpp>
#include <opencv2/opencv.hpp>

```

Include dependency graph for FaceRecognizer.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [FaceRecognizer](#)

Macros

- `#define __FACES_DEFAULT_DIR__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/assets/faces"`
- `#define __CASCADE_DEFAULT_MODEL__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/models/haarcascade_frontalface_default.xml"`
- `#define __FACE_RECOGNIZER_DEFAULT_MODEL__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/models/face_recognizer(final version).yaml"`
- `#define __CAMERA_DEFAULT_FPS__ 30`
- `#define __CONFIDENCE_DEFAULT_SCORE__ 0.5`

7.31.1 Detailed Description

[FaceRecognizer](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-12

Copyright

Copyright (c) 2025 FaceRecognizer.hpp

7.31.2 Macro Definition Documentation

7.31.2.1 __CAMERA_DEFAULT_FPS__

```
#define __CAMERA_DEFAULT_FPS__ 30
```

7.31.2.2 __CASCADE_DEFAULT_MODEL__

```
#define __CASCADE_DEFAULT_MODEL__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/models/haarcascade_frontalface_default.xml"
```

7.31.2.3 __CONFIDENCE_DEFAULT_SCORE__

```
#define __CONFIDENCE_DEFAULT_SCORE__ 0.5
```

7.31.2.4 __FACE_RECOGNIZER_DEFAULT_MODEL__

```
#define __FACE_RECOGNIZER_DEFAULT_MODEL__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/models/face_recognizer(final version).yaml"
```

7.31.2.5 __FACES_DEFAULT_DIR__

```
#define __FACES_DEFAULT_DIR__ "/home/zouari_omar/Documents/Daily/Projects/Astra/project/assets/faces"
```

7.32 FaceRecognizer.hpp

[Go to the documentation of this file.](#)

```
00001
00010
00011 ///  
00012 #ifndef  
00013 #define  
00014 #define  
00015 #define  
00016 #define  
00017 #define  
00018 #define  
00019  
00020 ///  
00021 ///  
00022 #include  
00023  
00024 ///  
00025 #include  
00026 #include  
00027  
00028 ///  
00029 #include  
00030 #include  
00031 #include  
00032  
00033 ///  
00034 class  
00035 public:  
00036     ~FaceRecognizer();
```

```

00037 void trainEmbeddedModel(const std::string &, const std::string &, const std::string &);
00038 void captureFrame(QLabel *);
00039 std::string recognize(QLabel *);
00040 void load();
00041
00042 private:
00043 cv::VideoCapture cap; // Video capture object to access the camera
00044 cv::Mat currentFrame; // Holds the current captured frame
00045 cv::CascadeClassifier faceCascade; // Haar Cascade for face detection
00046 cv::Ptr<cv::face::LBPHFaceRecognizer> recognizer; // LBPH face recognizer
00047 std::map<int, std::string> indexer; // Map to store labels and corresponding names
00048
00049 QImage matToQImage(const cv::Mat &mat) const;
00050 std::pair<std::vector<cv::Mat>, std::vector<int>> loadData(const std::string &dataDir);
00051 void loadIndexer(const std::string &dataDir);
00052 void loadEmbeddedModel(const std::string &);
00053 void loadCascadeModel(const std::string &);
00054 }; // FaceRecognizer class
00055
00056 #endif // __FACE_RECOGNITION_HPP__

```

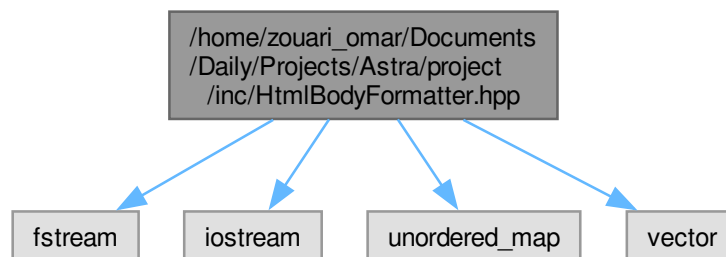
7.33 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/← HtmlBodyFormatter.hpp File Reference

```

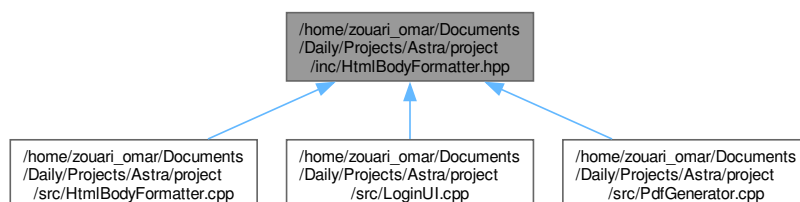
#include <fstream>
#include <iostream>
#include <unordered_map>
#include <vector>

```

Include dependency graph for HtmlBodyFormatter.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [HtmlBodyFormater](#)
HtmlBodyFormater class.

7.34 HtmlBodyFormatter.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __HTML_BODY_FORMATER_HPP__
00013 #define __HTML_BODY_FORMATER_HPP__
00014
00030
00031 /// Include prototype declaration part
00032 // * Include std C++ header
00033 #include <fstream>
00034 #include <iostream>
00035 #include <unordered_map>
00036 #include <vector>
00037
00038 /// Class(es) prototype declaration part
00039
00044 class HtmlBodyFormater {
00045 public:
00046     explicit HtmlBodyFormater(const std::string &, const std::unordered_map<std::string, const
std::string> &_param = {});
00047     std::string get_inner_html();
00048
00049 private:
00050     void format_inner_html();
00051     std::string inner_html;
00052     const std::unordered_map<std::string, const std::string> param;
00053 }; // HtmlBodyFormater class
00054
00055 #endif // __HTML_BODY_FORMATER_HPP__

```

7.35 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ LoginUI.hpp File Reference

[LoginUI](#) header file.

```

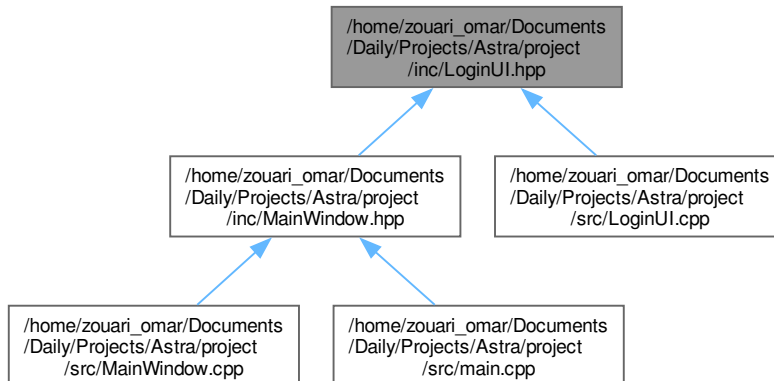
#include <QtConcurrent/QtConcurrent>
#include <QtCore/QDebug>
#include <QtCore/QPropertyAnimation>
#include <QtCore/Qtimer>
#include <QtGui/QMovie>
#include <QtWidgets/QGraphicsOpacityEffect>
#include <QtWidgets/QMainWindow>
#include <QtWidgets/QMessageBox>
#include <QtWidgets/QProgressDialog>
#include "Employees.hpp"
#include "FaceRecognizer.hpp"
#include "../ui/ui_login.h"

```

Include dependency graph for LoginUI.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [LoginUI](#)

Namespaces

- namespace [Ui](#)
[Ui](#) namespace.

Macros

- `#define __LOGIN_GIF_ANIMATION__ 5000`
- `#define __START_LG_Q_LABEL_X__ 210`
- `#define __END_LG_Q_LABEL_X__ 330`
- `#define __FULL_VISIBLE__ 1.0`
- `#define __FULL_TRANSPARENT__ 0.0`

7.35.1 Detailed Description

[LoginUI](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-18

Copyright

Copyright (c) 2025 LoginUI.hpp

7.35.2 Macro Definition Documentation

7.35.2.1 __END_LG_Q_LABEL_X__

```
#define __END_LG_Q_LABEL_X__ 330
```

7.35.2.2 __FULL_TRANSPARENT__

```
#define __FULL_TRANSPARENT__ 0.0
```

7.35.2.3 __FULL_VISIBLE__

```
#define __FULL_VISIBLE__ 1.0
```

7.35.2.4 __LOGIN_GIF_ANIMATION__

```
#define __LOGIN_GIF_ANIMATION__ 5000
```

7.35.2.5 __START_LG_Q_LABEL_X__

```
#define __START_LG_Q_LABEL_X__ 210
```

7.36 LoginUI.hpp

[Go to the documentation of this file.](#)

```
00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __LOGIN_UI_HPP__
00013 #define __LOGIN_UI_HPP__
00014 #define __LOGIN_GIF_ANIMATION__ 5000 // 5sec
00015 #define __START_LG_Q_LABEL_X__ 210 // Start 1 QLabel x position = 210px
00016 #define __END_LG_Q_LABEL_X__ 330 // End 1 QLabel x position = 210px
00017 #define __FULL_VISIBLE__ 1.0 // Fully visible (opacity)
00018 #define __FULL_TRANSPARENT__ 0.0 // Fully transparent (opacity)
00019
00020 /// Include prototype declaration part
00021 /// Include std headers (Qt)
00022 #include <QtConcurrent/QtConcurrent>
00023 #include <QtCore/QDebug>
00024 #include <QtCore/QPropertyAnimation>
00025 #include <QtCore/Qtimer>
00026 #include <QtGui/QMovie>
00027 #include <QtWidgets/QGraphicsOpacityEffect>
00028 #include <QtWidgets/QMainWindow>
00029 #include <QtWidgets/QMessageBox>
00030 #include <QtWidgets/QProgressDialog>
00031
00032 /// Include custom header(s)
00033 #include "Employees.hpp"
00034 #include "FaceRecognizer.hpp"
00035
00036 /// Include generated ui_.h files
00037 #include "../ui/ui_login.h"
00038
00039 /// Namespace(s) prototype declaration part
00040
00045 QT_BEGIN_NAMESPACE
```

```

00046 namespace Ui {
00047 class Login;
00048 }; // Namespace Ui
00049 QT_END_NAMESPACE
00050
00051 ///< Class prototype declaration part
00052
00053 class LoginUI : public QMainWindow {
00054     Q_OBJECT
00055 public:
00056     explicit LoginUI(QWidget *parent = nullptr);
00057     ~LoginUI();
00058     const SqlParam get_employee();
00059 private:
00060     Ui::Login *ui;
00061     QTimer *gifTimer, *camTimer;
00062     QMovie *currentMovie;
00063     int currentGifIndex;
00064     const QStringList gifPaths;
00065     std::string generated_password;
00066     SqlParam employee;
00067     FaceRecognizer *faceRecognition;
00068
00069     ///< Events
00070     bool eventFilter(QObject *, QEvent *) override;
00071     bool forget_password_events(QObject *, QEvent *);
00072     bool login_btn_events(QObject *, QLabel *, QEvent *);
00073
00074     ///< Effects
00075     void updateGif();
00076     void QGroupBoxFadeOutEffect(QGroupBox *, QGroupBox *);
00077     QPropertyAnimation *FadeEffect(QGroupBox *, const QVariant, const QVariant);
00078     void change_hideShowBtnIcon(QLineEdit *, QPushButton *, const QLineEdit::EchoMode mode =
00079         QLineEdit::Password, const QString path =
00080         "/home/zouari_omar/Documents/Daily/Projects/Astra/project/assets/login_imgs/closedEye.png");
00081
00082     ///< Helpers
00083     void enableResetPassword(const bool &);
00084     void clearResetPassword();
00085
00086 private slots:
00087     ///< Login sub-interface group box slots
00088     void on_pushButton_clicked();
00089     void on_hide_show_btn_clicked();
00090     void on_reset_clicked();
00091     void on_faceId_clicked();
00092     void on_fp_returnBtn_clicked();
00093
00094     ///< Forget password sub-interface group box slots
00095     void on_hide_show_btn_2_clicked();
00096     void on_returnBtn_clicked();
00097     void on_sendEmailBtn_clicked();
00098
00099 signals:
00100     void loginSuccessful(); ///< Signals fns
00101     void loginSuccessful(); ///< Signal to notify successful login
00102 }; // Login class
00103
00104 #endif // __LOGIN_UI_HPP__

```

7.37 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ MainWindow.hpp File Reference

[MainWindow](#) header file.

```

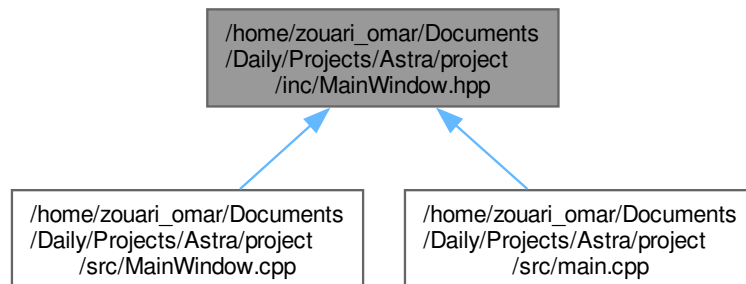
#include <QtWidgets/QMainWindow>
#include <QtWidgets/QStackedWidget>
#include "../inc/EmployeesUI.hpp"
#include "../inc/LoginUI.hpp"

```

Include dependency graph for MainWindow.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class **MainWindow**

Macros

- #define __SCREEN_WIDTH__ 1920
- #define __SCREEN_HIGHT__ 1080

7.37.1 Detailed Description

MainWindow header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date _____

2025-02-20

Copyright

Copyright (c) 2025 MainWindow.hpp

7.37.2 Macro Definition Documentation

7.37.2.1 __SCREEN_HIGHT__

```
#define __SCREEN_HIGHT__ 1080
```

7.37.2.2 __SCREEN_WIDTH__

```
#define __SCREEN_WIDTH__ 1920
```

7.38 MainWindow.hpp

[Go to the documentation of this file.](#)

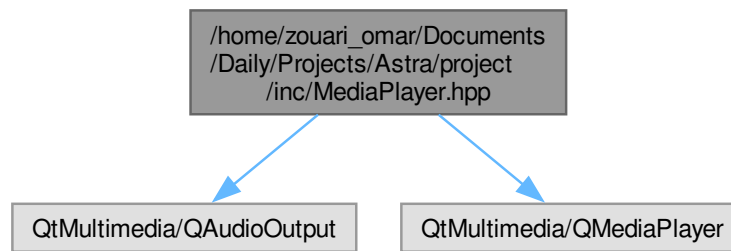
```
00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __INC_HPP__
00013 #define __INC_HPP__
00014 #define __SCREEN_WIDTH__ 1920
00015 #define __SCREEN_HIGHT__ 1080
00016
00017 /// Include prototype declaration part
00018 /// Include std libs (Qt)
00019 #include <QtWidgets/QMainWindow>
00020 #include <QtWidgets/QStackedWidget>
00021
00022 /// Include custom header(s) (.hpp)
00023 #include "../inc/EmployeesUI.hpp"
00024 #include "../inc/LoginUI.hpp"
00025
00026 /// Class prototype declaration part
00027 class MainWindow : public QMainWindow {
00028 public:
00029     MainWindow();
00030     ~MainWindow();
00031
00032 private:
00033     QStackedWidget *stackedWidget;
00034     LoginUI *loginUI;
00035     EmployeesUI *employee_ui;
00036 }; // MainWindow class
00037
00038 #endif // __INC_HPP__
```

7.39 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/↵ MediaPlayer.hpp File Reference

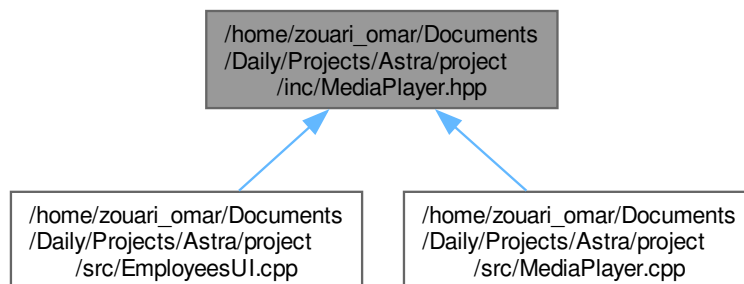
[MediaPlayer](#) header file.

```
#include <QtMultimedia/QAudioOutput>
#include <QtMultimedia/QMediaPlayer>
```

Include dependency graph for MediaPlayer.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [MediaPlayer](#)
MediaPlayer class.

7.39.1 Detailed Description

[MediaPlayer](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-18

Copyright

Copyright (c) 2025 MediaPlayer.hpp

7.40 MediaPlayer.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __MEDIA_PLAYER_HPP__
00013 #define __MEDIA_PLAYER_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include std Qt header(s)
00017 #include <QtMultimedia/QAudioOutput>
00018 #include <QtMultimedia/QMediaPlayer>
00019
00024 class MediaPlayer {
00025 public:
00026     static void playNotif(const QString &, const int &volume = 1);
00027 }; // MediaPlayer class
00028
00029 #endif // __MEDIA_PLAYER_HPP__

```

7.41 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/PasswordGenerator.hpp File Reference

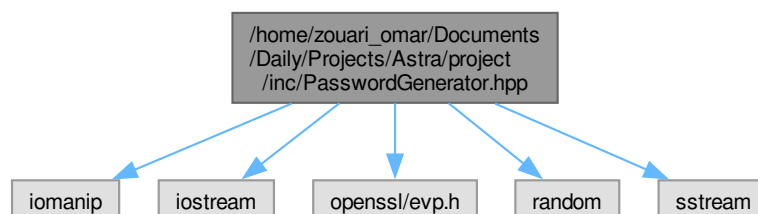
[PasswordGenerator](#) header file.

```

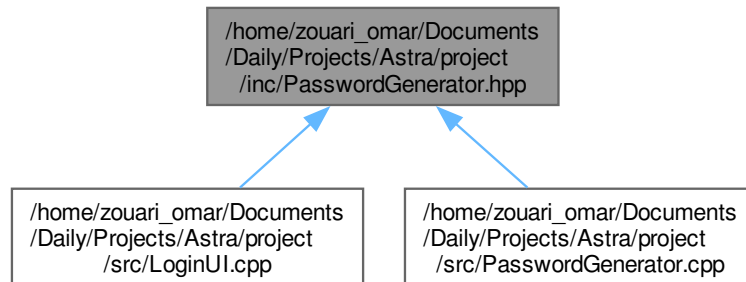
#include <iomanip>
#include <iostream>
#include <openssl/evp.h>
#include <random>
#include <sstream>

```

Include dependency graph for PasswordGenerator.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [PasswordGenerator](#)

Macros

- `#define PASSWORD\_GENERATOR\_DEFAULT\_PATTERN "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZHIJKLMNOPQRSTUVWXYZ_+"`
- `#define ERR\_SHA256\_CREATE std::cerr << "Error: Can't create a new EVP_MD_CTX!" << std::endl`
- `#define ERR\_SHA256\_INIT std::cerr << "Error: Can't initialize SHA-256!" << std::endl`
- `#define ERR\_SHA256\_UPDATE std::cerr << "Error: Can't update SHA-256 hash!" << std::endl`
- `#define ERR\_SHA256\_FINAL std::cerr << "Error: Can't finalize SHA-256 hash!" << std::endl`

7.41.1 Detailed Description

[PasswordGenerator](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-01

Copyright

Copyright (c) 2025 PasswordGenerator.hpp std::random_device Mersenne twister algorithm How to Generate SHA256 Hash v
SHA256_Init, SHA256_Update and SHA256_Final are deprecated in OpenSSL 3.0 openssl doxygen

7.41.2 Macro Definition Documentation

7.41.2.1 ERR_SHA256_CREATE

```
#define ERR_SHA256_CREATE std::cerr << "Error: Can't create a new EVP_MD_CTX!" << std::endl
```

7.41.2.2 ERR_SHA256_FINAL

```
#define ERR_SHA256_FINAL std::cerr << "Error: Can't finalize SHA-256 hash!" << std::endl
```

7.41.2.3 ERR_SHA256_INIT

```
#define ERR_SHA256_INIT std::cerr << "Error: Can't initialize SHA-256!" << std::endl
```

7.41.2.4 ERR_SHA256_UPDATE

```
#define ERR_SHA256_UPDATE std::cerr << "Error: Can't update SHA-256 hash!" << std::endl
```

7.41.2.5 PASSWORD_GENERATOR_DEFAULT_PATTERN

```
#define PASSWORD_GENERATOR_DEFAULT_PATTERN "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789!@#$_+"
```

7.42 PasswordGenerator.hpp

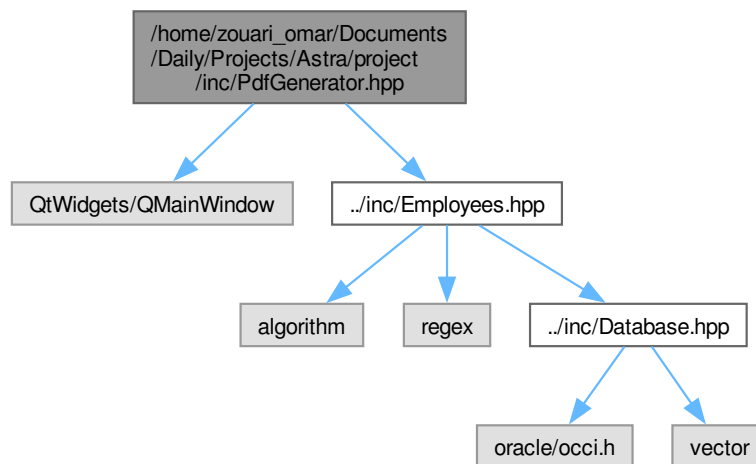
[Go to the documentation of this file.](#)

```
00001
00015
00016 /// Pre-Processor prototype declaration part
00017 #ifndef __PASSWORDGENERATOR_HPP__
00018 #define __PASSWORDGENERATOR_HPP__
00019 #define PASSWORD_GENERATOR_DEFAULT_PATTERN
00020 "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789!@#$_%^&*()_+"
00021 /// Include prototype declaration part
00022 /// * Include std C++ headers
00022 #include <iomanip>
00023 #include <iostream>
00024 #include <openssl/evp.h>
00025 #include <random>
00026 #include <sstream>
00027
00028 /// Error/Success Messages declaration part
00029 #define ERR_SHA256_CREATE std::cerr << "Error: Can't create a new EVP_MD_CTX!" << std::endl
00030 #define ERR_SHA256_INIT std::cerr << "Error: Can't initialize SHA-256!" << std::endl
00031 #define ERR_SHA256_UPDATE std::cerr << "Error: Can't update SHA-256 hash!" << std::endl
00032 #define ERR_SHA256_FINAL std::cerr << "Error: Can't finalize SHA-256 hash!" << std::endl
00033
00034 /// Class prototype declaration part
00035 class PasswordGenerator {
00036 public:
00037     static std::string generate(const int &length = 6, const std::string &pattern =
00038     PASSWORD_GENERATOR_DEFAULT_PATTERN);
00038     static std::string sha256sum(const std::string &);
00039 }; /// PasswordGenerator class
00040
00041 #endif /// __PASSWORDGENERATOR_HPP__
```

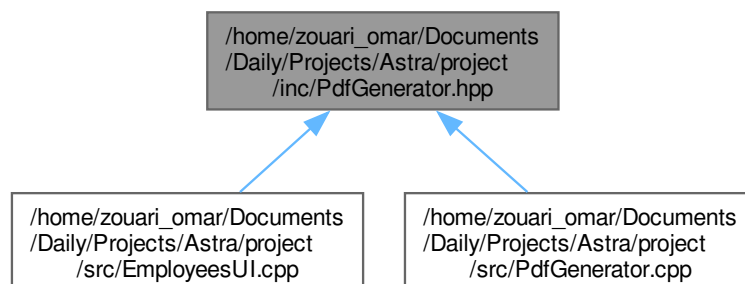

7.43 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/PdfGenerator.hpp File Reference

[PdfGenerator](#) header file.

```
#include <QtWidgets/QMainWindow>
#include "../inc/Employees.hpp"
Include dependency graph for PdfGenerator.hpp:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [PdfGenerator](#)

7.43.1 Detailed Description

[PdfGenerator](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-10

Copyright

Copyright (c) 2025 PdfGenerator.hpp

7.44 PdfGenerator.hpp

[Go to the documentation of this file.](#)

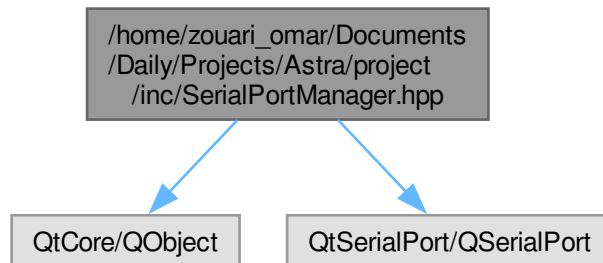
```
00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __PDF_GENERATOR_HPP__
00013 #define __PDF_GENERATOR_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include std Qt header(s)
00017 #include <QtWidgets/QMainWindow>
00018
00019 /// Include custom header(s)
00020 #include "../inc/Employees.hpp"
00021
00022
00023 /// Class(es) prototype declaration part
00024
00025 class PdfGenerator {
00026 public:
00027     static const unsigned int generatePdf(const QString &filePath, const SqlParam &);
00028 }; // PdfGenerator class
00029
00030 #endif // __PDF_GENERATOR_HPP__
```

7.45 /home/zouari_omar/Documents/Daily/Projects/Astra/project/inc/SerialPortManager.hpp File Reference

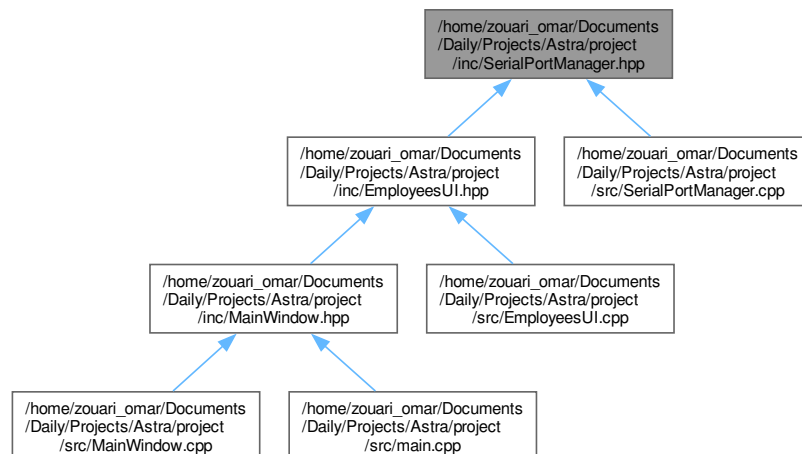
[SerialPortManager](#) header file.

```
#include <QtCore/QObject>
#include <QtSerialPort/QSerialPort>
```

Include dependency graph for SerialPortManager.hpp:



This graph shows which files directly or indirectly include this file:



Classes

- class [SerialPortManager](#)
SerialPortManager class.

7.45.1 Detailed Description

[SerialPortManager](#) header file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-25

Copyright

Copyright (c) 2025 SerialPortManager.hpp

7.46 SerialPortManager.hpp

[Go to the documentation of this file.](#)

```

00001
00010
00011 /// Pre-Processor prototype declaration part
00012 #ifndef __SERIAL_PORT_MANAGER_HPP__
00013 #define __SERIAL_PORT_MANAGER_HPP__
00014
00015 /// Include prototype declaration part
00016 /// Include std Qt header(s)
00017 #include <QtCore/QObject>
00018 #include <QtSerialPort/QSerialPort>
00019
00020 /// Class prototype declaration part
00021
00026 class SerialPortManager : public QObject {
00027     Q_OBJECT
00028 public:
00029     SerialPortManager(QObject *);
00030     ~SerialPortManager();
00031     void details() const;
00032     void listen();
00033
00034 private:
00035     QObject *ui;
00036     QSerialPort *ino;
00037     const quint16 inoVendorId, inoProductId;
00038     QString serialBuffer, parsed_data;
00039     QByteArray serialData;
00040     QString inoPortName;
00041     void connect();
00042
00043     const bool is_inoAvailable();
00044     void setupSerialPort();
00045     inline void openSerialPort();
00046
00047 private slots:
00048     void on_readSerial();
00049 }; // SerialPortManager class
00050
00051 #endif // __SERIAL_PORT_MANAGER_HPP__

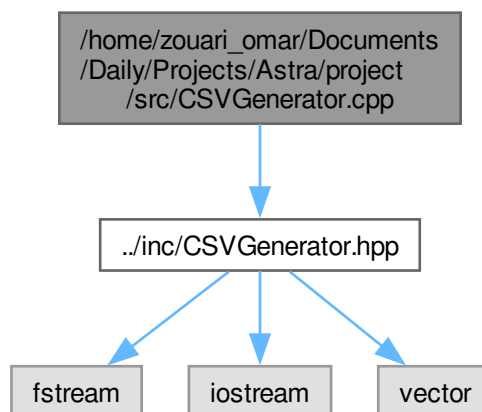
```

7.47 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/↵ CSVGenerator.cpp File Reference

[CSVGenerator](#) source file.

```
#include "../inc/CSVGenerator.hpp"
```

Include dependency graph for CSVGenerator.cpp:



7.47.1 Detailed Description

[CSVGenerator](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-16

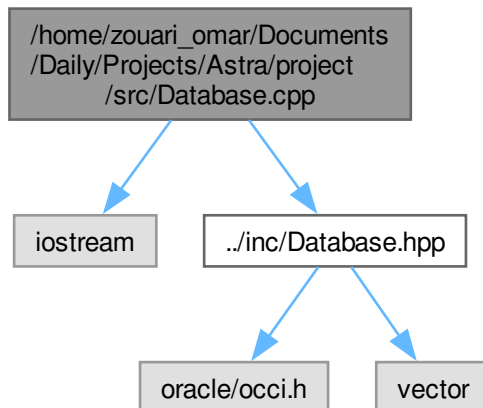
Copyright

Copyright (c) 2025 CSVGenerator.cpp

7.48 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/Database.cpp File Reference

[Database](#) source file.

```
#include <iostream>
#include "../inc/Database.hpp"
Include dependency graph for Database.cpp:
```



7.48.1 Detailed Description

[Database](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-13

Copyright

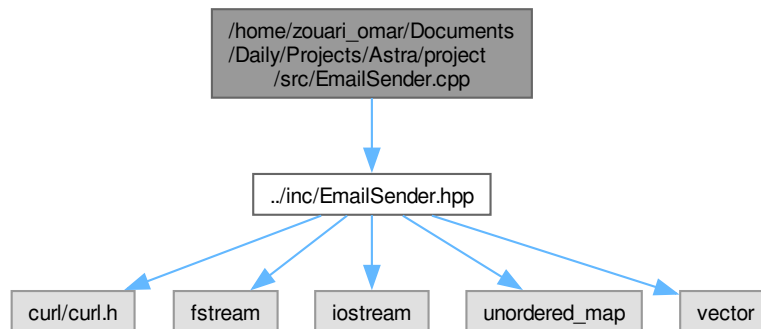
Copyright (c) 2025 oracle

7.49 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/EmailSender.cpp File Reference

EmailSender source file.

```
#include "../inc/EmailSender.hpp"
```

Include dependency graph for EmailSender.cpp:



7.49.1 Detailed Description

EmailSender source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-23

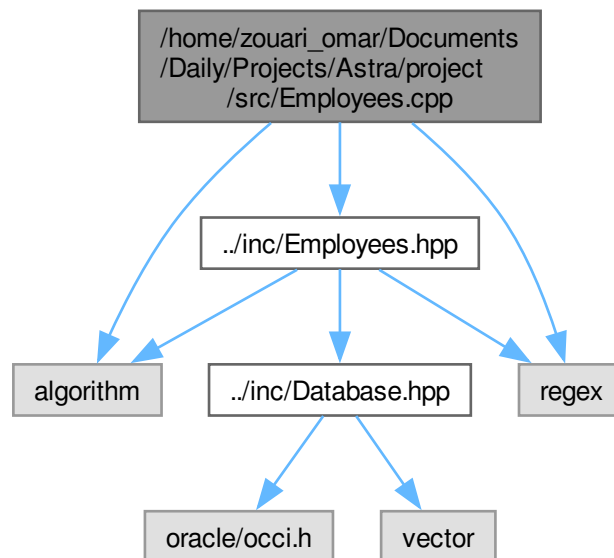
Copyright

Copyright (c) 2025 EmailSender.cpp smtp-mail.c "In the absence of a mime type!"

7.50 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/↵ Employees.cpp File Reference

[Employees](#) source file.

```
#include <algorithm>
#include <regex>
#include "../inc/Employees.hpp"
Include dependency graph for Employees.cpp:
```



7.50.1 Detailed Description

[Employees](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-01

Copyright

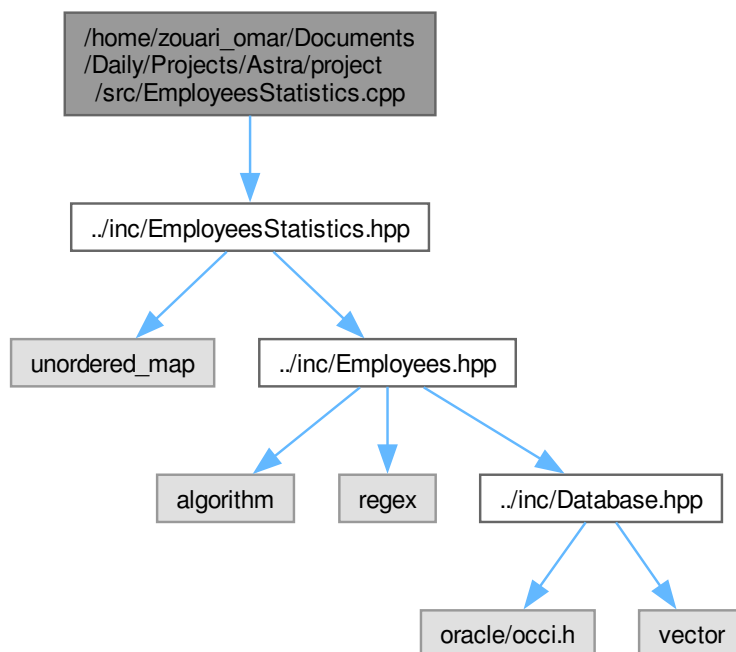
Copyright (c) 2025 Employees.cpp

7.51 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/ EmployeesStatistics.cpp File Reference

[EmployeesStatistics](#) source file.

```
#include "../inc/EmployeesStatistics.hpp"
```

Include dependency graph for EmployeesStatistics.cpp:



7.51.1 Detailed Description

[EmployeesStatistics](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-01

Copyright

Copyright (c) 2025 EmployeesStatistics.cpp

7.52 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/↵ EmployeesUI.cpp File Reference

[EmployeesUI](#) source file.

```
#include "../inc/EmployeesUI.hpp"
#include "../inc/CSVGenerator.hpp"
#include "../inc/EmployeesStatistics.hpp"
#include "../inc/EmployeesUICharts.hpp"
#include "../inc/MediaPlayer.hpp"
#include "../inc/PdfGenerator.hpp"
```

Include dependency graph for EmployeesUI.cpp:



Functions

- void [insertRow_employees_table](#) ()

7.52.1 Detailed Description

[EmployeesUI](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-08

Copyright

Copyright (c) 2025 EmployeesUI.cpp

7.52.2 Function Documentation

7.52.2.1 insertRow_employees_table()

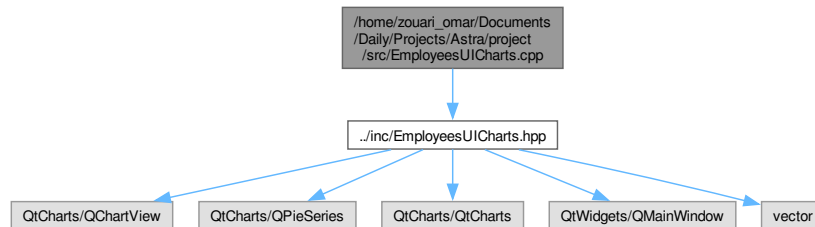
```
void insertRow_employees_table ()
```

7.53 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/ EmployeesUICharts.cpp File Reference

[EmployeesUICharts](#) source file.

```
#include "../inc/EmployeesUICharts.hpp"
```

Include dependency graph for EmployeesUICharts.cpp:



7.53.1 Detailed Description

[EmployeesUICharts](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-05

Copyright

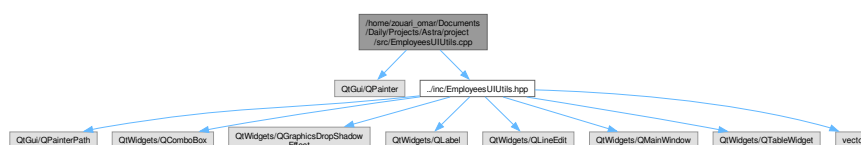
Copyright (c) 2025 EmployeesUICharts.cpp

7.54 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/ EmployeesUIUtils.cpp File Reference

[EmployeesUIUtils](#) source file.

```
#include <QtGui/QPainter>
#include "../inc/EmployeesUIUtils.hpp"
```

Include dependency graph for EmployeesUIUtils.cpp:



7.54.1 Detailed Description

[EmployeesUIUtils](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-11

Copyright

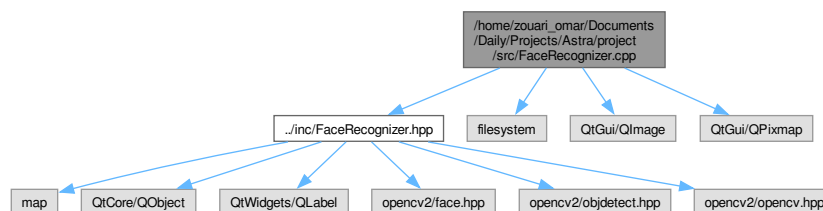
Copyright (c) 2025 EmployeesUIUtils.cpp

7.55 [/home/zouari_omar/Documents/Daily/Projects/Astra/project/src/](#)↵ FaceRecognizer.cpp File Reference

[FaceRecognizer](#) source file.

```
#include "../inc/FaceRecognizer.hpp"  
#include <filesystem>  
#include <QtGui/QImage>  
#include <QtGui/QPixmap>
```

Include dependency graph for FaceRecognizer.cpp:



7.55.1 Detailed Description

FaceRecognizer source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-12

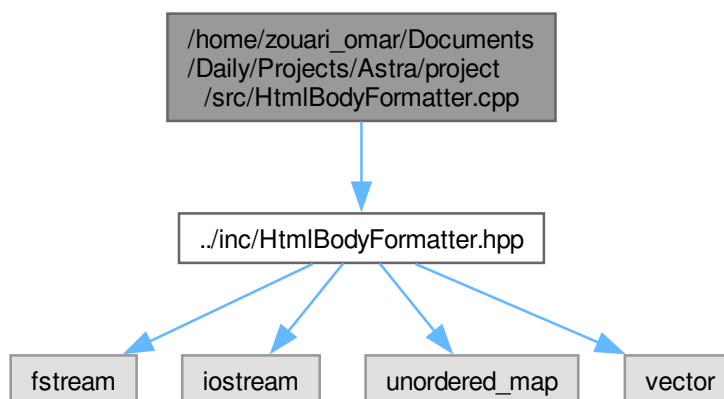
Copyright

Copyright (c) 2025 FaceRecognizer.cpp

7.56 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/HtmlBodyFormatter.cpp File Reference

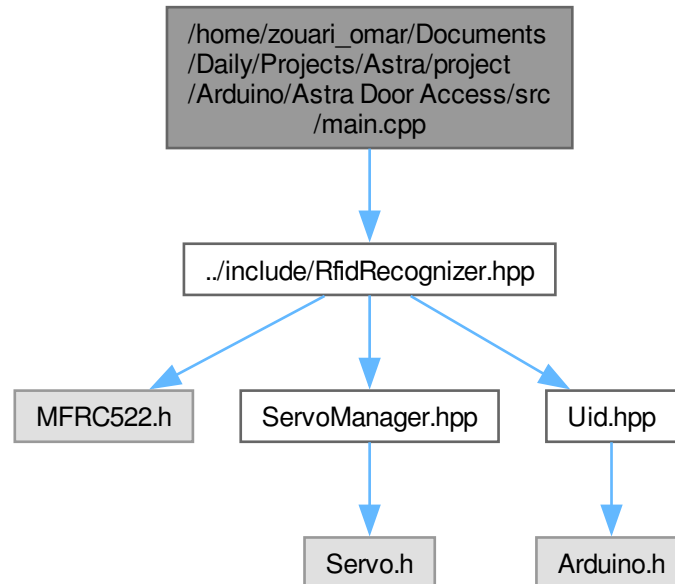
```
#include "../inc/HtmlBodyFormatter.hpp"
```

Include dependency graph for HtmlBodyFormatter.cpp:




```
#include "../include/RfidRecognizer.hpp"
```

Include dependency graph for main.cpp:



Functions

- void `setup()`
Setup Arduino Uno R3 part.
- void `loop()`
life time Arduino Uno R3 actions part

Variables

- `RfidRecognizer rfid`

7.58.1 Detailed Description

Main Arduino Uno R3 lancer file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-11

Copyright

Copyright (c) 2025

Note

Give permission to the arduino board - `sudo chmod 666 /dev/ttyACM0`

Upload the code cmd - `pio run --target upload`

Listen to the serial monitor - `stty -F /dev/ttyACM0 raw 9600 && cat /dev/ttyACM0`

Generate `compile_commands.json` - `pio run -t compiledb Arduino-MFRC522-RFID-READER main.cpp`

7.58.2 Function Documentation

7.58.2.1 loop()

```
void loop ()
```

life time Arduino Uno R3 actions part

Returns

void

7.58.2.2 setup()

```
void setup ()
```

Setup Arduino Uno R3 part.

Returns

void

7.58.3 Variable Documentation

7.58.3.1 rfid

```
RfidRecognizer rfid
```


The main source file (app lancer)

Functions

- ### 7.59.1 Detailed Description

The main source file (app lancer)

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date _____

2025-02-10

Copyright

Copyright (c) 2025 main.cpp

7.59.2 Function Documentation

7.59.2.1 main()

```
int main (
    int argc,
    char ** argv)
```

The Main Program Function.

Parameters

<i>argc</i>	int
<i>argv</i>	{char **}

Returns

int

7.60 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/↵ MainWindow.cpp File Reference

[MainWindow](#) source file.

```
#include "../inc/MainWindow.hpp"
```

Include dependency graph for MainWindow.cpp:



7.60.1 Detailed Description

[MainWindow](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-02-21

Copyright

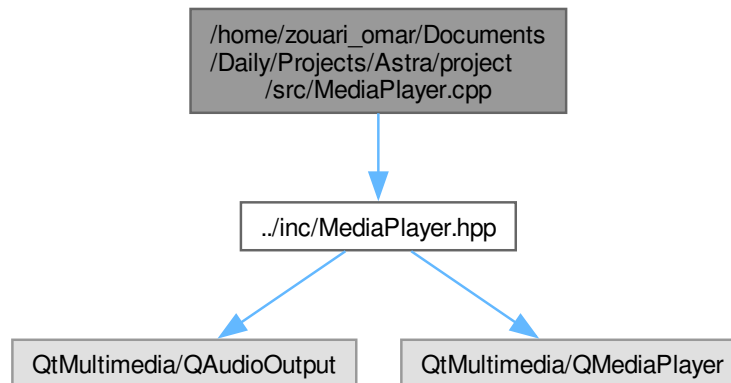
Copyright (c) 2025 MainWindow.cpp

7.61 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/↵ MediaPlayer.cpp File Reference

[MediaPlayer.](#)

```
#include "../inc/MediaPlayer.hpp"
```

Include dependency graph for MediaPlayer.cpp:



7.61.1 Detailed Description

[MediaPlayer.](#)

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-18

Copyright

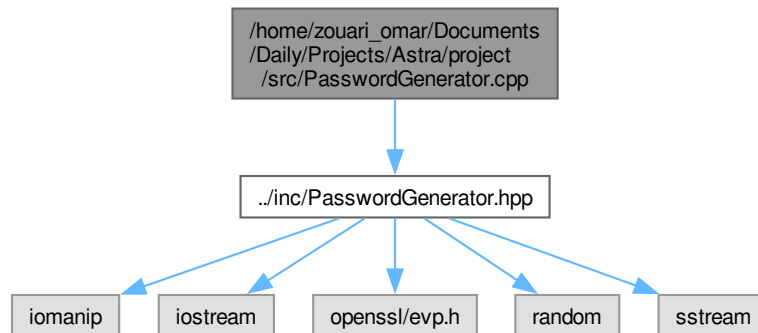
Copyright (c) 2025 MediaPlayer.cpp

7.62 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/PasswordGenerator.cpp File Reference

[PasswordGenerator](#) source file.

```
#include "../inc/PasswordGenerator.hpp"
```

Include dependency graph for PasswordGenerator.cpp:



7.62.1 Detailed Description

[PasswordGenerator](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-03-01

Copyright

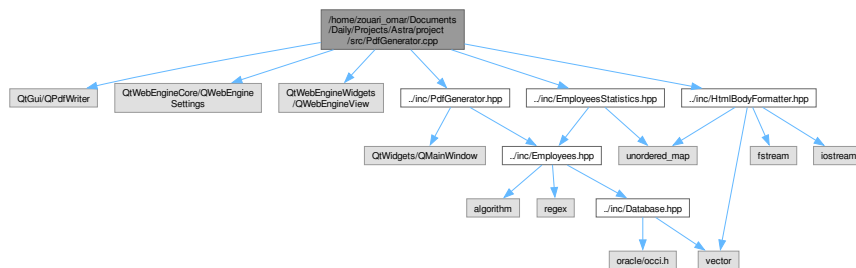
Copyright (c) 2025 PasswordGenerator.cpp

7.63 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/PdfGenerator.cpp File Reference

[PdfGenerator](#) source file.

```
#include <QtGui/QPdfWriter>
#include <QtWebEngineCore/QWebEngineSettings>
#include <QtWebEngineWidgets/QWebEngineView>
#include "../inc/EmployeesStatistics.hpp"
#include "../inc/HtmlBodyFormatter.hpp"
#include "../inc/PdfGenerator.hpp"
```

Include dependency graph for PdfGenerator.cpp:



7.63.1 Detailed Description

[PdfGenerator](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-11

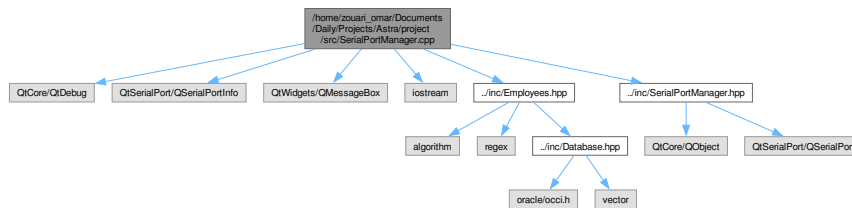
Copyright

Copyright (c) 2025 PdfGenerator.cpp

7.64 /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/SerialPortManager.cpp File Reference

[SerialPortManager](#) source file.

```
#include <QtCore/QtDebug>
#include <QtSerialPort/QSerialPortInfo>
#include <QtWidgets/QMessageBox>
#include <iostream>
#include "../inc/Employees.hpp"
#include "../inc/SerialPortManager.hpp"
Include dependency graph for SerialPortManager.cpp:
```



7.64.1 Detailed Description

[SerialPortManager](#) source file.

Author

@ZouariOmar (zouariomar20@gmail.com)

Version

0.1

Date

2025-04-25

Copyright

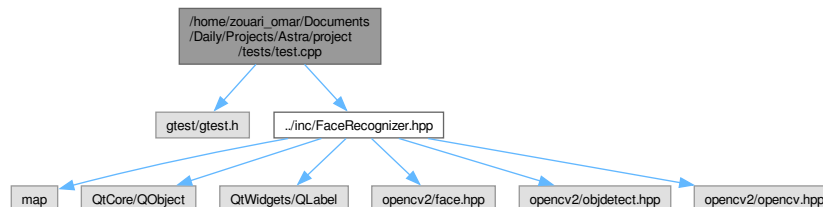
Copyright (c) 2025 SerialPortManager.cpp

7.65 /home/zouari_omar/Documents/Daily/Projects/↵ Astra/project/models/README.md File Reference

7.66 /home/zouari_omar/Documents/Daily/Projects/Astra/project/tests/↵ README.md File Reference

7.67 /home/zouari_omar/Documents/Daily/Projects/↵ Astra/project/tests/test.cpp File Reference

```
#include <gtest/gtest.h>
#include "../inc/FaceRecognizer.hpp"
Include dependency graph for test.cpp:
```



Functions

- `int main (int argc, char **argv)`
The Main Testing Function.
- `TEST (TestFn00, Test00)`
Simple test case that demonstrates basic output.

7.67.1 Function Documentation

7.67.1.1 main()

```
int main (
    int argc,
    char ** argv)
```

The Main Testing Function.

This is the entry point for running all the Google Tests in the project.

Parameters

<code>argc</code>	Number of command-line arguments.
<code>argv</code>	Array of command-line argument strings.

Returns

`int` Returns the result of running all tests (0 if all tests passed, non-zero if any failed).

7.67.1.2 TEST()

```
TEST (
    TestFn00 ,
    Test00 )
```

Simple test case that demonstrates basic output.

This test is used to ensure that a basic output function works as expected.

Chapter 8

Examples

8.1 Simple

example using [EmailSender](#) object

example using [EmailSender](#) object

```
std::vector<string> cc = {"<zouariomar20@gmail.com>"};
std::vector<string> attachments = {" /home/zouari_omar/Documents/Daily/Projects/Astra/project/src/main.cpp",
    "/home/zouari_omar/Downloads/test.png"};
EmailData data(
    "<omar.zouari@esprit.tn>", "Test Subject", "<html><body><h1>Hello Omar</h1><p>This is a test
    email.</p></body></html>", cc, attachments);
EmailAuth auth;
EmailSender email(auth);
email.send(data);
```


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